Technical Training

Heat Pump Diagnostics

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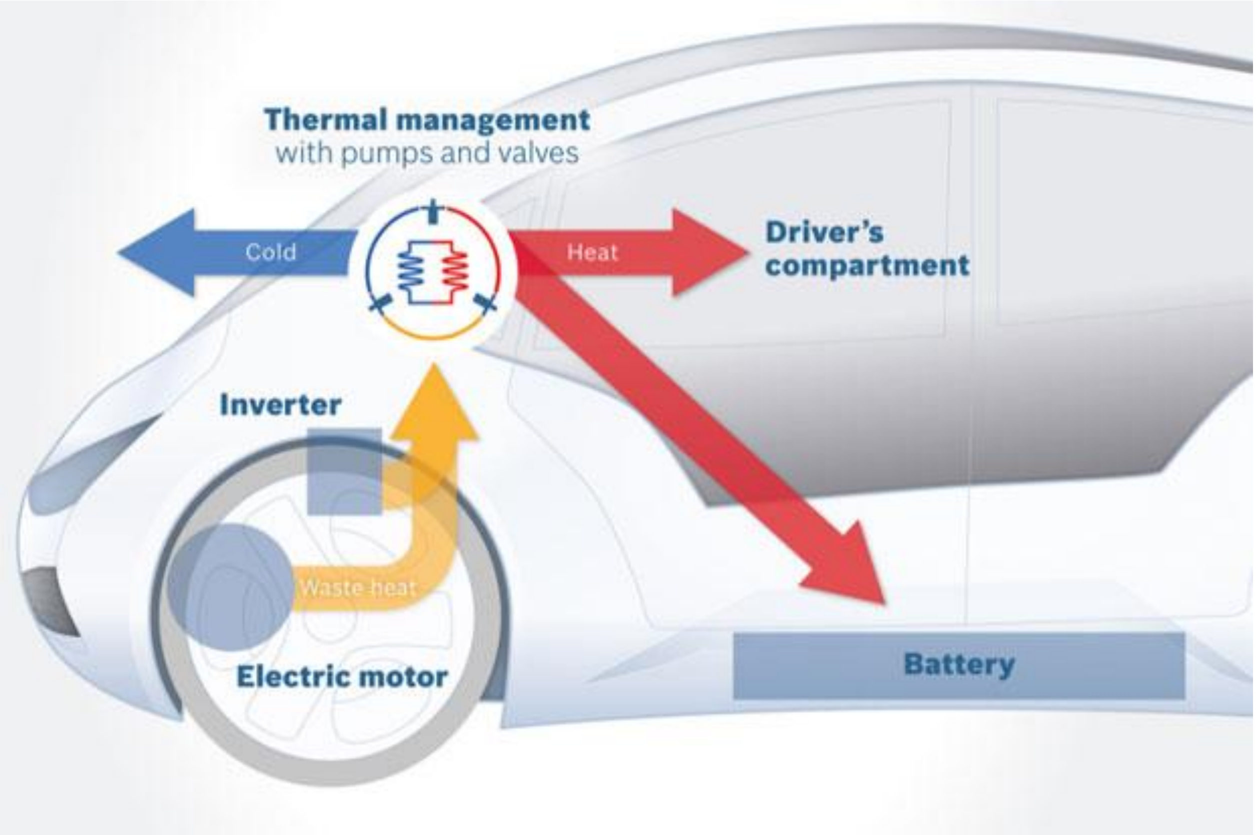
ACADÉMIE Vast auto

Training Objectives:



- Explain the different components of a heat pump system
- Explain the operating principle of this type of system
- Explain the work methodology for a heat pump system
- Explain the diagnostic approach for this type of system

Halocarbons

Item	Canada – Federal	Québec – Provincial
Regulation	Federal Halocarbon Regulations (2022)	Halocarbons Regulation (Q-2, r.29)
Halocarbon release	❌ Prohibited	❌ Prohibited
Mandatory recovery	✓ Yes (servicing and decommissioning)	✓ Yes (servicing and decommissioning)
Leak testing	✓ Required	✓ Required
Qualified personnel	✓ Mandatory (certification)	✓ Mandatory (certification)
Recordkeeping	✓ Mandatory	✓ Mandatory
Main objective	Reducing emissions and protecting the climate	Protecting the environment and the ozone layer

Halocarbons: violations and penalties

Obligation	Example of Violation	Type	Technician	Company	Sections
🎓 **Certification**	Not carrying/displaying certification	SAP	**250 $**	**1 000 $**	61.1 (2°)
🎓 **Certification**	Working without qualification	SAP	**500 $**	**2 500 $**	61.3
🎓 **Certification**	Working without certification	PENAL	**1 000 $ to 100 000 $**	**3 000 $ to 600 000 $**	46 → 62
🌬 **Release to the atmosphere**	Release of halocarbon into the atmosphere	SAP	**2 000 $**	**10 000 $**	61.7
🌬 **Release to the atmosphere**	Uncontrolled release or leak	PENAL	**12 500 $ to 1 000 000 $**	**37 500 $ to 6 000 000 $**	5 → 67.2
🔍 **Leak test**	Test not performed	SAP	**500 $**	**2 500 $**	61.3 (1°)
🔍 **Leak test**	Repeated or serious omission	PENAL	**2 500 $ to 250 000 $**	**7 500 $ to 1 500 000 $**	22 → 64
📒 **Logbook**	Missing or unmaintained logbook	SAP	**250 $**	**1 000 $**	61.1 (4°)
📒 **Logbook**	Logbook not kept or not submitted	SAP	**350 $**	**1 500 $**	61.2
📒 **Logbook**	Serious or repeated non-compliance	PENAL	**2 000 $ to 100 000 $**	**6 000 $ to 600 000 $**	59 → 63

The applicable regulation (sections 64 and 22)

64. Commet une infraction et est passible, dans le cas d'une personne physique, d'une amende de 2 500 $ à 250 000 $ ou, dans les autres cas, d'une amende de 7 500 $ à 1 500 000 $, quiconque:

1° contrevient à l'article 7, au premier ou au deuxième alinéa de l'article 9, à l'article 22, 43, 50 ou 51;

2° fait défaut de faire évaluer la quantité d'halocarbures rejetée lors d'une fuite, conformément au deuxième alinéa de l'article 11.

D. 1091-2004, a. 64; D. 676-2013, a. 8; D. 201-2020, a. 67.

22. Le propriétaire d'un appareil visé au paragraphe 6, 7 ou 9 de l'article 18 doit s'assurer que l'ensemble des composantes qui renferment ou qui sont destinées à renfermer un halocarbure est soumis à une épreuve d'étanchéité au moins une fois par année, avec au plus 15 mois entre chaque épreuve.

Cette épreuve d'étanchéité doit être effectuée à l'aide d'un détecteur de fuites électronique ayant une sensibilité d'au moins 5 g par année pour le type d'halocarbure utilisé.

Le propriétaire d'un appareil visé au premier alinéa ayant été réparé à la suite de la détection d'une fuite doit de nouveau soumettre l'appareil à une épreuve d'étanchéité entre le 30e et le 60e jour après qu'il ait été remis en fonction.

D. 1091-2004, a. 22; D. 201-2020, a. 20; D. 986-2023, a. 13.

https://www.legisquebec.gouv.qc.ca/fr/document/rc/q-2,%20r.%2029

Electronic leak detectors (SAE J2913)

Use only electronic leak detectors that comply with SAE J2913 for use with R-1234yf

The refrigerant recovery logbook (Québec)

Environnement, Lutte contre les changements climatiques, Faune et Parcs
Québec

Registre des travaux de récupération, d'entretien et de démantèlement

Règlement sur les halocarbures (art.59)

Appareil de climatisation de véhicule ou de réfrigération de transport

1.	**Identification**		
NOM DE L'INTERVENANT:			
NUMÉRO D'ATTESTATION DE QUALIFICATION ENVIRONNEMENTALE:			
NOM DE L'EMPLOYEUR:			
ADRESSE:			
VILLE:		CODE POSTAL:	

2.	**Type d'appareil**
TYPE DE VÉHICULE:	
DESCRIPTION DU VÉHICULE (Marque, modèle et année):	
NUMÉRO DE SÉRIE:	
NUMÉRO D'IMMATRICULATION:	

3.	**Récupération de l'halocarbure (s'il y a lieu)**		
A	RÉSULTAT DU TEST D'ÉTANCHÉITÉ DU CONTENANT DE RÉCUPÉRATION:	☐ RÉUSSI	☐ ÉCHOUÉ
B	HALOCARBURE RÉCUPÉRÉ:	TYPE:	QUANTITÉ (Kg)
	L'HALOCARBURE RÉCUPÉRÉ EST-IL CONSERVÉ PAR LE PROPRIÉTAIRE? Si oui, remplir la section C	☐ OUI	☐ NON
C	NOM DU PROPRIÉTAIRE:		
	ADRESSE:		
	VILLE:	CODE POSTAL:	

4.	**Nature des travaux**		
A	DESCRIPTION SOMMAIRE DES TRAVAUX:		
B	**Ajout d'un halocarbure (s'il y a lieu)**		
	RÉSULTAT DU TEST D'ÉTANCHÉITÉ:	☐ RÉUSSI	☐ ÉCHOUÉ
C	HALOCARBURE AJOUTÉ:	TYPE:	QUANTITÉ (Kg)

J'atteste que les renseignements fournis dans ce rapport sont exacts:

Nom de la personne autorisée (lettres moulées):	
Signature de la personne autorisée:	
Date:	

Ministère de l'Environnement, de la Lutte contre les changements climatiques, de la Faune et des parcs

5 years / 2 copies

System Identification

J639 = SAE standard covering safety procedures for pressurized air conditioning systems

J2842 = SAE standard covering rules governing evaporator design

J2845 = SAE standard covering repair standards and the equipment to be used

Automotive refrigerants: comparison

Feature	R-134a	R-1234yf	R-744
Chemical Type	HFC	HFO	CO_2 (carbon dioxide)
Application	Vehicles ~1994–2024	Transition started in 2011 Mandatory from 2024	New high-pressure systems
PRG / GWP	≈ 1430	≈ 4	1
Flammability	Non-flammable	Slightly flammable	Non-flammable
Operating Pressure	Medium	Medium	Very high
Typical High-Side Pressure	150–250 psi	140–240 psi	1000–2000 psi
Energy Efficiency	Good	Very similar to R-134a	Very good in certain conditions
Oil Used	PAG, POE	PAG, POE, specific PVE	POE oil / specific CO_2 oil
Recharge Machine	Standard R-134a	Dedicated R-1234yf machine	High-pressure CO_2 machine
Service Fittings	Standard R-134a	Different for safety	Different high-pressure fittings
Environmental Impact	High	Very low	Very low
Regulation	Being progressively phased out	Current standard	Emerging technology

Refrigerant retrofit: prohibited

In either direction, no matter the reason

- **Converting a system from one refrigerant to another is prohibited**
  – Q-2, r.29, s. 64: modifying or converting a unit = 2 500 $ to 250 000 $ (technician)
  – S. 67.3: non-compliant or unauthorized refrigerant = 1 000 $ to 100 000 $ (technician)

- **R-134a remains available**
  – Vehicles designed for R-134a continue to be serviced with R-134a

Warning "Even if you know some people are doing it": a retrofit is still a violation, no matter what the market does. Both the technician AND the company are exposed to the fines.

Penalties for retrofitting (Q-2, r.29)

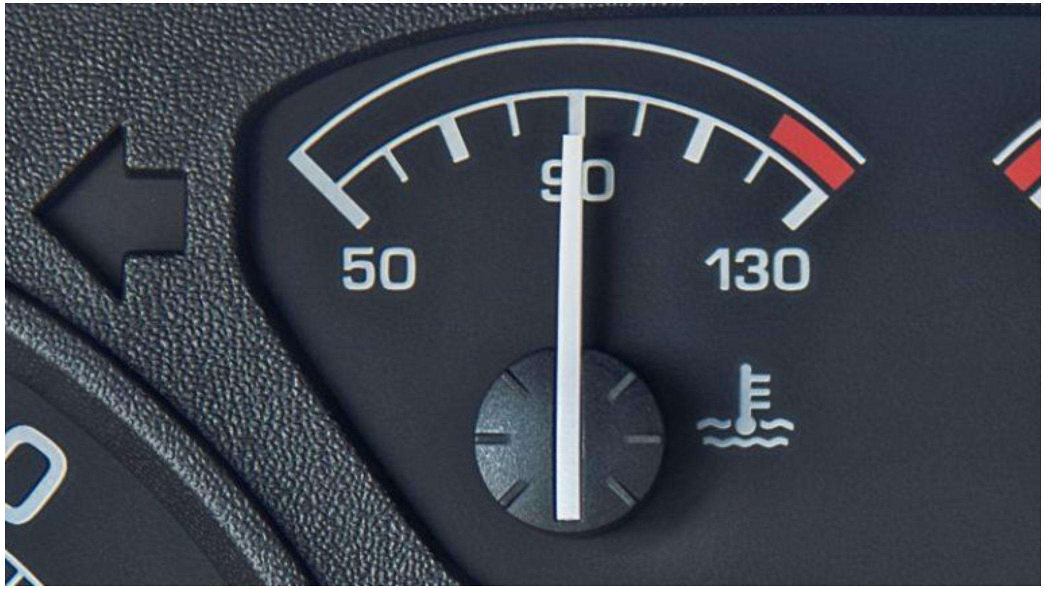
Type of violation	Section of the regulation	Technician (individual)	Company
Modifying or converting a unit in contravention of the regulation	S. 64	**2 500 $ to 250 000 $**	**7 500 $ to 1 500 000 $**
Using a non-compliant or unauthorized refrigerant in a unit	S. 67.3	**1 000 $ to 100 000 $**	**3 000 $ to 600 000 $**

What are the thermal systems in an internal combustion vehicle?

- Coolant, radiator, water pump, heater core...
- Refrigerant, air conditioning
- Oil cooler
- Etc..

What are the thermal **NEEDS** of an internal combustion vehicle?

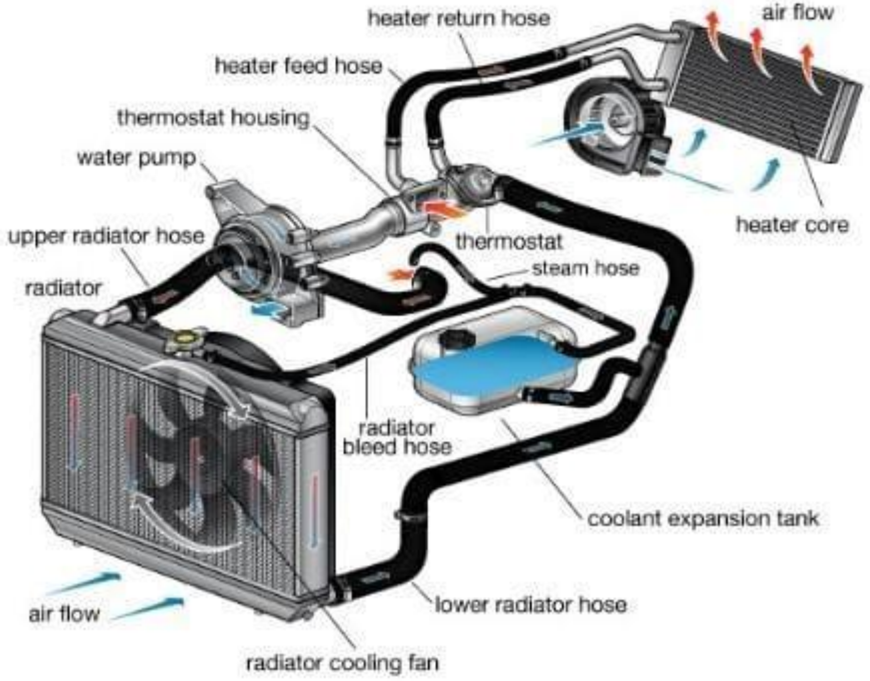
- Keep engine coolant at its ideal temperature, so between 90 and 105 °C

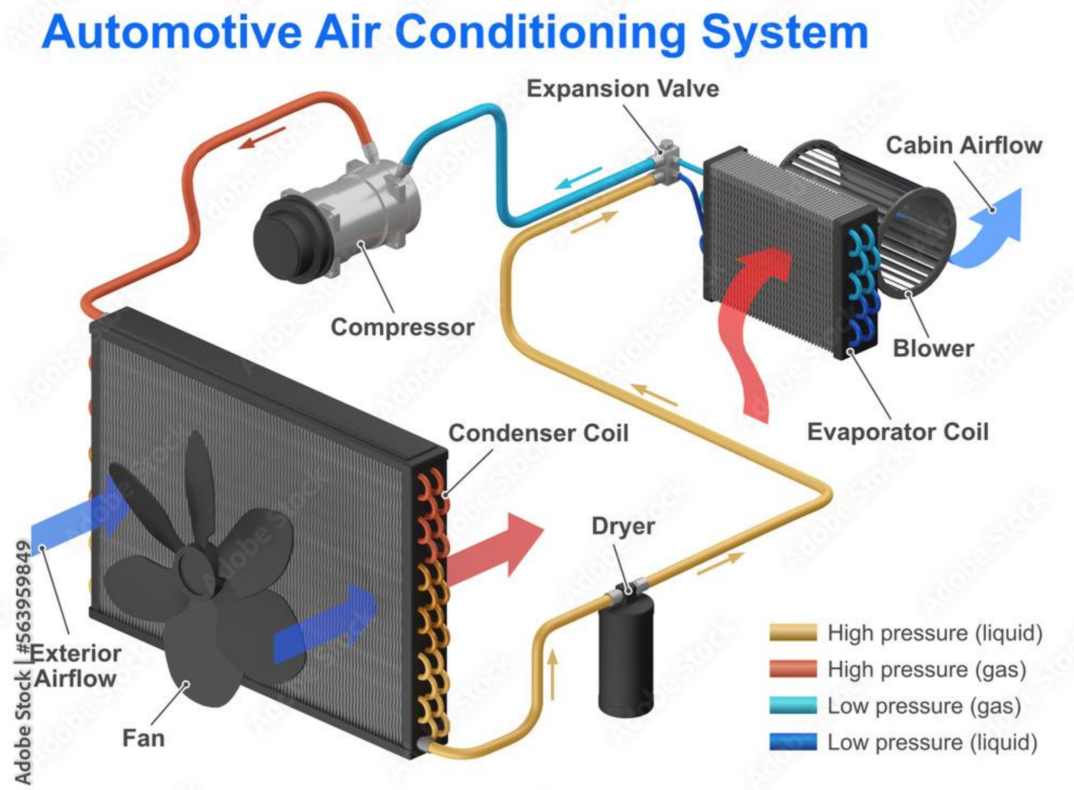


- Keep engine oil at its ideal temperature, so between 70 and 95 °C

What are the thermal COMFORT functions of an internal combustion vehicle?

- Heating the passengers
  - Uses the waste heat produced by the combustion engine (60-70%)
  - Transferred into the cabin through the coolant system and a blower
- Cooling the passengers
  - Refrigerant, air conditioning compressor





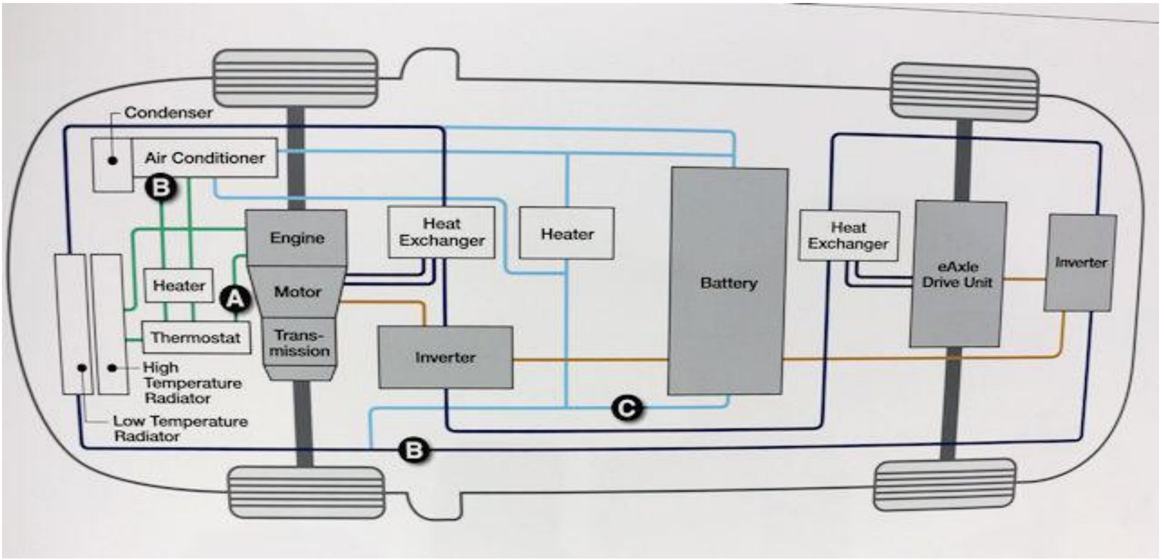
What are the thermal NEEDS of a PHEV?

Combustion need

- Keep engine coolant at its ideal temperature, so between 90 and 105 °C
- Keep engine oil at its ideal temperature, so between 70 and 95 °C

EV need

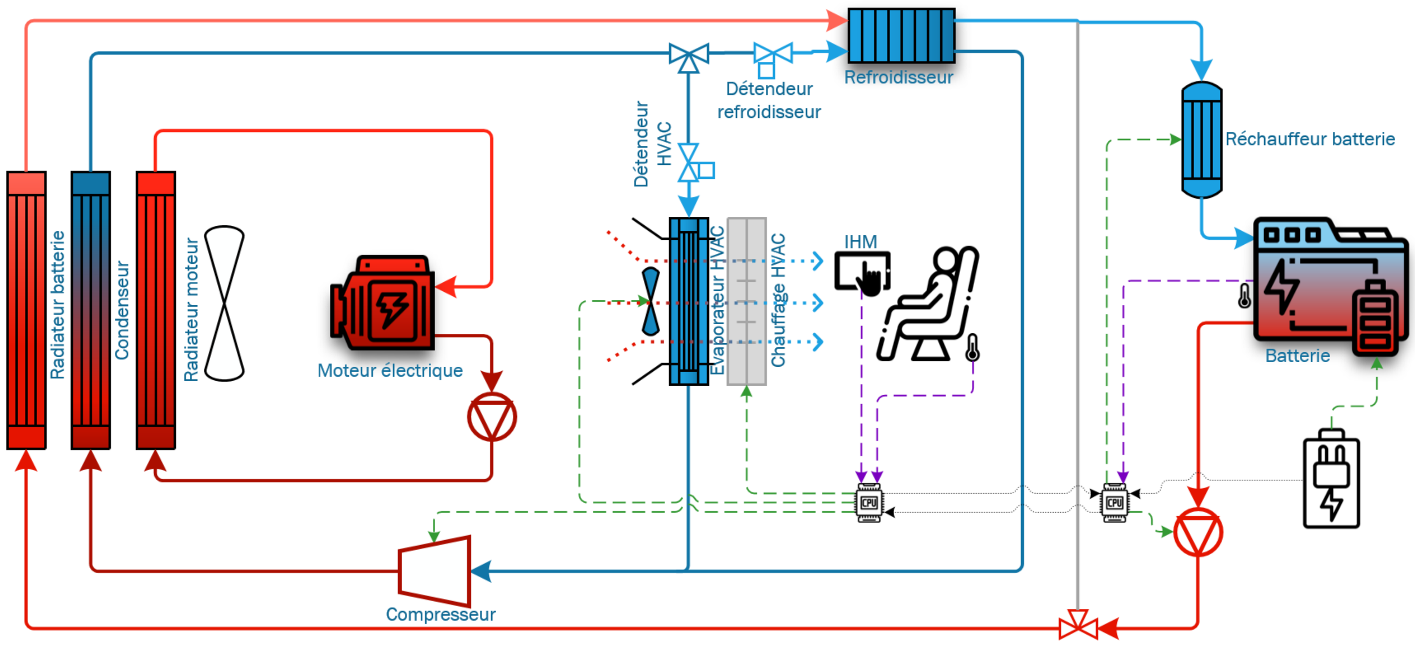
- Maintain the ideal temperature for all HV components,
- HV battery 15-35 °C (ideal ~25 °C), electric motor (below its thermal limit), onboard charger, HV-LV converter)



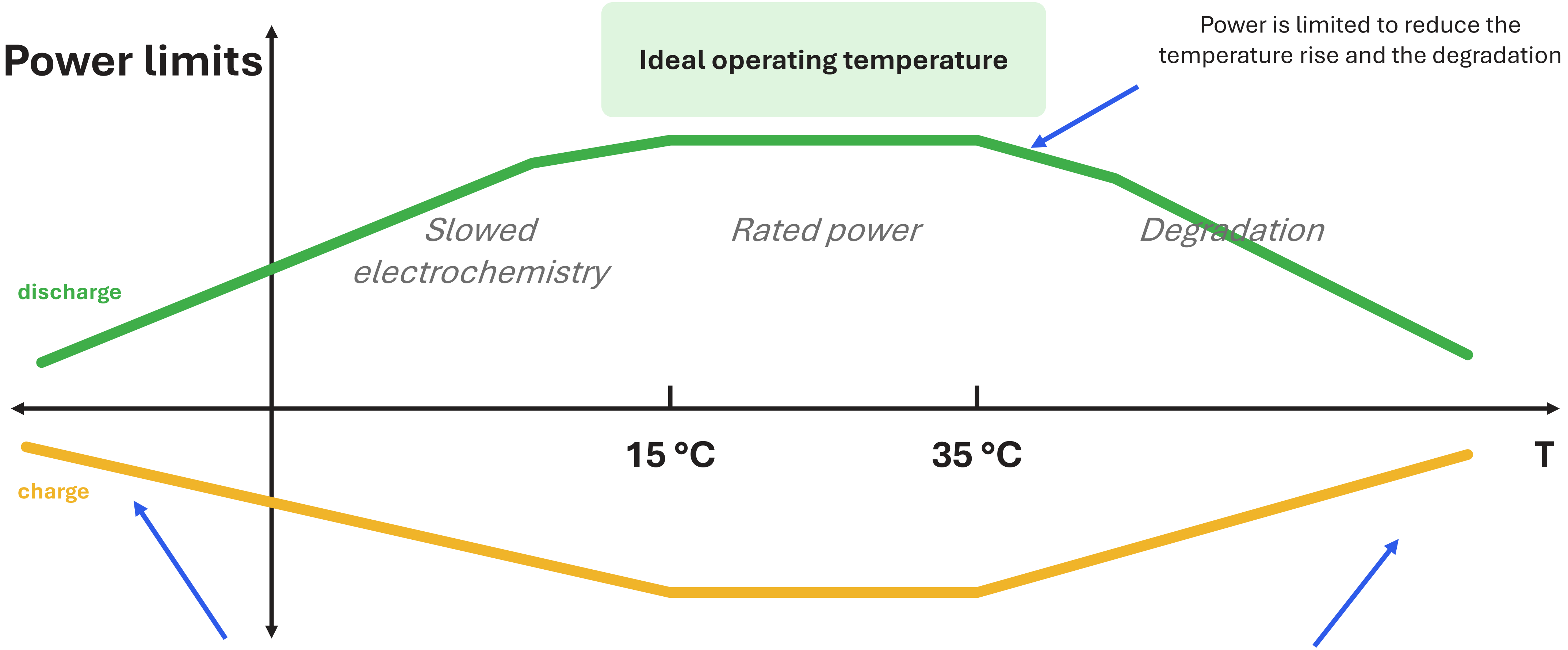
What are the thermal NEEDS of an EV?

- Maintain the ideal temperature for all HV components (HV battery, electric motor, onboard charger, HV-LV converter)

> 30°C



What are the thermal **NEEDS** of an EV?



Dictates the power available on a cold start

Also limits electric range

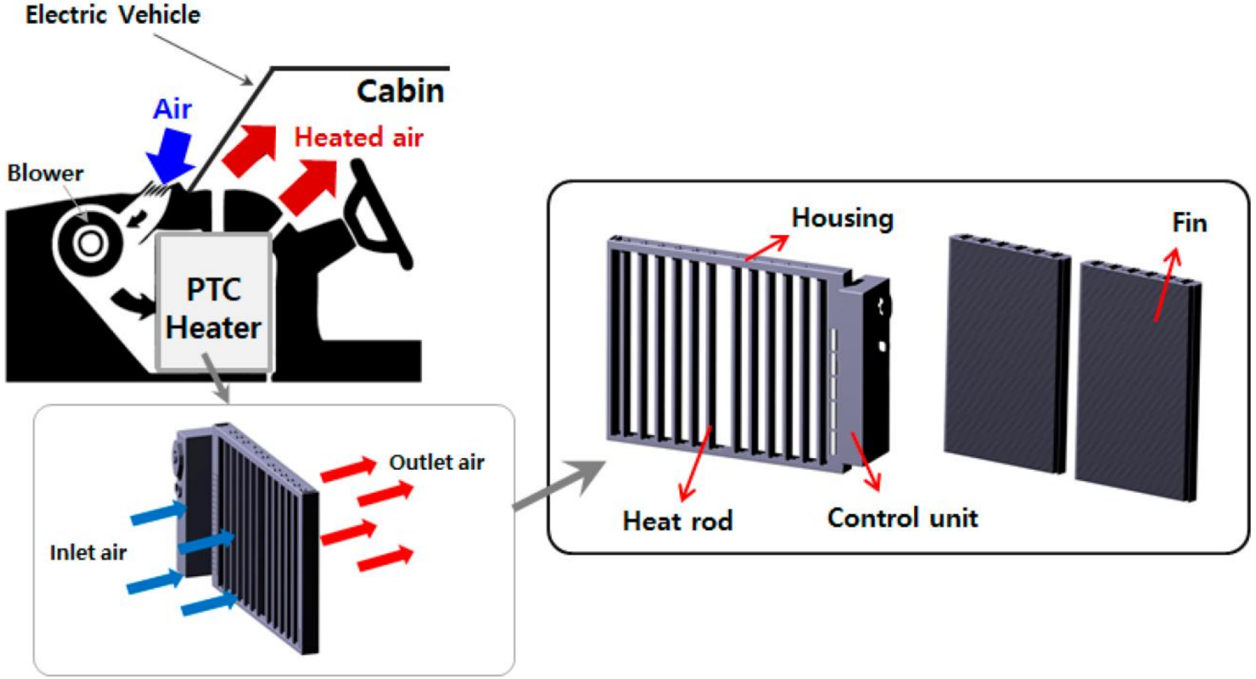
Dictates sizing based on the rate of power and energy loss

How do we meet the thermal NEEDS of an EV?

- Glycol/coolant
  - Cooled by the radiator
  - Heated by a PTC heating element
- Refrigerant/air conditioning
  - Introduction of the chiller
  - Introduction of the heat pump

🔑 **KEY POINT** A li-ion battery likes a narrow window: 15-35 °C, ideal around 25 °C. Above 40 °C degradation accelerates; below 0 °C, fast charging risks lithium plating. The whole thermal system exists to keep it inside that zone.

What is a PTC heating element?



- A PTC heater is an electric heater with a positive temperature coefficient.
- Simple definition: PTC stands for **Positive Temperature Coefficient**. It means that the hotter the element gets, the higher its electrical resistance, which automatically limits the current.

•Cold: low resistance → current flows easily → it heats up fast
•Hot: resistance rises → current drops → the temperature stabilizes
Result: **self-regulating heating**, with no risk of overheating.

Glycol/coolant with a PTC heating element

The PTC heating element: an electric heating element used to warm the coolant in an EV or hybrid. It works like a smart heating element :

- When the coolant temperature is low, it produces more heat, so it draws more power

- When the coolant temperature rises, it automatically reduces its output, with no risk of overheating

In an EV, the PTC heating element quickly warms up the high-voltage battery, the cabin or the electronic components.

Why a heat pump?

HEAT PUMP SYSTEMS ARE ESSENTIAL ON ELECTRIC VEHICLES:

- Heat pumps help EVs use energy more efficiently for heating and cooling, which directly affects range
- Thermal management protects the life and the performance of the HV battery, especially in cold weather
- Heat pumps are now common on many EVs. Technicians therefore have to be ready to work on these systems
- Heat pumps bring in several components (HV compressor, EXV, chiller, coolant loop, refrigerant loop and PTC)

HEAT PUMP EXPLANATION

What is a heat pump?

A heat pump in an EV moves heat rather than producing it.

It can heat or cool:

- The high-voltage battery, to hold it at its optimal temperature.
- The electronic components, to prevent overheating.
- The cabin, for passenger comfort.

It is energy efficient because it uses heat that is already available (outside air or internal heat) instead of burning a lot of electricity to create heat.

HEAT PUMP EXPLANATION

Why does it matter?

Because it uses less energy than traditional heating systems, a heat pump preserves more battery range, especially in winter

HEAT PUMP EXPLANATION

How does it work?

Even when the outside temperature is low, the heat pump can extract heat from the outside air (or from another source such as the ground or water) and transfer it inside.

The system works on the principle of heat transfer through a refrigerant:

1. The refrigerant absorbs heat from outside (even when it is cold, there is always thermal energy).
2. It is compressed, which raises its temperature.
3. That heat is then released into the cabin.

And if needed, the cycle can change its path, which cools the cabin the same way a conventional air conditioner does.

HEAT PUMP EXPLANATION

Why does it matter?

- **Energy efficiency** : unlike conventional electric resistance heaters, which turn electricity directly into heat (at roughly 100% efficiency), the heat pump moves existing heat from one place to another, which produces 2 to 4 times more heat per unit of electricity consumed.

    - PTC heater: 1 kWh of electricity = 1 kWh of heat (efficiency ≈ 100 %)
    - Heat pump: 1 kWh of electricity = 2 to 4 kWh of heat depending on temperature (COP > 1)

HEAT PUMP EXPLANATION

- **Battery preservation** : in winter, traditional heating can drain the battery quickly and cut the vehicle's range. The heat pump limits that extra draw, extending the distance the vehicle can cover between charges.

- **Versatility**: it heats, but it can also cool the cabin and certain electronic components, while staying efficient even in cold weather.

DRAWBACKS OF THE HEAT PUMP?

Are there any drawbacks to using a heat pump system?

The heat pump is very efficient and environmentally friendly, but it does have a few drawbacks and limits, mostly depending on use and conditions.

1. **Efficiency limited by extreme cold**
   - At very low temperatures (e.g. **-20 °C** or below), the heat pump captures less heat from the outside air.
   - It may need a backup heater (PTC heater).

   Result: energy consumption can rise sharply in winter.

2. **Performance varies with humidity and with the heat source**
   - If the outside air is very dry or very humid, the heat pump can be less efficient.
   - Under some conditions, the heat flow may not be enough to warm things up quickly.

3. **Maintenance and complexity**
   - Multiple components: expansion valve, evaporator, condenser...
   - More parts → more chances of failure or of specific servicing.
   - Repairs can be expensive.

Heat Pump COP vs Refrigerant

	R134A	R1234YF	R744
-10°C	1.6–2.4	1.5–2.3	1.7–2.6
-20°C	1.2–1.8	1.1–1.7	1.3–2.0
-30°C	1.0–1.4	1.0–1.3	1.1–1.6

	R134A + Coolant PTC	R1234YF + Coolant PTC	R744 + Coolant PTC
-10°C	1.4–2.1	1.3–2.0	1.6–2.4
-20°C	1.1–1.6	1.0–1.5	1.3–1.9
-30°C	0.9–1.3	0.9–1.2	1.1–1.6

	R134A + Coolant PTC + Chiller + Recovery	R1234YF + Coolant PTC + Chiller + Recovery	R744 + Coolant PTC + Chiller + Recovery
-10°C	1.6–2.4	1.5–2.3	1.6–2.5
-20°C	1.5–2.2	1.4–2.1	1.5–2.3
-30°C	1.4–2.0	1.3–1.9	1.4–2.1

How does all of this work?

Latent heat

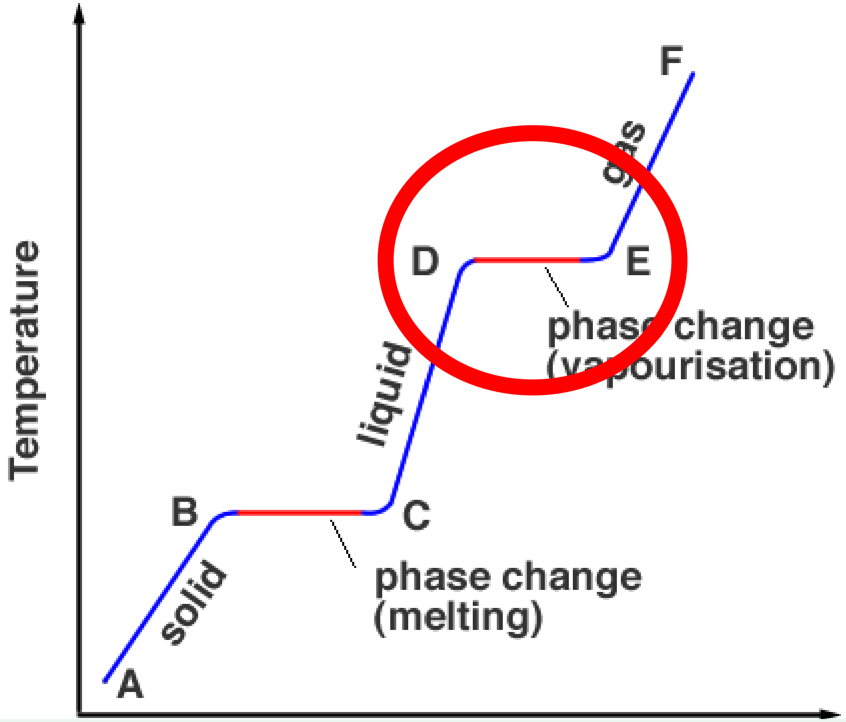
What is latent heat?

- **Latent heat** is a thermodynamic concept describing the energy needed to change the state of a substance without changing its temperature. In other words, it is the heat absorbed or released during a phase change such as melting, vaporization or solidification.

In heat pumps, this is what makes refrigerants so powerful:

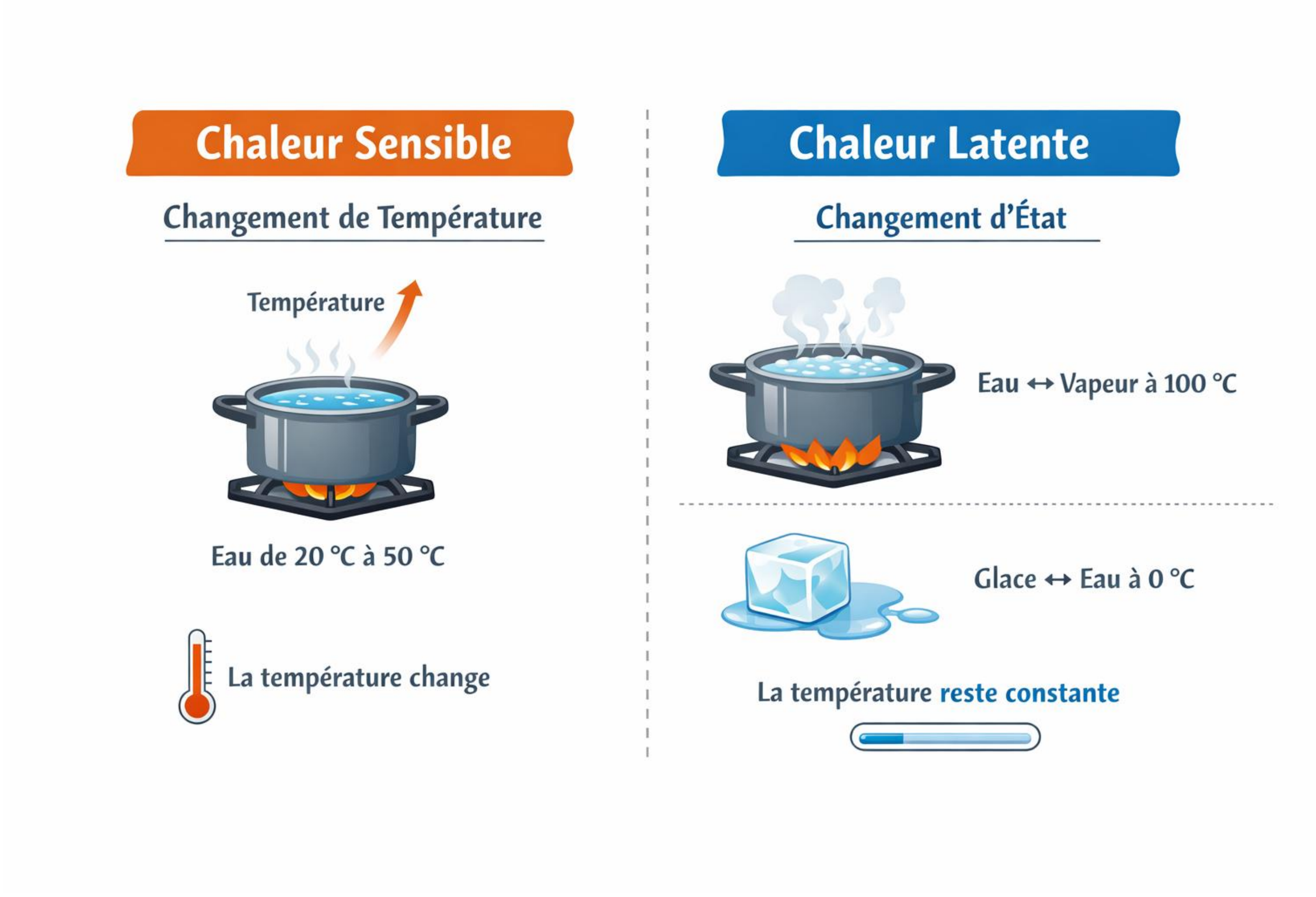
- Refrigerants absorb a lot of heat (energy) when they evaporate and release it when they condense, all while staying at the same temperature

- That "invisible" energy transfer is what lets electric vehicles heat and cool efficiently



💡 **TIP** Sweat evaporating cools the skin without changing the temperature of the water: that is latent heat. The heat pump uses exactly that phenomenon, but in a closed, reversible circuit.

Sensible heat and latent heat



Latent heat and the heat pump

1. Principle: using the latent heat of the refrigerant

The heat pump uses a **refrigerant** that changes state (liquid ↔ gas) in a closed circuit.
At every change of state, the refrigerant **absorbs** or **releases** a large amount of energy **without changing temperature**: that is exactly ***latent heat***.

2. The two key stages where latent heat comes into play

A) A) Evaporation (outside) → the refrigerant captures latent heat.

The refrigerant arrives as a **low-pressure liquid**. It **evaporates** in contact with the outside air (even when it is cold!). As it evaporates, it **absorbs a large amount of latent heat**.
That is how the heat pump recovers the heat present in the environment.
Even at –10 °C, the air holds enough energy to evaporate the refrigerant.

B) Condensation (inside) → the refrigerant releases latent heat.

After compression, the refrigerant is a **hot gas**. It flows through the **condenser** where it turns back into a **liquid**. As it condenses, it **releases the latent heat** absorbed outside, plus the heat added by compression.
That released heat is what warms the cabin of an EV.

Latent heat and the heat pump

3. Why does latent heat make the heat pump so efficient?

Because liquid ↔ gas transitions can transfer **a huge amount of energy**:

- The latent heat of vaporization is very high.
- For every 1 kWh of electricity used by the compressor, the heat pump can move **2 to 4 kWh** of heat → that is the **COP**.

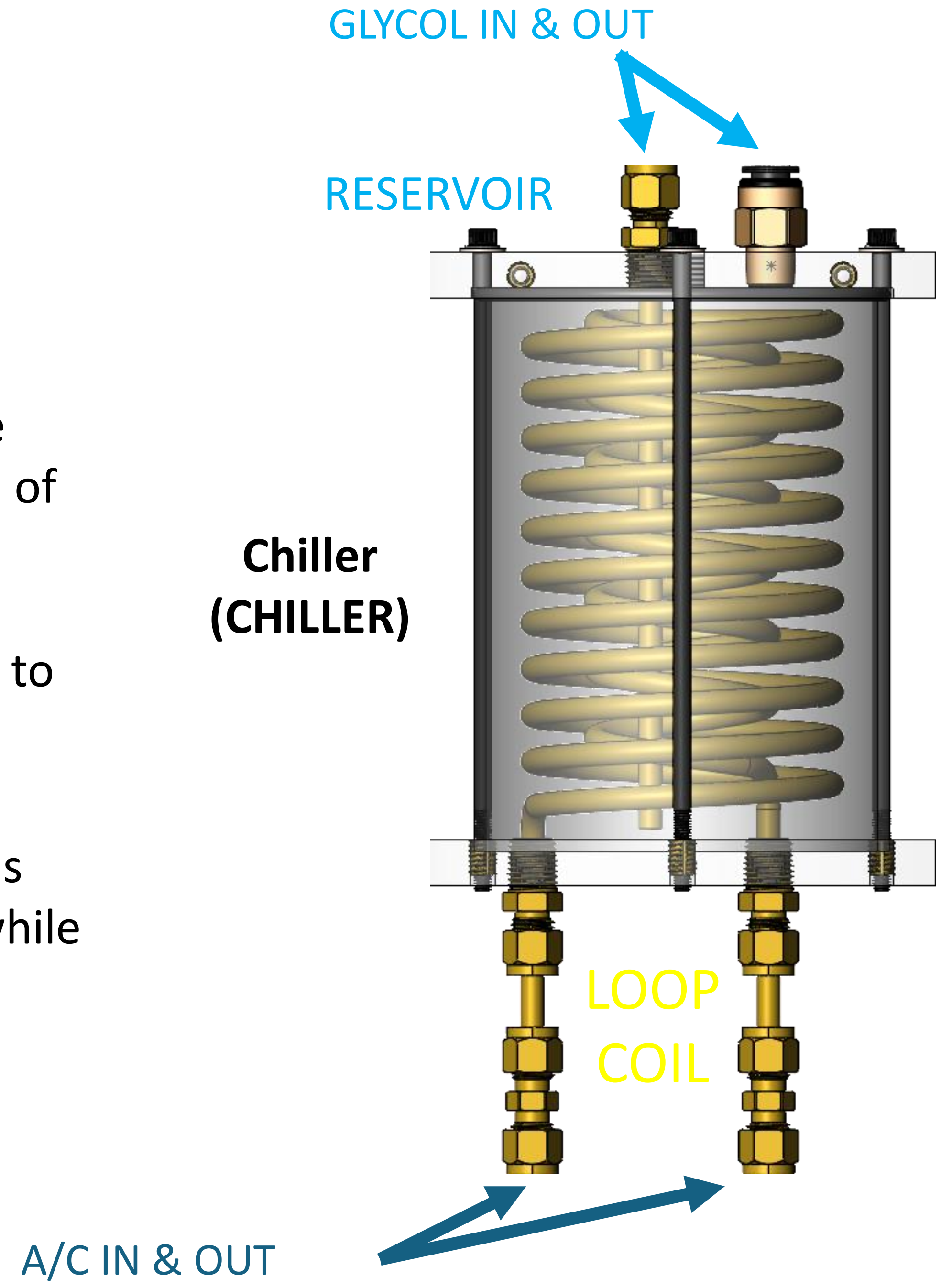
Without latent heat, the heat pump would perform far worse.

THE CHILLER EXPLAINED

THE CHILLER: a key component that transfers heat between two systems (coolant and refrigerant) inside an EV or hybrid:

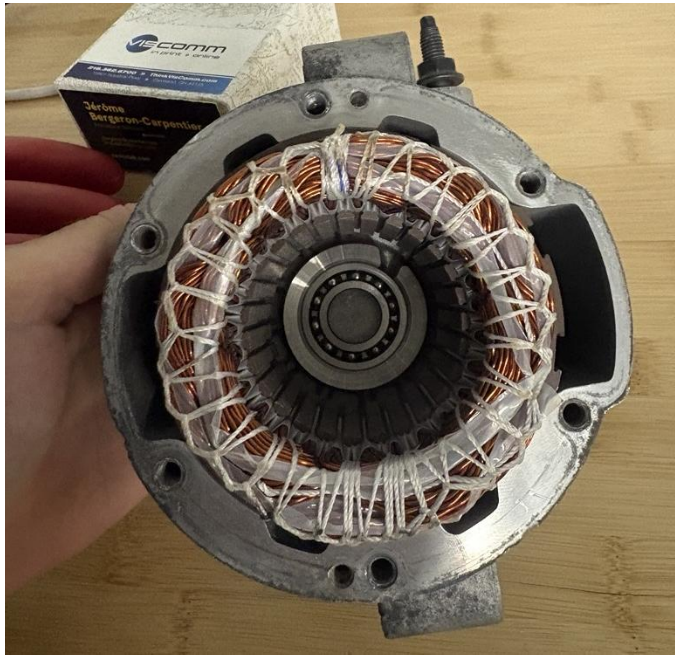
- the refrigerant loop (used to heat/cool the HV battery and the cabin) and the coolant loop (used to manage the temperature of the battery, the cabin and the electronic components)
- Think of the chiller as a bridge that moves heat from one loop to the other

It helps the vehicle stay warm in winter, cool in summer, and holds the HV battery at the right temperature (15-35 °C, ideal ~25 °C) while using energy efficiently



Mode "lossy" deliberate loss

Heat recovery by running the electric motors with deliberate losses (current injected without producing useful torque → heat in the windings)



The vapour-compression refrigeration cycle

Overview of the 4 stages of the cycle

- **The cycle uses a refrigerant circulating in a closed loop that changes phase**
  - Liquid → gas: absorbs heat (evaporation = cooling)
  - Gas → liquid: releases heat (condensation = heating)
- **The 4 stages of the cycle:**
  - Stage 1: COMPRESSION → the gas is compressed, its temperature rises
  - Stage 2: CONDENSATION → the hot gas gives up its heat and becomes liquid
  - Stage 3: EXPANSION → the liquid expands, the temperature drops sharply
  - Stage 4: EVAPORATION → the cold liquid absorbs ambient heat and turns back into gas
- **The direction of the cycle determines the operating mode:**
  - HEATING mode: evaporator outside, condenser in the cabin
  - AIR CONDITIONING mode: evaporator in the cabin, condenser outside
- **Key principle: the compressor is the 'engine' of the cycle**
  - No compressor = no pressure difference = no heat transfer
  - Compressor speed determines the thermal capacity of the system

Stage 1 – Compression

Heart of the cycle: the electric compressor

- **Inlet state: refrigerant gas, low pressure, low temperature**
  - R-1234yf example: suction pressure ~30 to 60 PSI, temperature ~30 °F to 60 °F
- **The compressor draws in the gas and raises its pressure**
  - Scroll compressor: the most common type today
  - Two nested scrolls compress the gas continuously and quietly
  - Variable speed: 800 to 9,000 rpm depending on thermal demand
- **Outlet state: refrigerant gas, high pressure, high temperature**
  - R-1234yf example: discharge pressure ~200 to 370 PSI, temperature ~130 °F to 180 °F
- **The mechanical work supplied to the compressor turns into thermal energy**
  - Q_compression = W_compressor (electrical energy consumed)
  - It is the only external energy input in the system
- **Technical precautions:**
  - Always keep enough oil in the compressor
  - Avoid drawing in liquid refrigerant (liquid slugging → mechanical damage)
  - The compressor sits on the HV circuit (400–800 V) → electrical isolation is essential

Stage 2 – Condensation

The hot refrigerant releases its heat

- **The hot high-pressure refrigerant gas enters the condenser**
- **In HEATING mode (heat pump):**
  - The condenser is the heat exchanger on the CABIN side
  - The refrigerant gives up its heat to the air blown into the cabin
  - The cabin air is warmed → thermal comfort for the occupants
- **In AIR CONDITIONING mode:**
  - The condenser is the outside exchanger (front of the vehicle)
  - The refrigerant gives up its heat to the outside ambient air
  - An electric fan forces air through the exchanger
- **Phase change: hot vapour → hot liquid (at high pressure)**
  - It all happens at a nearly constant temperature (saturation temperature)
  - Energy released = latent heat of condensation of the refrigerant
  - R-1234yf example at 290 PSI: condensation at about 160 °F (71 °C)
- **Types of exchangers used in automotive applications:**
  - Air-to-refrigerant condenser: aluminum tubes and fins (outside, cabin)
  - Refrigerant-to-coolant condenser (chiller): brazed plates (battery)

Stage 3 – Expansion

Pressure and temperature drop

- **The high-pressure liquid refrigerant enters the electronic expansion valve (EXV)**
- **The EXV creates a controlled restriction (variable orifice)**
  - Pressure drops sharply from the high side to the low side
  - Example: from 290 PSI 160 °F (71 °C) → 43 PSI 45 °F (7 °C) for R-1234yf
- **Corresponding temperature drop (the refrigerant's P-T law)**
  - The temperature can fall by 20 °C, 30 °C or more
  - That temperature drop is what makes it possible to absorb heat outside
- **Start of vaporization: the liquid begins to turn into vapour**
- **The EXV modulates the flow to hold the optimal superheat at the evaporator outlet**
  - Superheat too low: risk of liquid returning to the compressor
  - Superheat too high: loss of cooling capacity
- **Types of expansion devices in automotive applications:**
  - EXV (electronically controlled expansion valve): the most common on recent EVs
  - TXV (thermostatic): used on some older vehicles
  - Orifice tube: simple systems, fixed flow (uncommon on modern vehicles)

Stage 4 – Evaporation

Absorbing ambient or waste heat

- **The cold low-pressure refrigerant enters the evaporator**
- **In HEATING mode (heat pump):**
  - The evaporator is the outside exchanger (front of the vehicle)
  - The refrigerant absorbs heat from the ambient air (even at -15 °C or -20 °C)
  - The 'chilled' outside air leaves the exchanger (frost may form)
- **In AIR CONDITIONING mode:**
  - The evaporator is in the cabin
  - The refrigerant absorbs BOTH heat and moisture from the cabin air
  - The cabin air is cooled and dehumidified
- **Phase change: cold liquid → vapour (at low pressure)**
  - Heat absorbed = latent heat of evaporation of the refrigerant
  - Evaporation temperature is tied to the low pressure (P-T chart)
  - R-1234yf example at 32 PSI: evaporation at about 32 °F (0 °C)
- **The gas produced returns to the compressor → the cycle starts again**
- **Alternative heat sources in heat pump mode:**
  - Waste heat from the motor / inverter (through the coolant circuit)
  - Waste heat from the battery

Automotive refrigerants

Evolution and regulation in Canada

- **R-12 (Freon / dichlorodifluoromethane) — high GWP, high ODP**
  - Used until 1994 on North American vehicles
  - Banned under the Montreal Protocol for destroying ozone
  - Still found on older vehicles; requires conversion or replacement
- **R-134a (tetrafluoroethane / HFC-134a) — GWP = 1,430**
  - North American standard from 1994 to 2024
  - Banned on new Canadian/U.S. models since 2024.
  - Still very common in North America (millions of vehicles in service)
- **R-1234yf (HFO-1234yf) — GWP = 4**
  - Mandatory on new NA vehicles since 2024 (SAE J2842 standard)
  - Slightly flammable (class A2L); requires SAE J2845 certified equipment
  - Compatible oil: PAG 46yf, 100yf, etc. (do not interchange with R-134a)
- **R-744 (CO_2) — GWP = 1**
  - Used sparingly by BMW, Korean makes, Audi; very high pressure (up to 2000 PSI)
  - Very rare on today's North American market; dedicated equipment required

Key component: the electronic expansion valve (EXV)

Precise control of refrigerant flow

- **Role: expand the high-pressure liquid refrigerant → low pressure and low temperature**
  and modulate the flow to optimize superheat at the evaporator outlet
- **How it works:**
  - Needle driven by a stepper motor or a solenoid
  - Position from 0 to 100% open (readable on an OEM scan tool)
  - Commanded by the vehicle's thermal module based on measured superheat
  - Response time: < 1 second; accuracy: ±0.5 °C of superheat
- **Advantages of the EXV over a thermostatic TXV:**
  - Independent control of each branch of the system
  - Can be calibrated with a scan tool (zero initialization)
  - Live data available (% open, calculated superheat)
- **Number of EXVs depending on system complexity:**
  - Simple systems (hybrids): 1 to 2 EXVs
  - Integrated systems (modern EVs): 3 to 6 EXVs (cabin, battery, outside, recovery)
- **Diagnostics:**
  - Check the % open live (it must vary with demand)
  - Check that pressure changes as you sweep the % open

Key component: the sensors

Monitoring and controlling the system

- **High-side pressure sensor (HP):**
  - Location: discharge line (between compressor and condenser)
  - Signal: 0.5 to 4.5 V (3-wire ratiometric sensor)
  - Range: 0 to 500 PSI (R-1234yf / R-134a)
- **Low-side pressure sensor (LP):**
  - Location: suction line (between evaporator and compressor)
  - Signal: same as the HP sensor
  - Range: 0 to 150 PSI (R-1234yf / R-134a)
- **Discharge temperature sensor:**
  - Location: compressor outlet
  - Protects the compressor against overheating (typical limit: 250-275 °F) (120-135 °C)
- **Ambient temperature sensor:**
  - Location: front of the vehicle (ahead of the outside exchanger)
  - Drives the mode-switching logic of the heat pump
- **Evaporator / defrost temperature sensor:**
  - Triggers the defrost cycle if T < (23-25 °F) (-3 °C to -5 °C) (depending on the manufacturer)
- **Battery temperature sensor:**
  - Several probes inside the high-voltage battery; manages battery heating/cooling mode
- **All sensors are monitored live → available through a scan tool**

Thermal management: ICE vehicle vs electric vehicle

Why the heat pump is essential on an EV

▶ Internal combustion vehicle (ICE):

– The combustion engine rejects ~65–70% of its energy as heat
– That waste heat is used free of charge to warm the cabin
– Engine coolant circuit → heater core → cabin blower
– Energy cost of cabin heating: near zero (the energy was lost anyway)

▶ Electric vehicle (EV):

– The electric motor is very efficient (~95%) → very little waste heat
– At -15 °C, an electric motor may produce only 100–300 W of heat
– To heat the cabin, energy has to be drawn from the high-voltage battery

▶ Impact of PTC heaters (Positive Temperature Coefficient):

– Electric resistance: 3 to 7 kW drawn continuously in cold weather
– On a 75 kWh battery: 30–50% range loss at -15 °C
– Real example: 400 km of range → 200–280 km in winter with PTC alone

▶ The heat pump solves that problem:

– COP 2.5 to 3.5 → delivers 2.5 to 3.5 kWh of heat for 1 kWh consumed
– Range loss cut to 10–25% (depending on temperature and vehicle)

Coefficient of Performance (COP)

Measuring heat pump efficiency

▶ **Definition:**

– COP_heating = Q_heating / W_electrical = heat delivered / energy consumed
– COP_cooling = Q_cooling / W_electrical

▶ **Sample calculation:**

– Compressor: 2 kW (W_elec)
– Heat absorbed outside: 4 kW (Q_ext)
– Heat delivered to the cabin: 6 kW (Q_cabin = 2 + 4)
– COP_heating = 6 / 2 = 3.0

▶ **Comparison of heating systems:**

– PTC heater (resistance): COP = 1.0 (1 kWh consumed = 1 kWh of heat)
– Heat pump at 0 °C: COP ≈ 2.5–3.5
– Heat pump at -15 °C: COP ≈ 1.5–2.2 (degraded by the low temperature)
– Heat pump at -25 °C: COP ≈ 1.0–1.5 (close to the limit)

▶ **Impact on EV range:**

– Without a heat pump (PTC alone) at -15 °C: 40–50% range loss
– With a heat pump at -15 °C: 15–25% range loss (a significant saving)
– Example: Tesla Model Y Long Range — 150 km of extra range in winter

▶ **Factors affecting COP:**

– Outside temperature (the biggest factor)
– Compressor speed · Fouled heat exchangers
– Refrigerant charge level · Oil quality

Measuring thermal output in the shop

Three simple tools: a thermometer, an anemometer and a ruler

▶ Why measure?

– Turn a subjective complaint ("it doesn't heat") into an objective, comparable number
– Quantify performance BEFORE taking anything apart

▶ The formula: $P = Qv \times 0.34 \times \Delta T$

– Qv: air flow (m^3/h) · 0.34: volumetric heat capacity of air ($Wh/m^3 \cdot °C$)
– ΔT: temperature at the vents − outside temperature
– Actual COP = thermal power ÷ electrical power consumed (read on the scan tool or the infotainment screen)

▶ The method, in 6 steps:

– 1. Vent area: measure each outlet vent (width × height, in metres) and add them up → m^2
– 2. Settings: heat at maximum, blower at maximum, vent mode, all vents open
– 3. Air speed: with the anemometer, measure the speed at the vents and average it (m/s)
– 4. Air flow: Qv = total area × average speed × 3,600 → m^3/h
– 5. Thermal power: $P = Qv \times 0.34 \times \Delta T$ → watts
– 6. COP: thermal power ÷ electrical power consumed by the compressor

Example: 2021 Toyota RAV4 Prime (heating mode)

Step	Measurement	Result
Vent area	4 identical vents of 9 × 4.5 cm (0.00405 m^2 each)	0.0162 m^2
Air speed	Average of the 4 vents with the anemometer	4.5 m/s
Air flow Qv	0.0162 × 4.5 × 3,600	262 m^3/h
ΔT	57 °C at the vents − 17 °C at the outside	40 °C
Thermal power	262 × 0.34 × 40	≈ 3,570 W
COP	3,570 W ÷ 1,550 W electrical	2.3

▶ Reading the result:

– COP between 2 and 3: healthy heat pump (depending on outside temperature)
– COP close to 1: the heat pump is not contributing, only the PTC is working → check the EXV, the mode valve, the refrigerant charge
– Low air flow (Qv): plugged cabin filter, weak blower, frosted or fouled evaporator
– Low ΔT with good flow: low refrigerant charge, stuck EXV, inefficient inside exchanger

💡 **TIP** The "summer = winter" rule: the power measured in cooling plus dehumidification in the summer matches the heating capacity available in the winter. Example: 2,000 W of cooling + 1,000 W of dehumidification ≈ 3,000 W of heating. So a heating system can be validated… in the middle of July.

Thermal challenges specific to EVs

Canadian winters — a challenge of their own

▶ **Extreme temperatures**
- extreme temperatures of -20 °C to -40 °C in winter
- These temperatures exceed the optimal range of most heat pumps (effective down to -20 °C)

▶ **The problem of the whole vehicle cooling down overnight**
- Cold battery: internal resistance rises → available power drops
- Oil viscosity rises → more compressor wear at start-up
- Solution: remote preconditioning (plugged in or on battery)

▶ **Three thermal challenges at once:**
- 1. Heat the cabin for occupant comfort
- 2. Heat the battery to restore performance
- 3. Defrost the outside exchangers (frost = loss of efficiency)

▶ **Heat pump systems handle these 3 challenges simultaneously**
- Multi-circuit architecture with multiple heat sources
- Dynamic priority based on demand (critical battery > cabin comfort)

▶ **Specific case: DC fast charging in cold weather**
- A fast charger requires a battery above 10–15 °C to accept high-power charging
- The heat pump preheats the battery to shorten charging time

The defrost cycle of the outside exchanger

Keeping the heat pump efficient in frosting conditions

▶ Why does frost form on the outside exchanger?

– In heat pump heating mode, the outside exchanger is the evaporator
– Its surface temperature drops below 0 °C (typically -5 °C to -15 °C)
– Humid air in contact with that surface → frost forms
– Critical frosting range: ambient temperature between -5 °C and +5 °C with relative humidity above 70%

▶ Impact of frost:

– Restricts air flow → reduces the ability to absorb heat
– In severe cases: complete blockage → low-side pressure collapses → compressor shuts down

▶ Detecting frost:

– Surface temperature sensor on the outside exchanger (< -3 °C → defrost triggered)
– Or: monitoring low-side pressure (an unexplained rapid drop → indirect detection)

▶ Defrost method 1: cycle reversal

– The cycle is temporarily reversed (A/C mode) → the outside exchanger becomes a hot condenser
– The frost melts in 2–5 minutes; the cabin is kept warm by the cabin chiller

▶ Defrost method 2: integrated electric resistance

– Less common; resistive wire on the exchanger; consumes more energy

⚠ **WARNING** A defrost cycle is not a fault: during one, cabin heating drops temporarily and vapour may escape at the front of the vehicle. That is normal operation: do not diagnose a refrigerant leak on that observation alone.

The electric compressor — in depth

Architecture, operation and integration in North American EVs

► Construction of a typical e-compressor (e.g. Hanon Systems, Mahle, Denso, Valeo)

– Sealed aluminum housings; high-pressure axial shaft seal
– Integrated PMSM motor (Permanent Magnet Synchronous Motor)
– Scroll mechanism: fixed scroll + orbiting scroll → smooth, continuous compression
– Integrated inverter: converts the HV battery supply (400–800 V DC) into three-phase AC

► Control and communication:

– Receives speed commands over the CAN bus from the thermal management module
– Data: actual speed, motor temperature, current draw, fault codes
– Automatic speed limiting if: discharge temp > limit · high-side pressure > threshold · loss of isolation

► Characteristics by North American manufacturer:

– Tesla: 400 V compressor (Model 3/Y) or 800 V (Plaid) — supplier: Dana/Sanden
– GM Ultium: 400 V or 800 V compressor (hummer, silverado, etc.) — supplier: Mahle or Hanon
– Ford F-150 Lightning / Mach-E: 400 V compressor — supplier: Denso or Hanon
– Hyundai IONIQ 5/6: 800 V compressor (E-GMP platform) — supplier: Hanon

► Compressor diagnostic tests:

– Insulation resistance (500 VDC megaohmmeter): must be > 10 MΩ between phases and ground
– HV supply voltage: check for no voltage drop
– Current draw: measure with an HV clamp meter (non-invasive)

The electric motor — thermal limits

Why the motor tolerates heat, unlike the battery

▶ The traction motor tolerates heat (unlike the battery)

– It has no narrow optimal window the way the battery does
– Goal: stay below its maximum thermal limits

▶ Thermal limits of the electric motor (PMSM)

– Windings (insulation): withstand up to ~180 °C (class 180-200 °C)
– Permanent magnets (neodymium): irreversible demagnetization from ~150-160 °C (permanent)
– Rule of thumb: every +10 °C above rating cuts insulation life in half

▶ Protection and consequences for the thermal architecture

– Thermal derating: the controller reduces current and torque before the limits are reached
– Motor = a loop that can run hot; battery = a loop that must stay cool (15-35 °C)
– That is why the battery has its own loop with a chiller, separate from the motor

🔑 **KEY POINT** Two opposite needs: the motor can run hot, the battery has to stay cool (15-35 °C). That is exactly why the battery has its own loop with a chiller, separate from the motor's.

Thermal management module

The brain of the heat pump system

▶ **Role: coordinate every thermal actuator in the vehicle in real time**

▶ **Actuators controlled by the thermal management module:**

– E-compressor (speed, on/off)
– EXV (% open)
– 3-way valves in the coolant circuit
– Electric coolant pumps (cabin, battery, motor)
– Fan(s) on the outside exchanger
– Auxiliary PTC heater (backup)
– Any other component the manufacturer chooses to add

▶ **Inputs (monitored sensors):**

– High-side and low-side pressures · Discharge, suction and ambient temperatures
– Battery temperatures (BMS) · Motor / inverter temperatures
– Requested cabin temperature · Vehicle speed

▶ **Communication:**

– CAN or CAN FD bus
– Receives requests from: BMS, HVAC module, ECM, BCM, etc.

▶ **Names by North American manufacturer:**

– Tesla: 'Thermal Control Unit (TCU)' · GM: 'BCM and BECM'
– Ford: 'HVAC, ACCM, HPVCM' · Hyundai/Kia: 'ACCM'
– Toyota : 'Air Conditioning Amplifier Assembly (ACAA)'

Direct battery cooling

Thermal protection and fast-charging optimization

▶ **Why cool the battery directly?**

– DC fast charging (DCFC) generates significant heat (internal losses)
– High charging power: 150–350 kW → cells can reach 45–55 °C
– Above 40 °C: accelerated cell degradation; the BMS cuts power

▶ **Refrigerant path (battery cooling mode):**

– Compressor → outside exchanger (condenser) → battery EXV → battery chiller (evaporator) → compressor

▶ **Advantage of direct cooling (refrigerant) over indirect cooling (coolant only):**

– Two to three times the cooling capacity
– Target temperature reached faster
– Allows longer charging sessions without power reduction

▶ **North American manufacturers using direct cooling:**

– Tesla (every model since 2012) · GM Ultium · Ford F-150 Lightning · Hyundai E-GMP

▶ **Typical trigger thresholds (example):**

– Battery temp > 30–35 °C: active cooling triggered by the BMS
– Battery temp > 40 °C: charging power automatically reduced

▶ **Link to diagnostics:**

– If the battery chiller fails → DCFC charging is power-limited
– Possible DTC in the BMS: 'Battery Cooling Fault'

Preconditioning the battery and the cabin

Best use before departure or before a DC fast charge

- **Definition: heating or cooling the battery and the cabin before use, ideally while plugged in**

- **Types of preconditioning:**
  - On a home charger (EVSE): free energy from the grid; does not affect range
  - On battery (not plugged in): uses up range; to be used sparingly

- **How it is triggered:**
  - Mobile app (Tesla, GM myChevrolet, FordPass, etc.)
  - Scheduled timer (automatic preconditioning)
  - GPS navigation to a DCFC station (automatic on recent Tesla, Ford, GM Ultium)

- **Measured benefits (Canadian winter conditions):**
  - Battery preconditioned to 20 °C vs a cold battery at -20 °C:
  - Range recovered: +10–20% (less cabin heating needed on the road)
  - DCFC charging power: +50–100% (accepts full power sooner)
  - Battery life: better (avoids charge cycles at too low a temperature)

> 💡 **TIP** The Quebec winter argument: schedule preconditioning the night before, with the vehicle plugged in. Battery and cabin at temperature at departure, maximum range and more efficient fast charging, even after a night at -30 °C.

Integration with DC fast charging (DCFC)

Thermal management during and after a high-power charging session

▶ The DCFC network in North America:

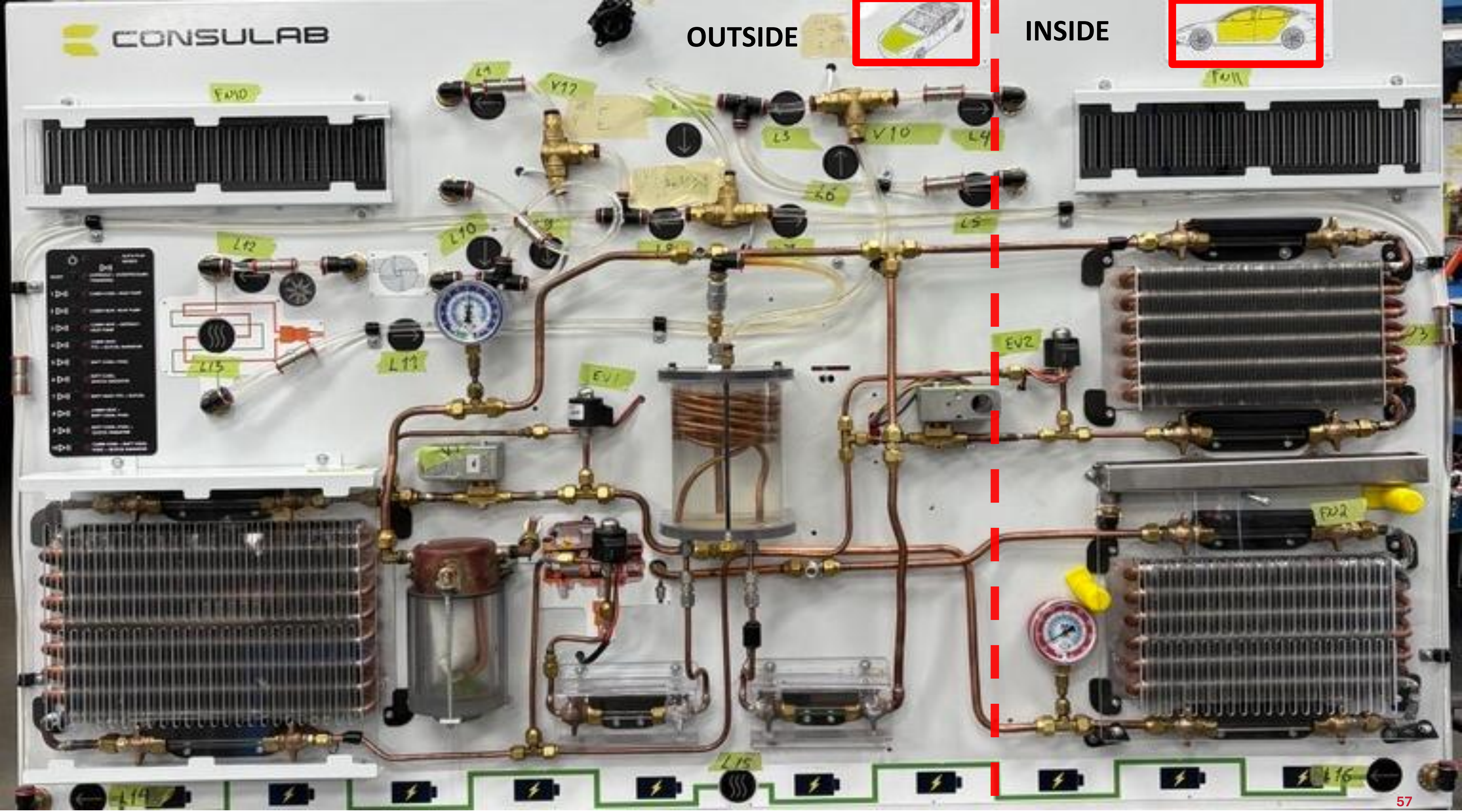
– Tesla Supercharger (V3/V4: 250–350 kW) · Circuit Électrique · ChargePoint · EVgo
– Ford Charge Network · GM Ultium Charge 360 · Toyota / Hyundai / Kia networks

▶ How a DCFC session unfolds thermally:

– On arrival: if the battery is cold (< 15–20 °C), charging is limited by the BMS (e.g. 50 kW instead of 250 kW)
– Active preconditioning: the heat pump warms the battery before arrival
– During charging: heat pump in battery cooling mode to hold the temperature below 35–40 °C
– End of charge (above 80%): power reduced by the BMS → less heat generated

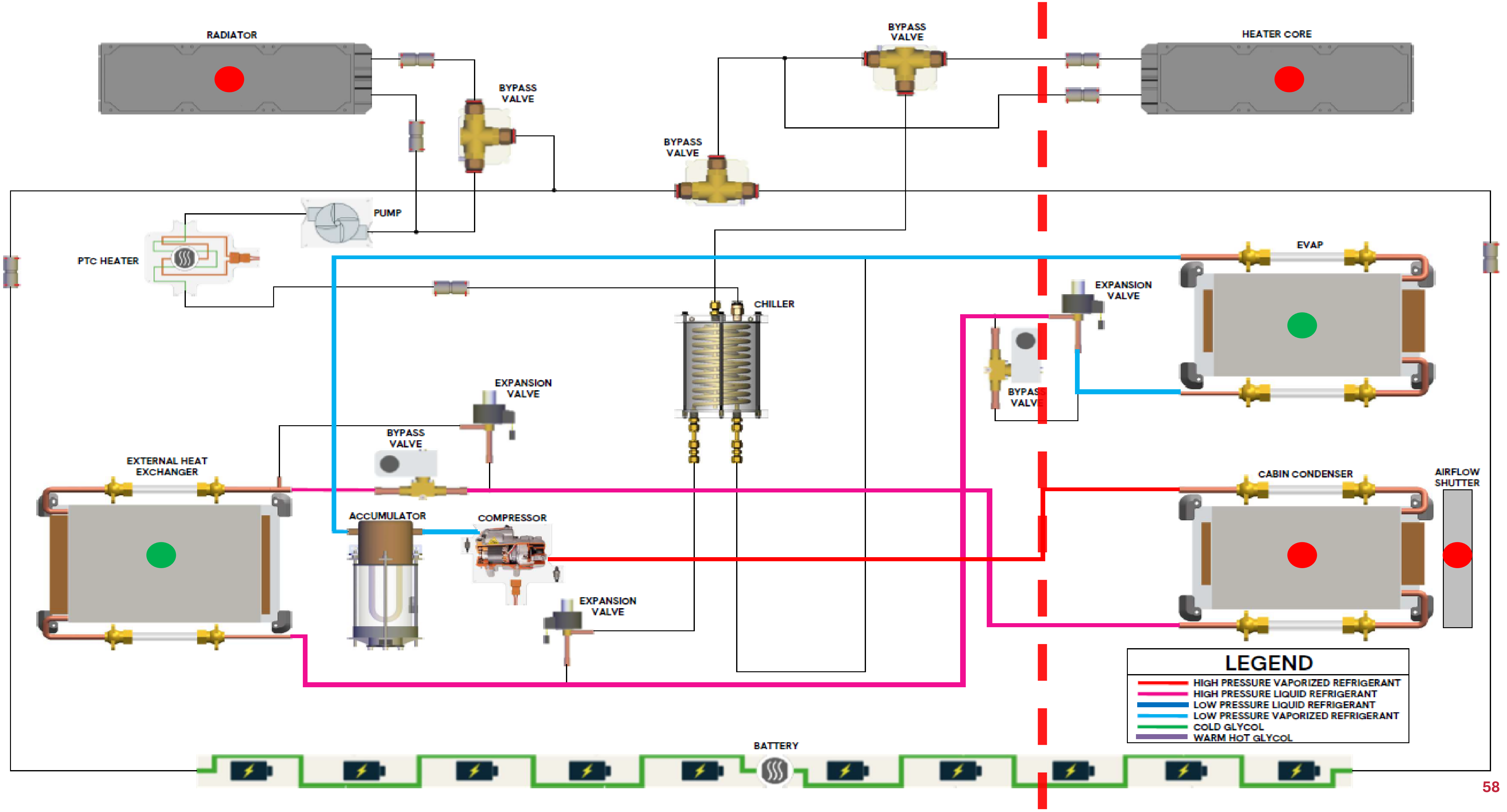
▶ Symptoms of a battery cooling system failure during DCFC:

– Charging power limited or slow even at a low state of charge
– Warning message: 'Battery Cooling System Fault' or 'Charging Power Reduced'
– Battery temperature climbing quickly on OBD data with no active cooling

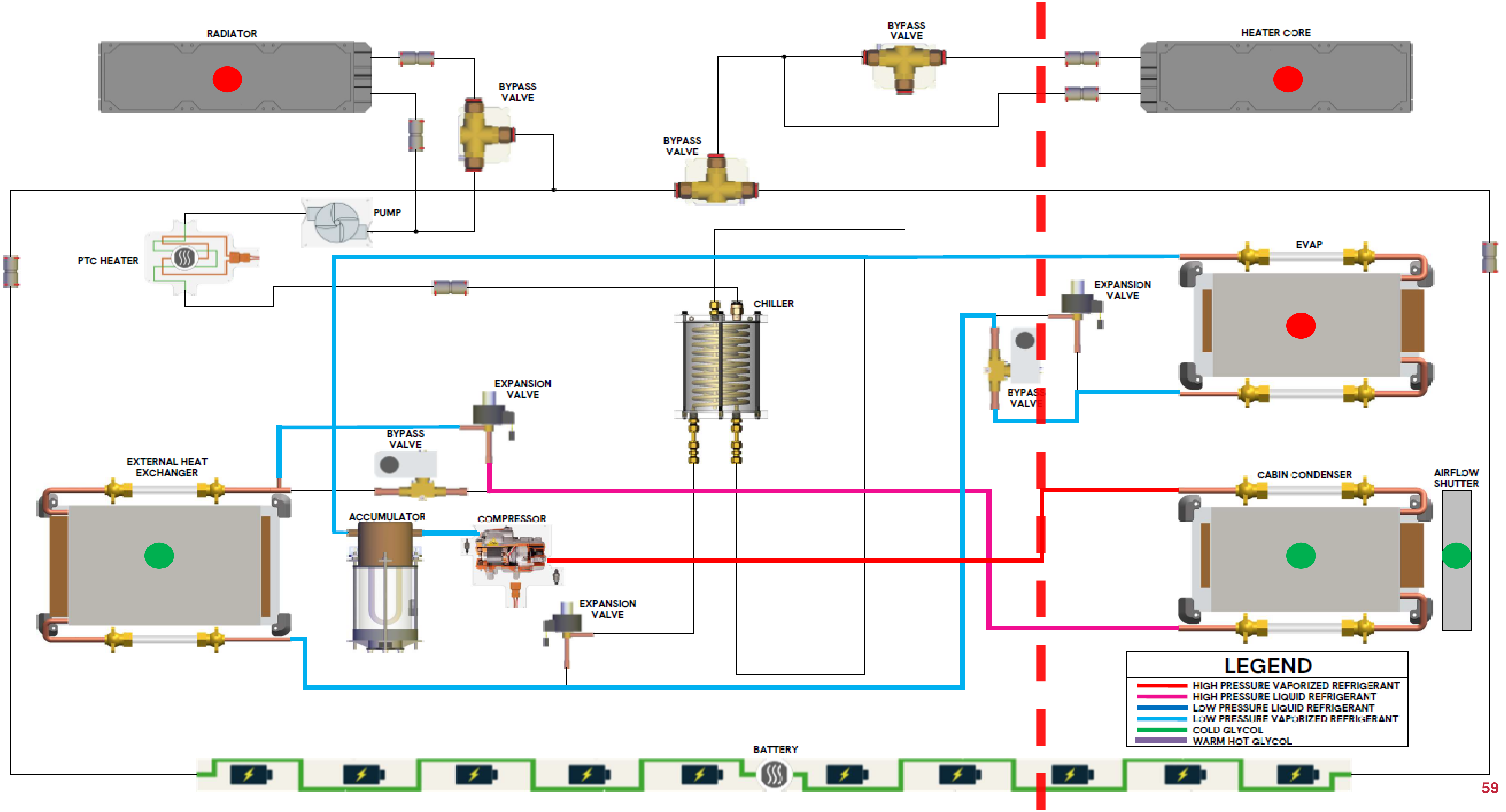


CONSULAB
OUTSIDE
INSIDE

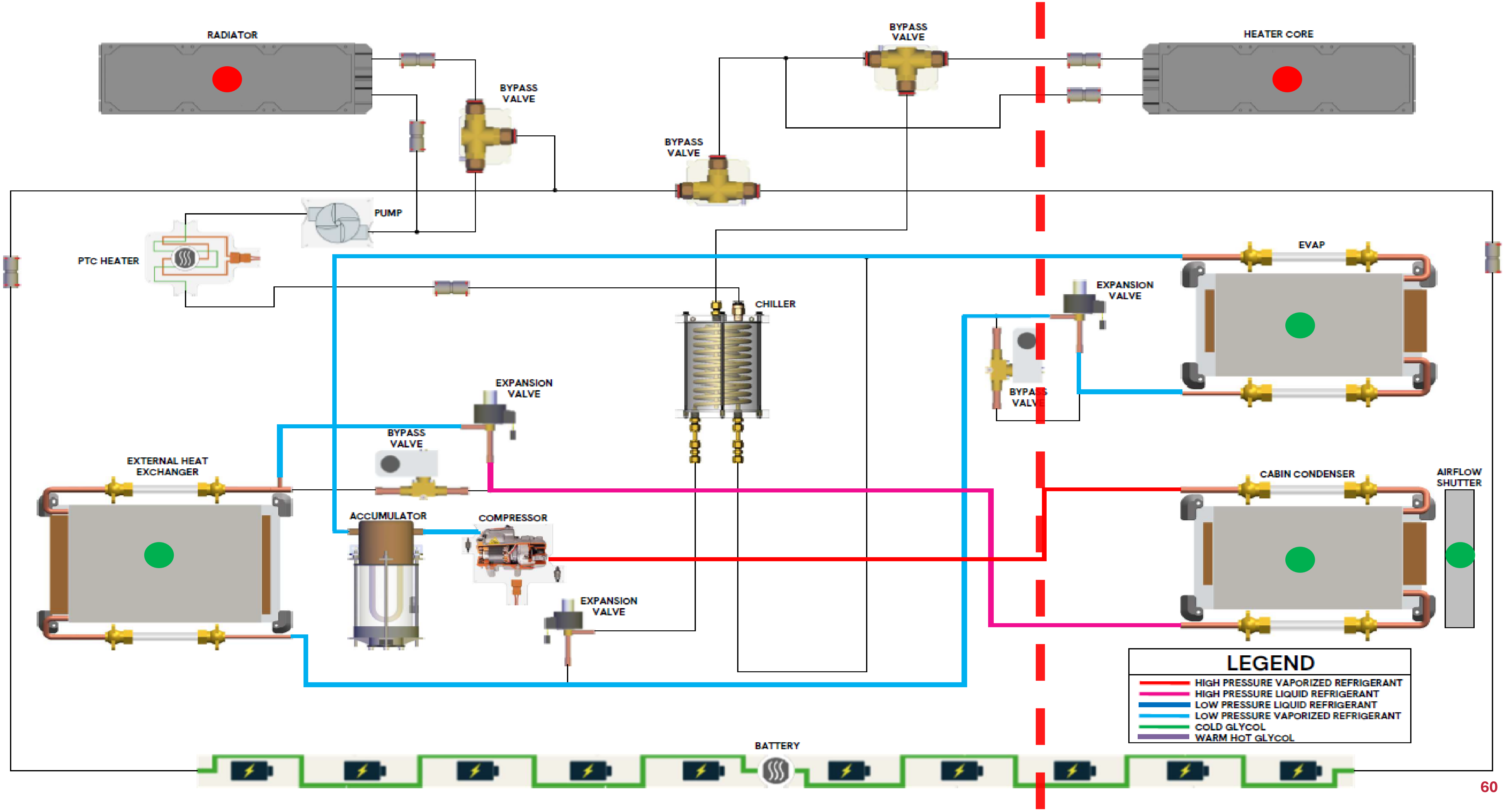
- MODE 1 = CABIN COOL : HEAT PUMP



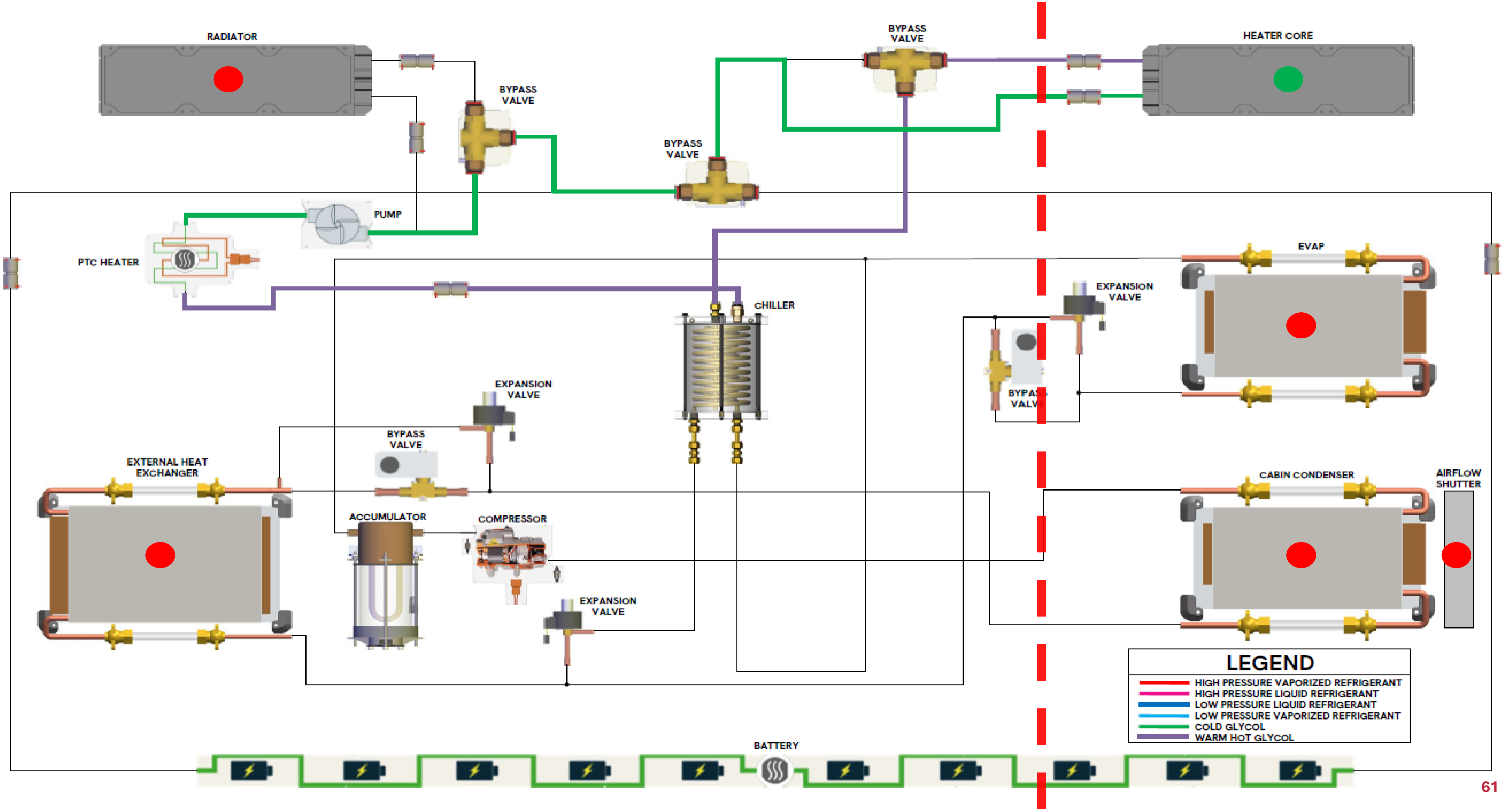
- MODE 2 = CABIN HEAT : HEAT PUMP



- MODE 3 = CABIN HEAT + DEFROST : HEAT PUMP

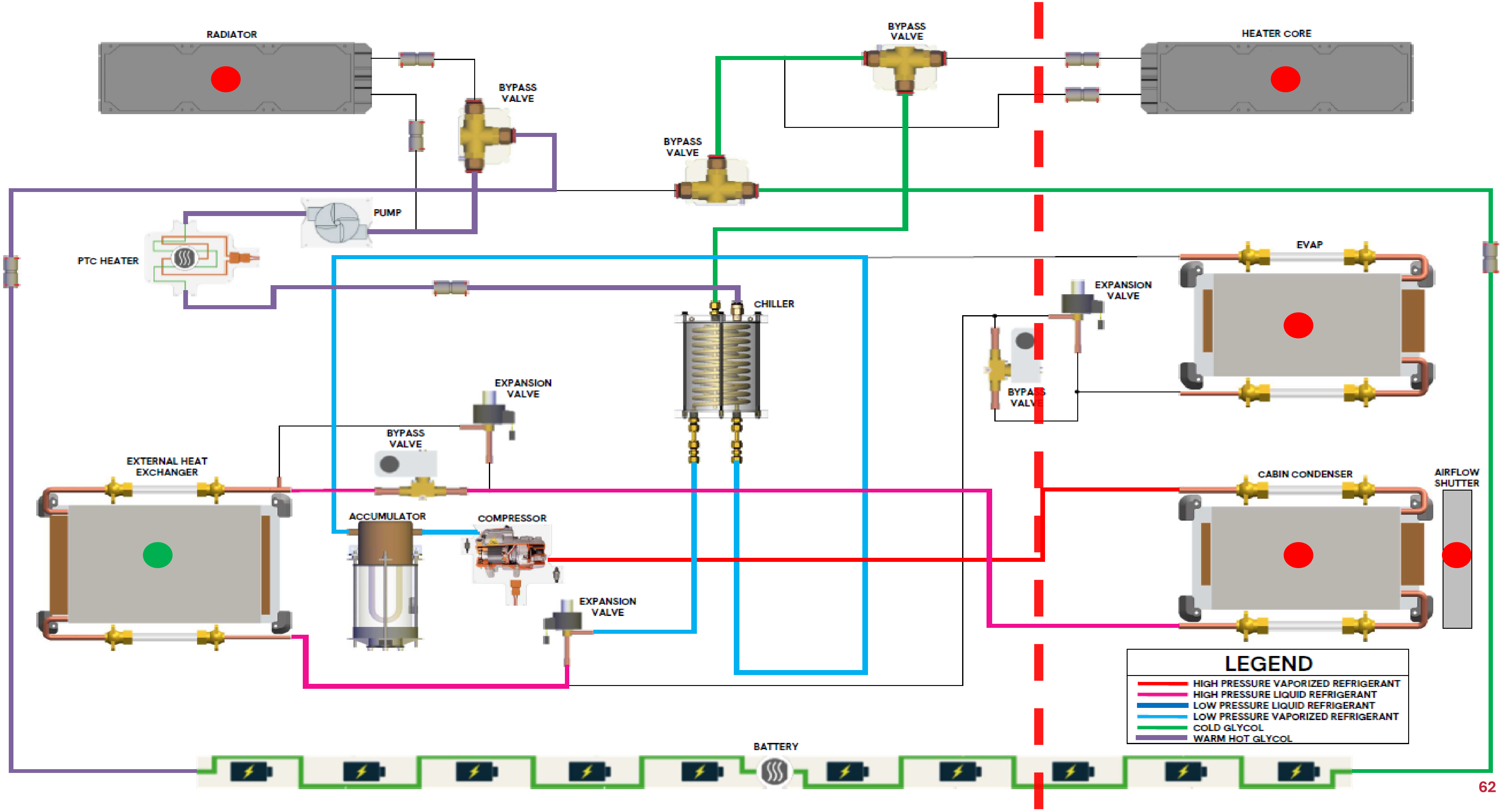


- MODE 4 = CABIN HEAT : PTC + GLYCOL RADIATOR

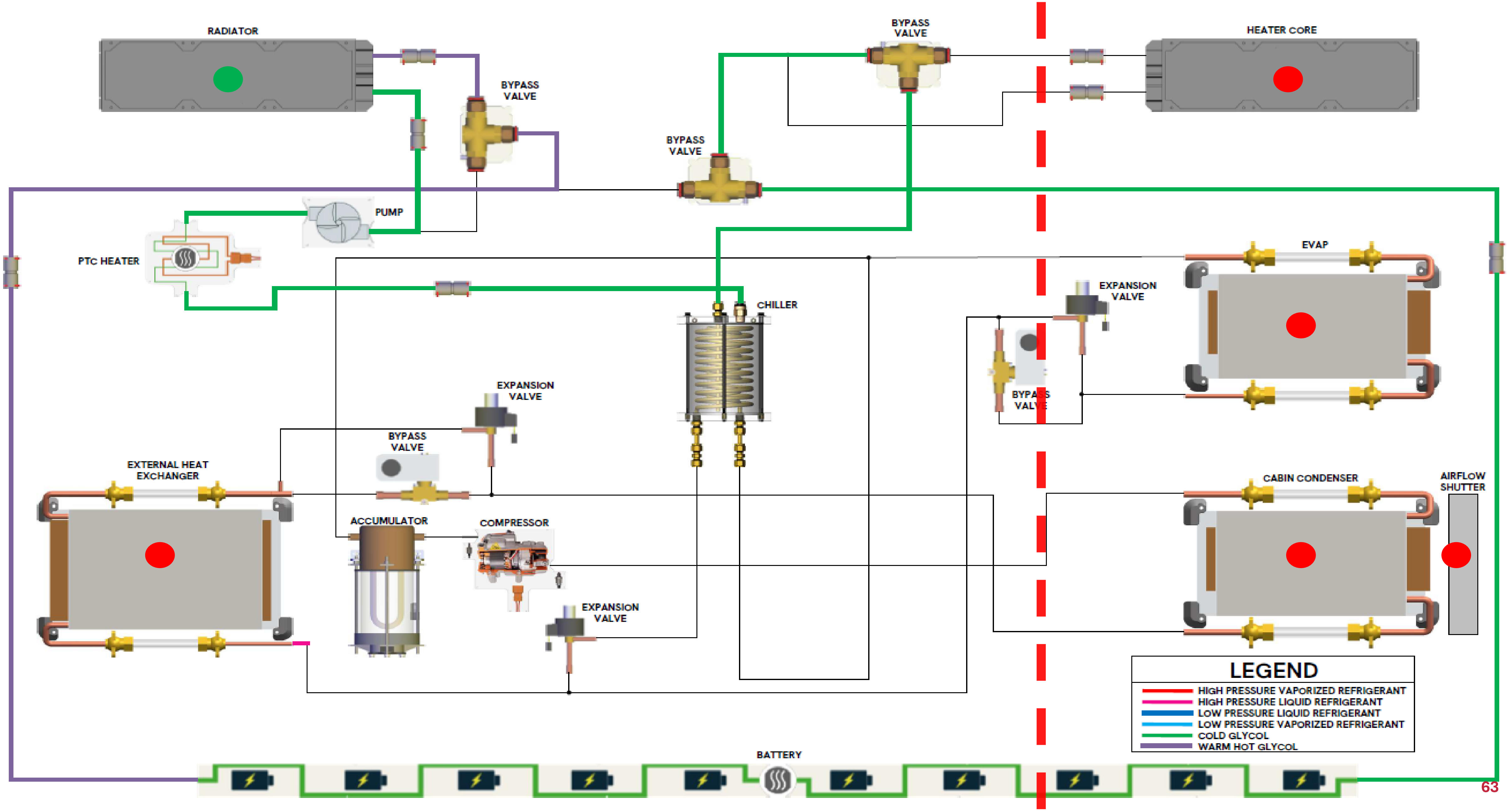


CONSULAB

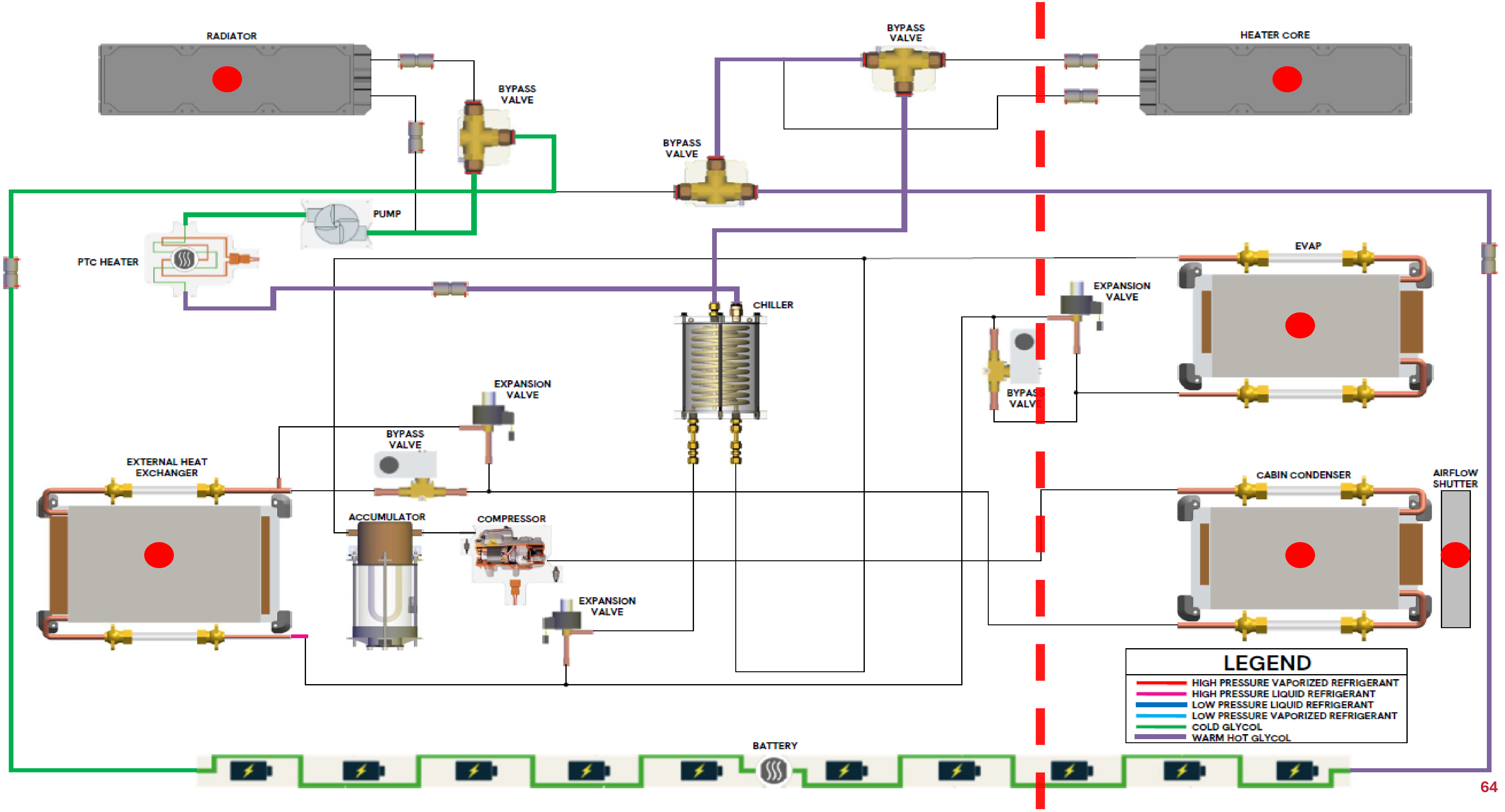
- MODE 5 : BATT COOL : HVAC



- MODE 6 = BATT COOL : GLYCOL RADIATOR

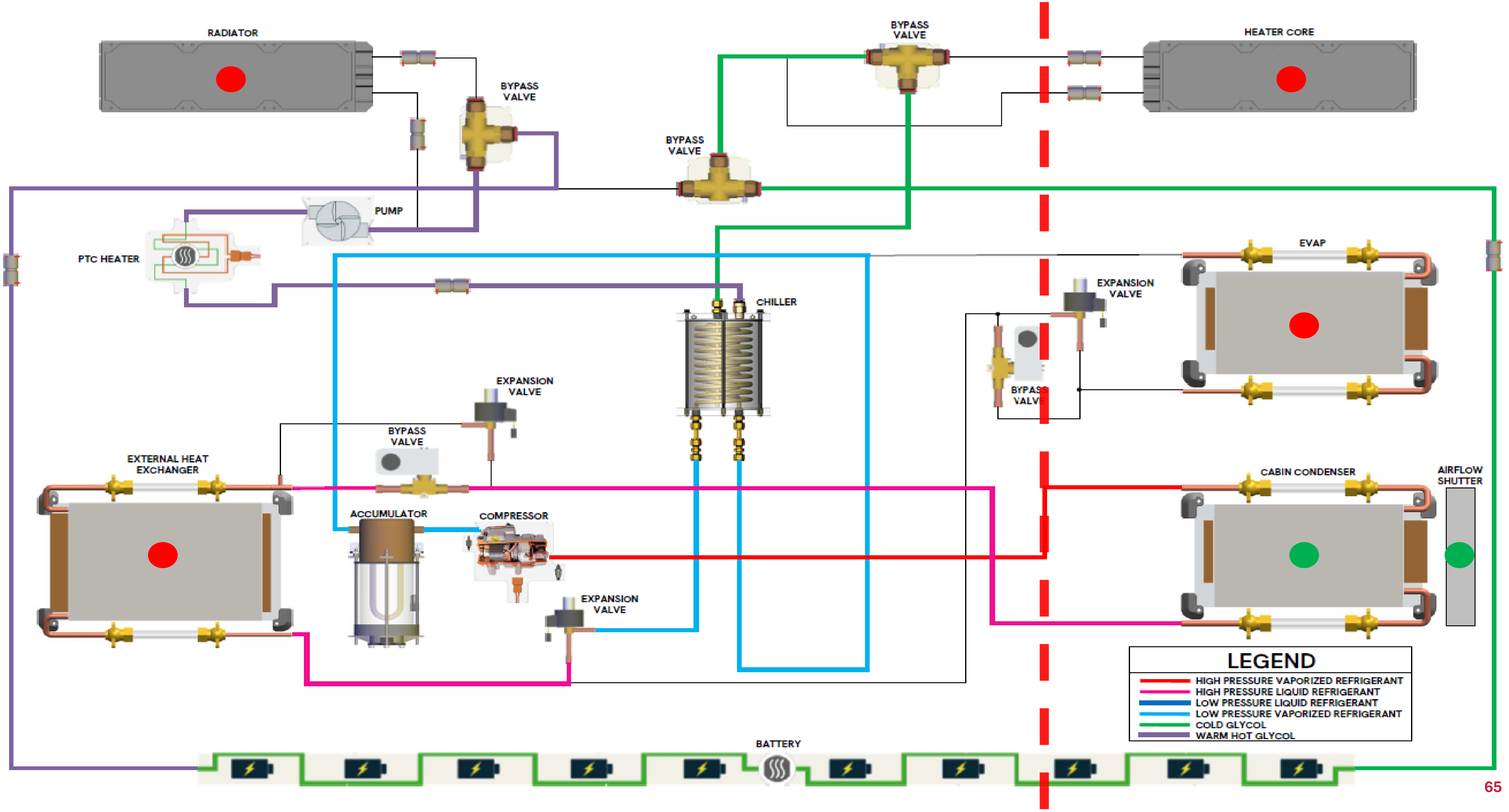


- MODE 7 = BATT HEAT : PTC + GLYCOL

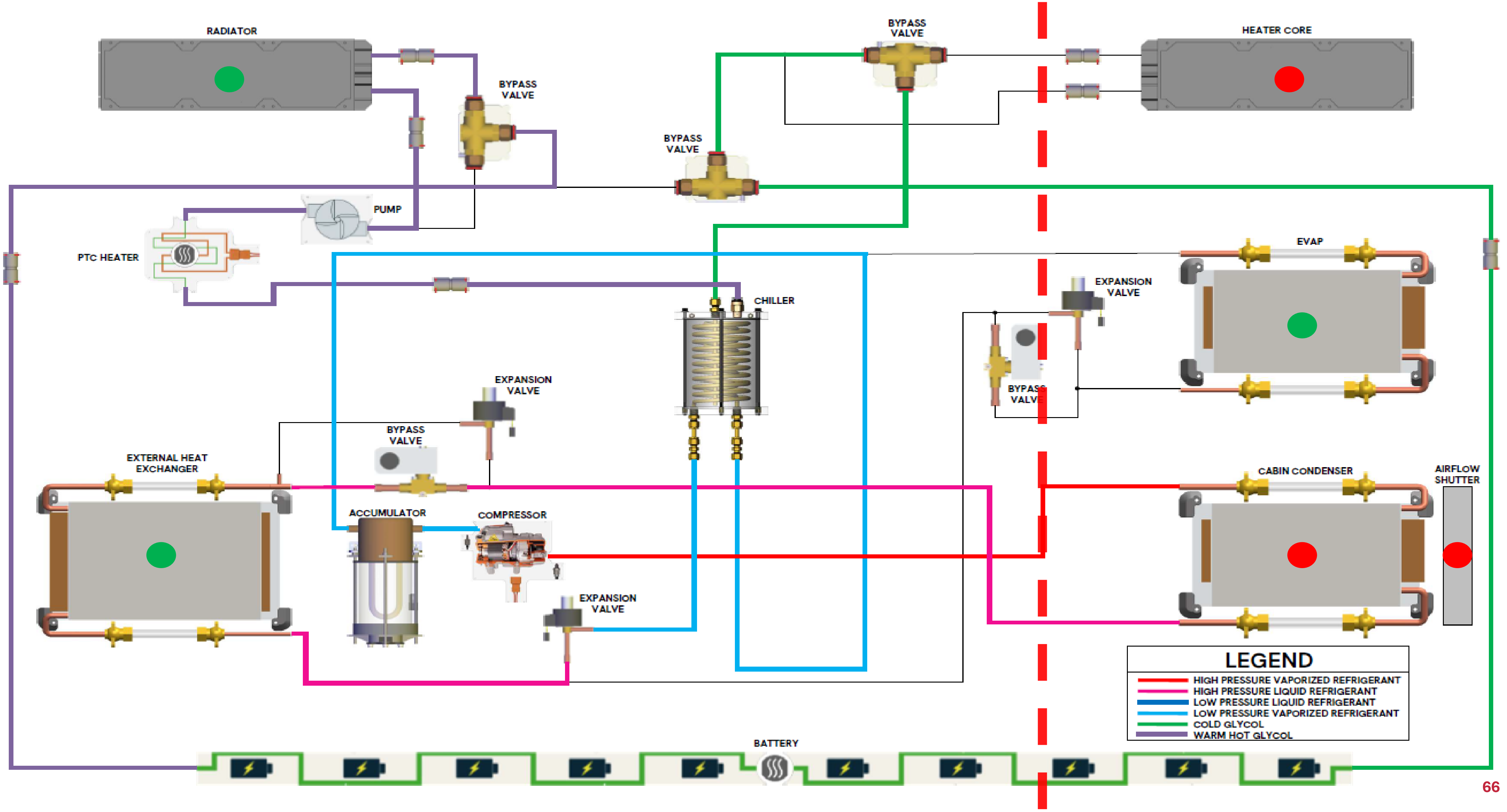


CONSULAB

- MODE 8 = CABIN HEAT AND BATT COOL : HVAC

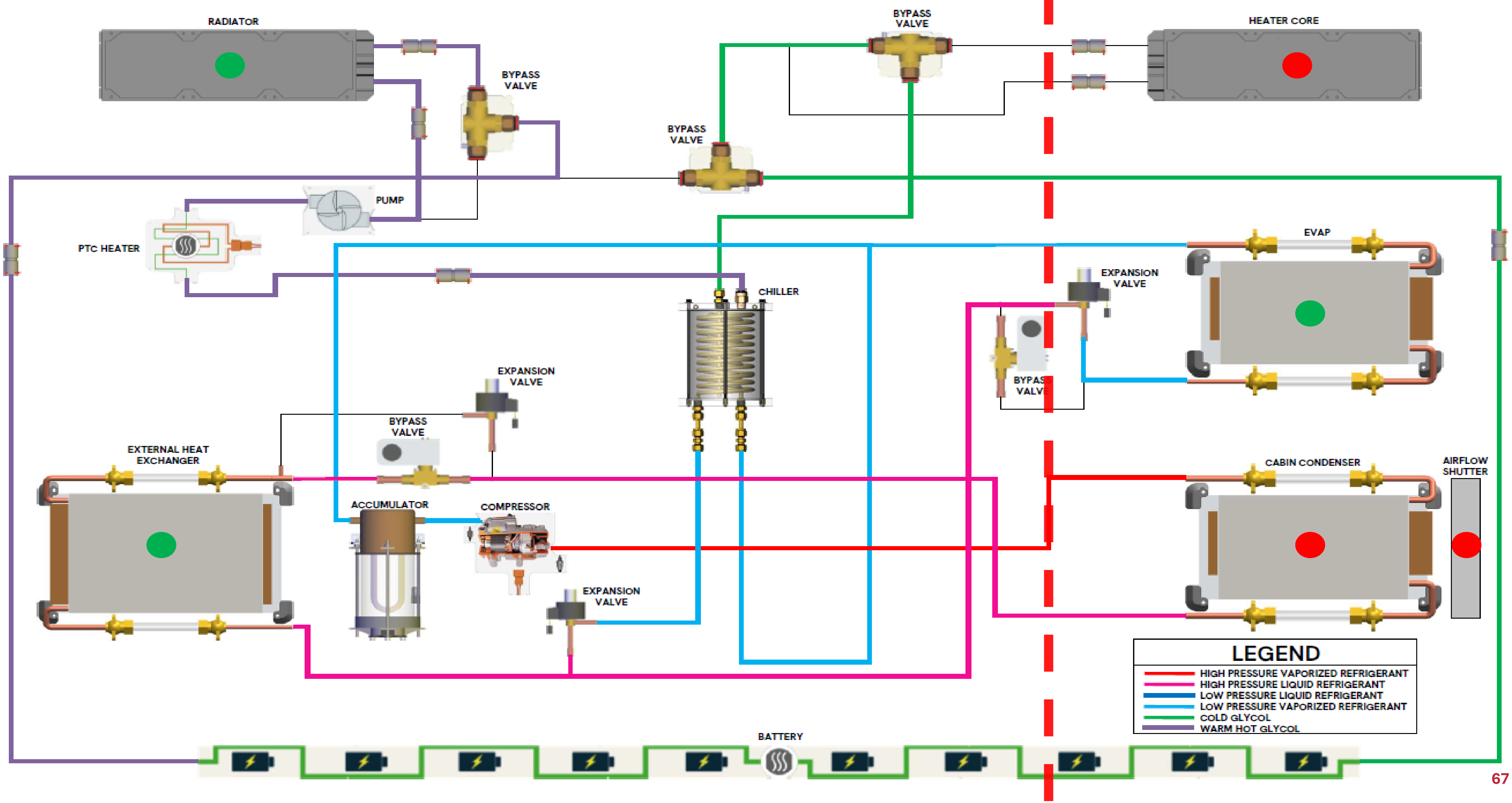


- MODE 9 = BATT COOL : HVAC + GLYCOL RADIATOR



CONSULAB

- MODE 10 = CABIN COOL + BATT COOL : HVAC + GLYCOL RADIATOR



CONSULAB

Tesla – Thermal system overview

The leader in integrated thermal management

▶ **Models concerned: Model 3 · Model Y · Model S · Model X · Cybertruck**

▶ **Evolution of the Tesla thermal system:**

– 2012 (Model S) : PAC basique + chaufferette PTC; refroidissement batterie indirect
– 2017 (Model 3): introduction of the 'Super Manifold' — combines pump, valve and reservoir
– 2021 (Model Y / refreshed 3): 'Octovalve' — a revolution in thermal architecture
– 2023 (Cybertruck): 48 V system + advanced thermal management for the 4 motors

▶ **Refrigerant:**

– R-134a or R-1234yf: always confirm with the VIN

▶ **Distinctive Tesla characteristics:**

– Fully software-driven control; OTA updates change the thermal strategy
– Limited diagnostic access for non-Tesla shops (Toolbox required)
– Very detailed data history in the Tesla Service portal

Tesla Model Y – Complete thermal architecture

The best-selling EV in Canada in recent years

▶ Thermal system (2021+):

– Compresseur : Hanon/Sanden ou Denso
– Octovalve: manages 6 coolant circuits
– 4 EXVs for the 4 branches of the refrigerant circuit
– Heat sources: outside air + front motor + rear motor + onboard charger

▶ Performances PAC selon Tesla :

– Heating at 0 °C: COP ≈ 3.0; saves ~100 km of range vs PTC alone
– Heating at -15 °C: COP ≈ 2.0; saves ~60–80 km vs PTC alone
– Lower limit of the air-source heat pump: -25 °C; the PTC takes over below that

▶ Refrigerant specifications:

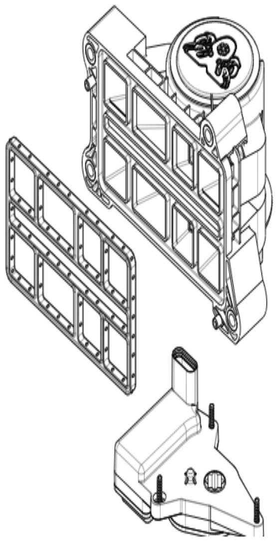
– Huile :

- **Denso** compresseur: ND-11 A/C oil POE84
- **Sanden/Hanon** compresseur: RB100EV A/C oil POE84

▶ Known issues:

– Refrigerant leaks at the fittings (O-rings)
– Octovalve actuator failure

Tesla – L'Octovalve



4. OCTOVALVE ASSEMBLY

1506859-00-D

CA$85.00

Quantity

- 1 +

Required for Repair: 1.

Part must be ordered in multiples of this quantity: 1.

Add to Cart

▶ What is the Octovalve?

– An 8-port fluid valve combining several 3-way valves in a single component
– Driven by 3 internal rotary actuators (stepper motors)
– Lets 6 separate coolant circuits be connected or disconnected

▶ The 6 circuits managed by the Octovalve:

– Battery circuit (cooling/heating) · Front motor circuit · Rear motor circuit
– Circuit chiller cabine · Circuit radiateur principal · Circuit chaufferette PTC

▶ Advantages of the Octovalve:

– Replaces 8–12 separate 3-way valves → fewer connections, fewer potential leaks
– Smooth, instant control by software; complex modes are easy to program
– Allows waste motor heat to be sent to the battery or the cabin with a single command

▶ Diagnosing the Octovalve:

– Angular position sensors → values visible in Tesla Toolbox
– Fault code if measured position ≠ commanded position: Octovalve Position Error
– Actuator test available in Tesla Toolbox: manual command of each rotor

⚠ **L'Octovalve se remplace** – Estimated replacement cost:
85$ CAD

Tesla – Diagnostic points and fault codes

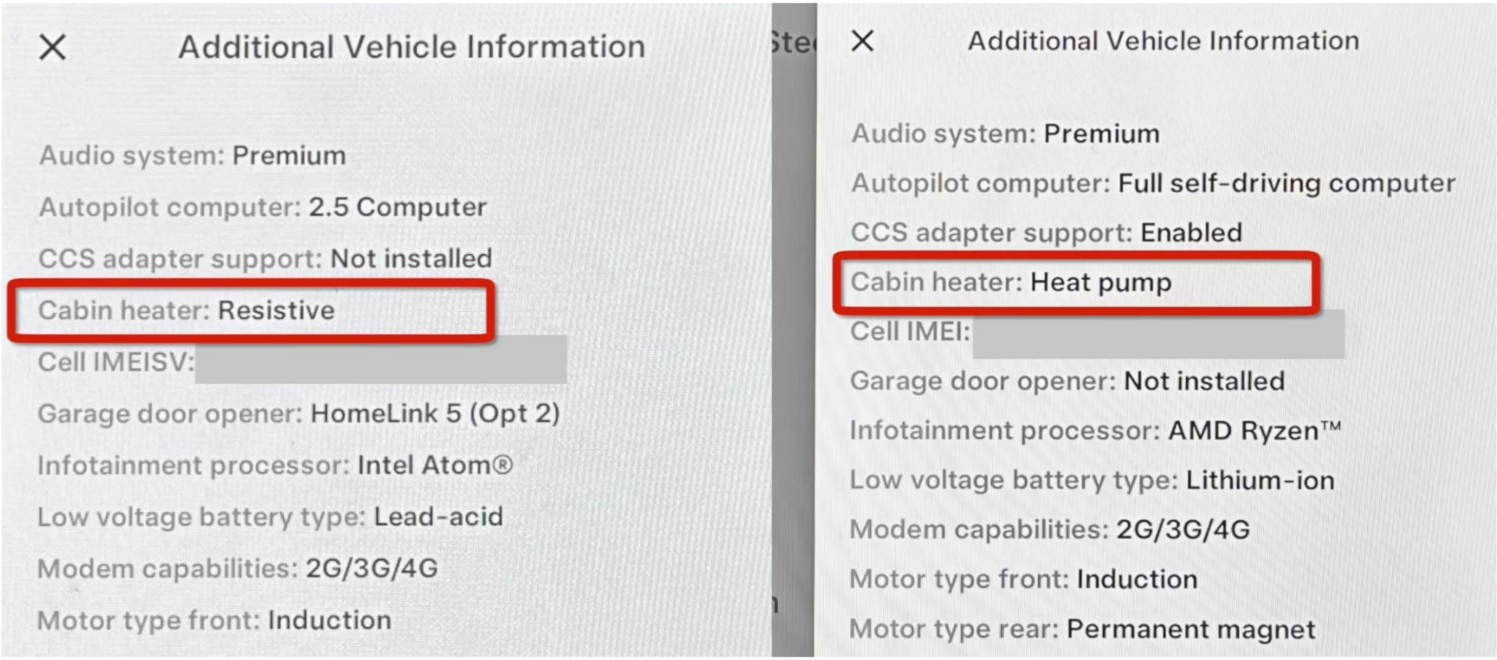
Tools, common DTCs and procedures

▶ Live data available in Toolbox:

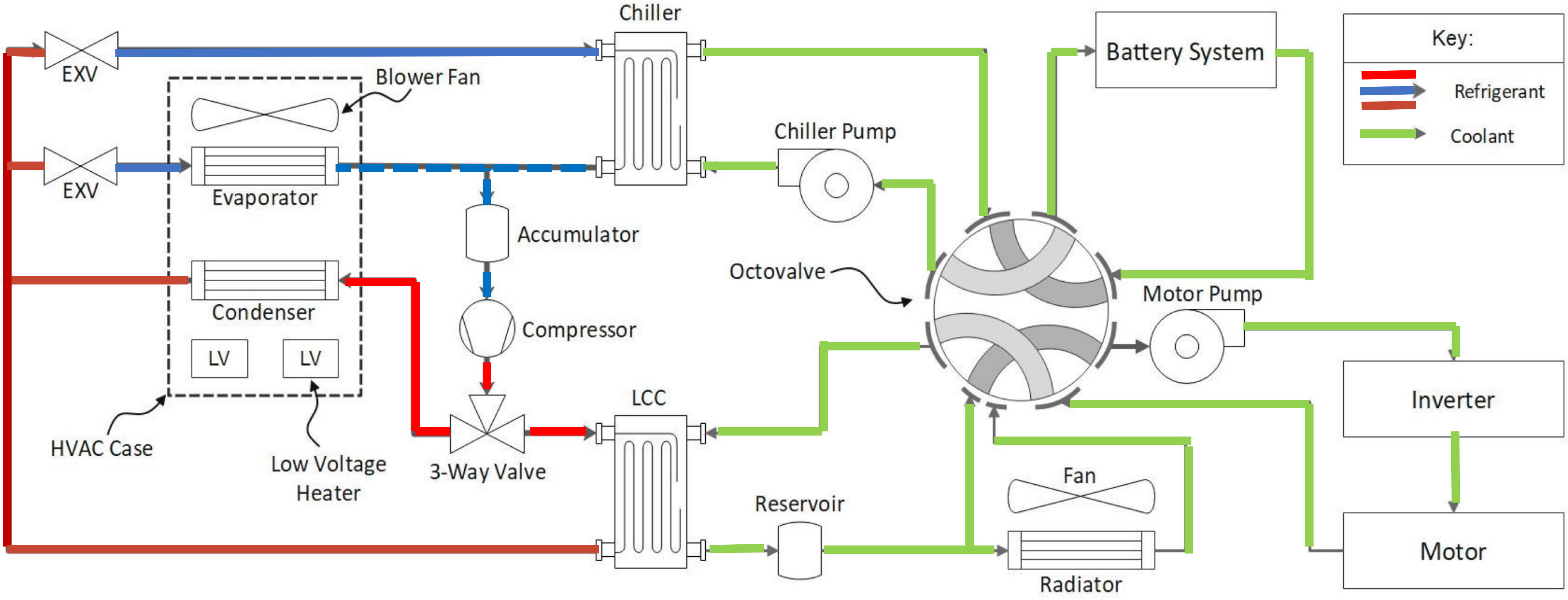
– Pressions HP/BP · T° refoulement compresseur · Vitesse compresseur (rpm)
– % open of the 4 EXVs · Octovalve position · Battery temp (min/max/average)

▶ Tesla-specific procedures:

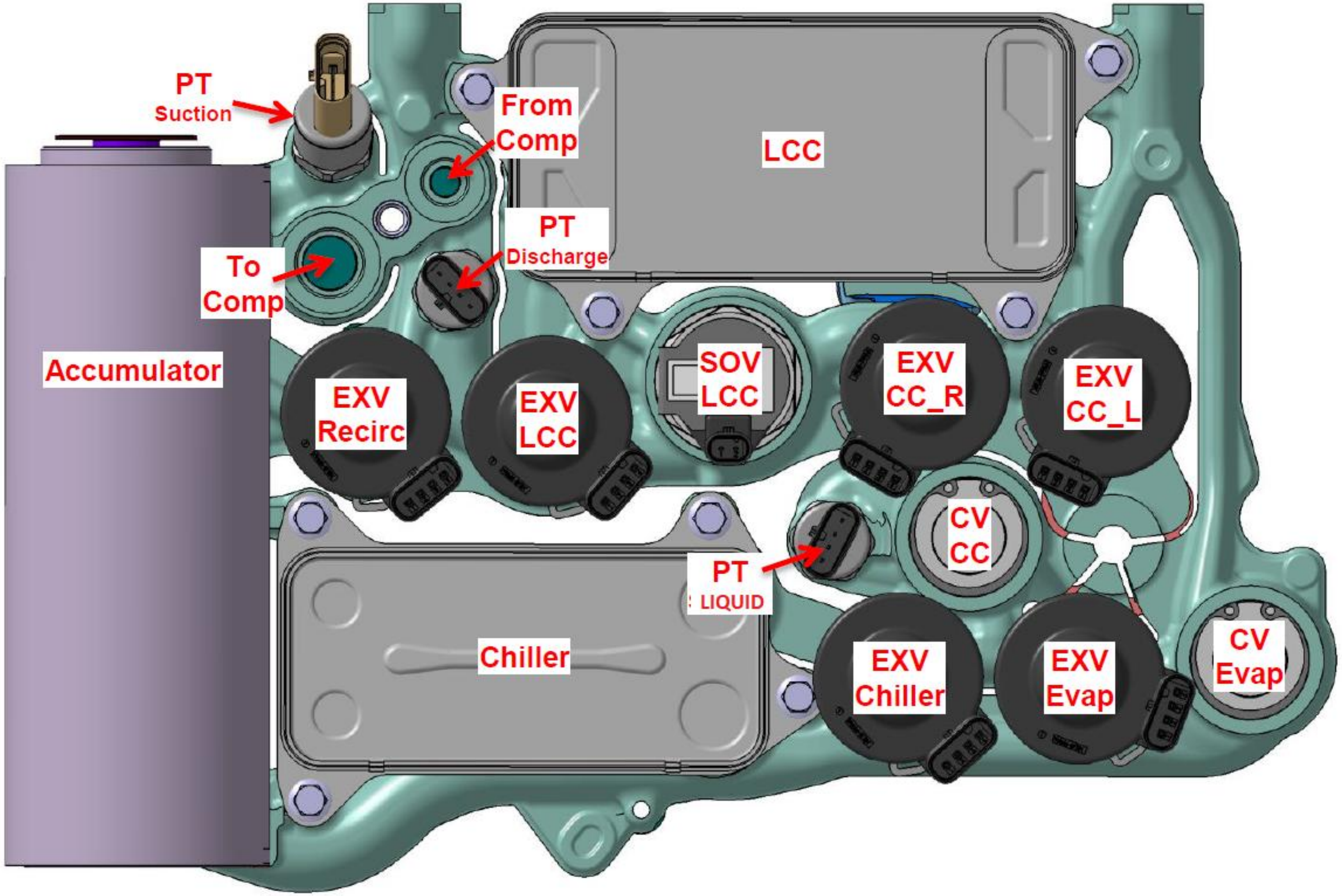
– Compressor replacement: 'Compressor Oil Fill' obligatoire via Toolbox
– Refrigerant recharge: 'Refrigerant Charge' procedure with the specified quantity
– After Octovalve replacement: 'Octovalve Calibration' in Toolbox



TESLA PAC Circuit – all models



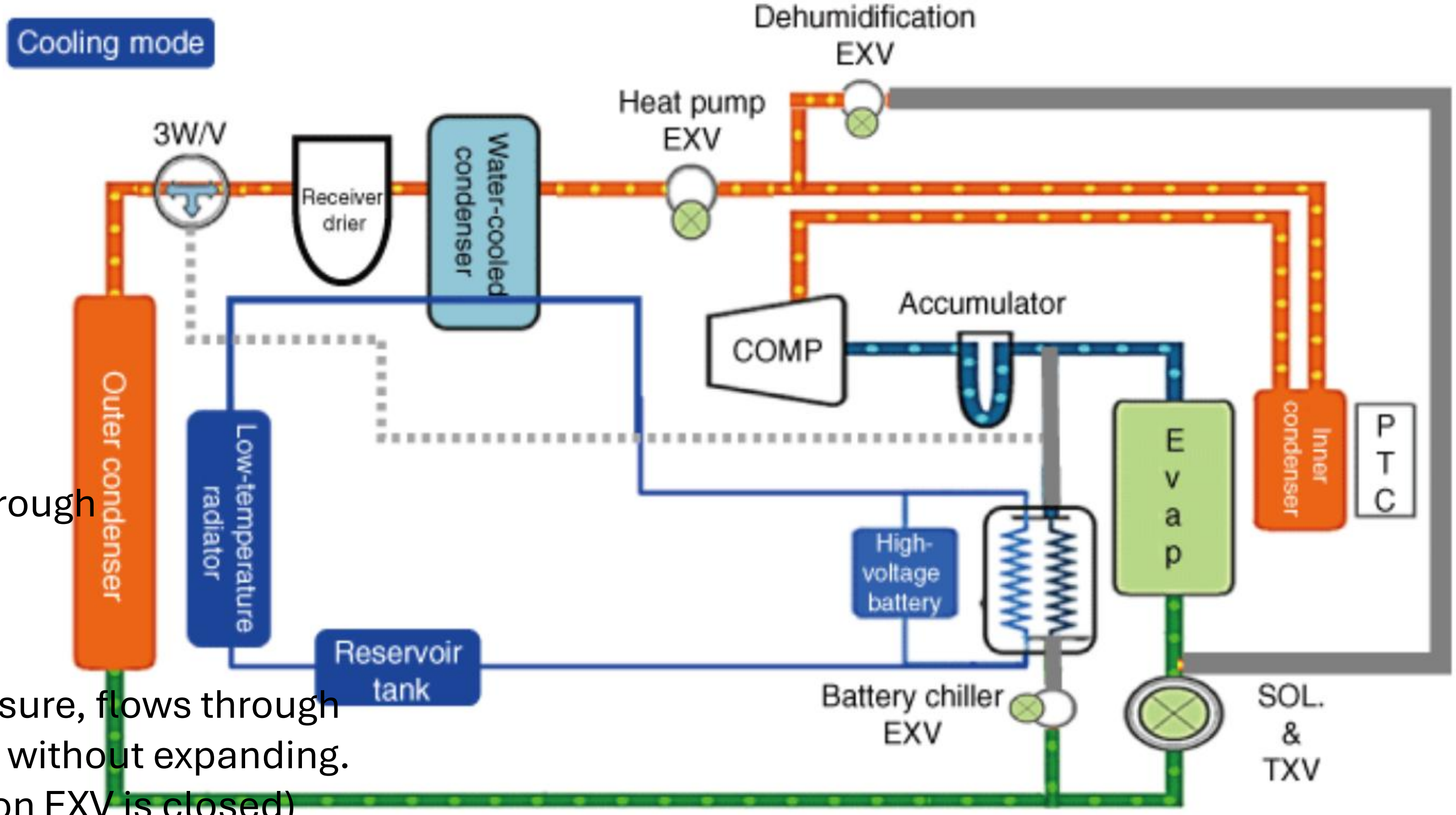
SUPER MANIFOLD



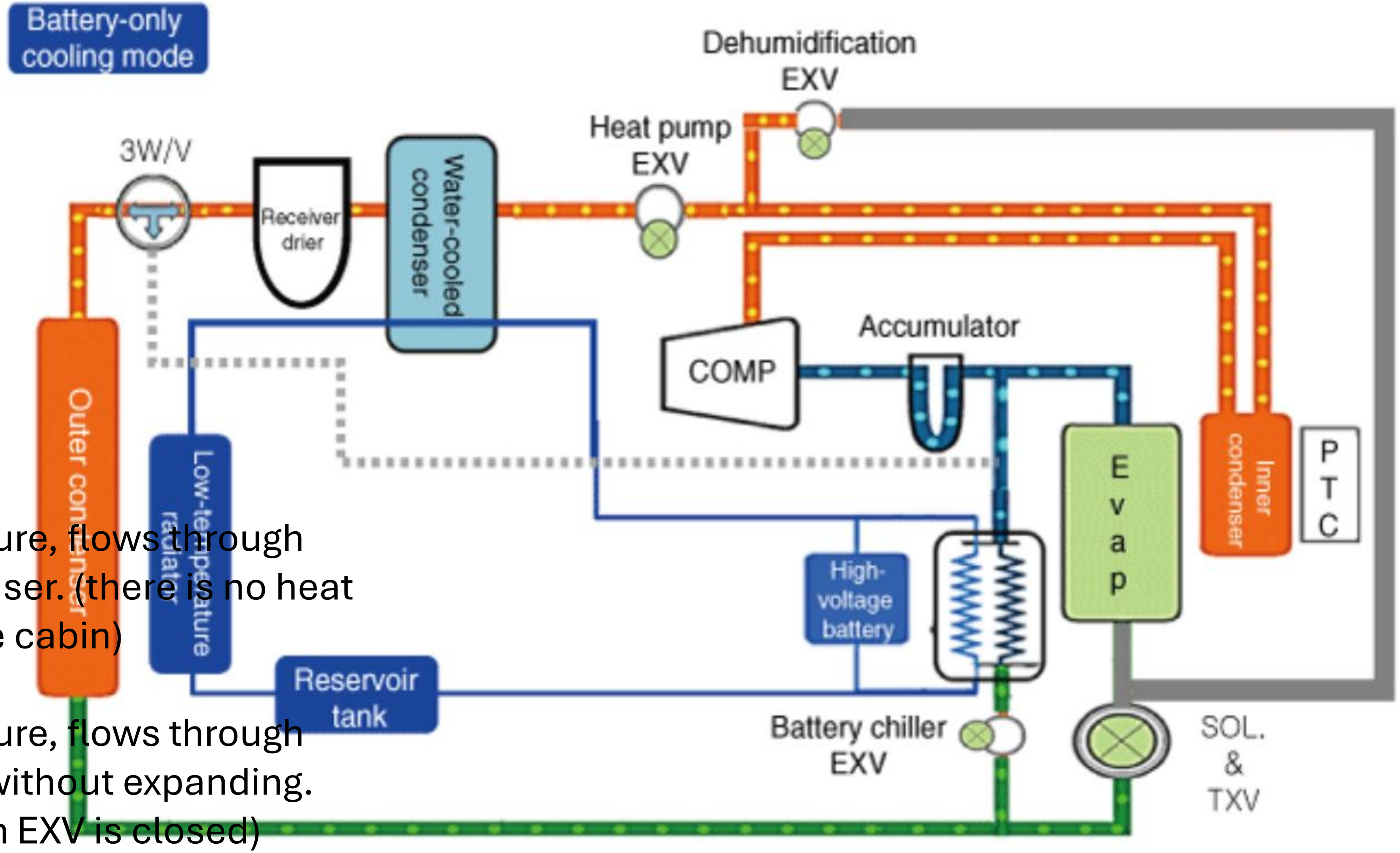
The six operating modes

The key: the outside condenser is reversible (condenser in A/C, evaporator in heating) · Adapted from CPCPA MAP-265 (2024)

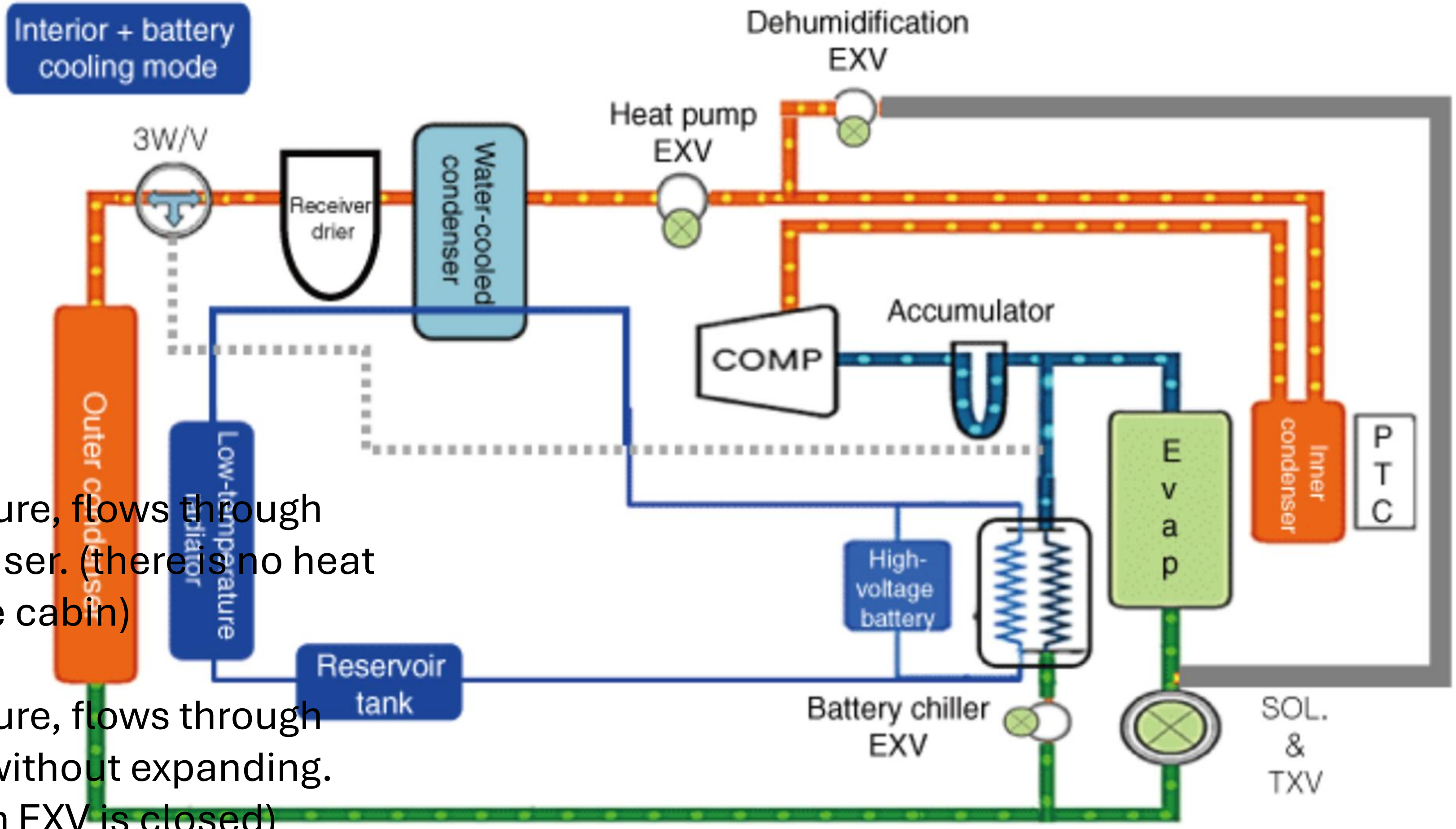
Mode	Heat path	Configuration clé
Climatisation	Cabin → outside	Blend door closed (inside condenser inactive); electric expansion valve open (no expansion); condensation at the outside condenser; expansion ahead of the inside evaporator
Chauffage	Outside → cabin	Blend door open; electric expansion valve restricting; the outside condenser becomes an evaporator; the 3-way valve bypasses the inside evaporator
Chauffage + PTC	Outside + resistive backup	Same circuit as in heating; the PTC comes in when it is very cold: ~1,000 W on low voltage, ~5,000 W on high voltage (depending on the manufacturer)
Dehumidification	Series circulation inside the cabin	Electric expansion valve closed, solenoid valve open: air is dried at the evaporator then reheated at the inside condenser; the outside condenser sees no refrigerant
Heating + dehumidification	Outside → cabin, in parallel	Refrigerant split in parallel through both inside exchangers: expansion valve restricting and solenoid valve open at the same time
Defrost	Loop → outside exchanger (cyclic)	Below 0 °C, frost blocks the outside exchanger; the system periodically sends hot high-pressure refrigerant back through it to melt the frost



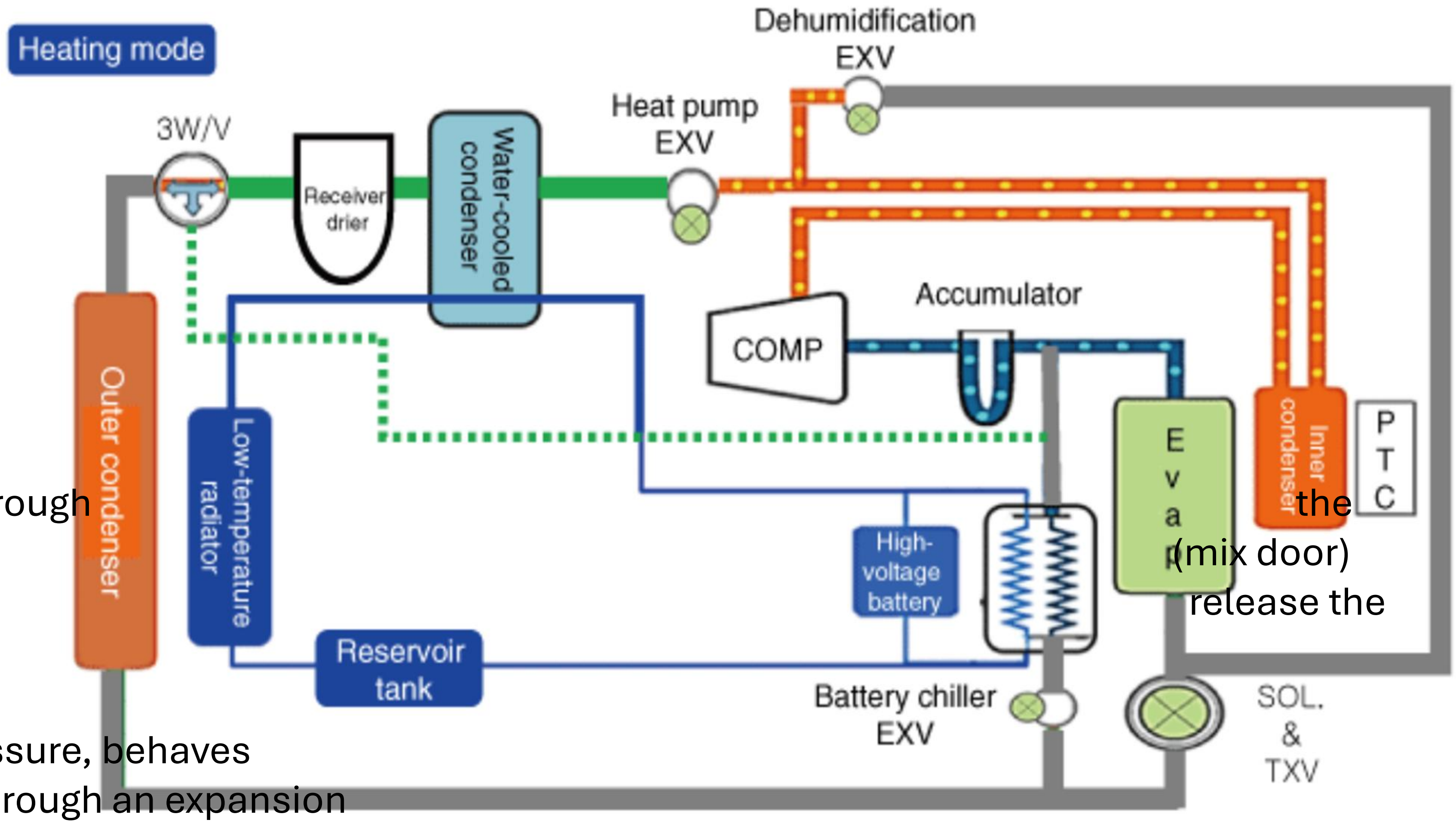
1.An electric compressor compresses the refrigerant.
2.The compressed refrigerant, at high temperature and high pressure, flows through the inside condenser. (there is no heat exchange with the cabin)
3.The compressed refrigerant, at high temperature and high pressure, flows through the heat pump EXV without expanding. (the dehumidification EXV is closed)
4.The compressed refrigerant, at high temperature and high pressure, gives up its heat to the water-cooled condenser and condenses.
5.The refrigerant 3-way valve opens toward the outside condenser.
6.As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.
7.The refrigerant expands through the SOL and the TXV (SOL open, high-voltage battery chiller EXV closed)
8.The expanded refrigerant, at low temperature and low pressure, absorbs heat from the cabin in the evaporator.
9.The refrigerant that absorbed heat inside the vehicle sends only gaseous refrigerant from the accumulator back to the electric compressor.



1. An electric compressor compresses the refrigerant.
2. The compressed refrigerant, at high temperature and high pressure, flows through the inside condenser. (there is no heat exchange with the cabin)
3. The compressed refrigerant, at high temperature and high pressure, flows through the heat pump EXV without expanding. (the dehumidification EXV is closed)
4. The compressed refrigerant, at high temperature and high pressure, gives up its heat to the water-cooled condenser and condenses.
5. The refrigerant 3-way valve opens toward the outside condenser.
6. As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.
7. The refrigerant expands in the battery chiller. (EXV duty-cycle control, SOL solenoid valves and TXV closed)
8. The battery chiller absorbs heat from the high-voltage battery and the refrigerant expands.



1. An electric compressor compresses the refrigerant.
2. The compressed refrigerant, at high temperature and high pressure, flows through the inside condenser. (there is no heat exchange with the cabin)
3. The compressed refrigerant, at high temperature and high pressure, flows through the heat pump EXV without expanding. (the dehumidification EXV is closed)
4. The compressed refrigerant, at high temperature and high pressure, gives up its heat to the water-cooled condenser and condenses.
5. The refrigerant 3-way valve opens toward the outside condenser.
6. As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.
7. The refrigerant expands through the SOL and the TXV (solenoid valve open); the refrigerant also expands in the battery chiller (EXV duty-cycle control).
8. The expanded refrigerant, at low temperature and low pressure, absorbs heat from the cabin in the evaporator and absorbs heat from the battery coolant in the battery chiller.



the
(mix door)
release the

1.An electric compressor compresses the refrigerant.

2.As the compressed refrigerant at high temperature and high pressure flows through inside condenser, the blend door opens on the heating side to heat.

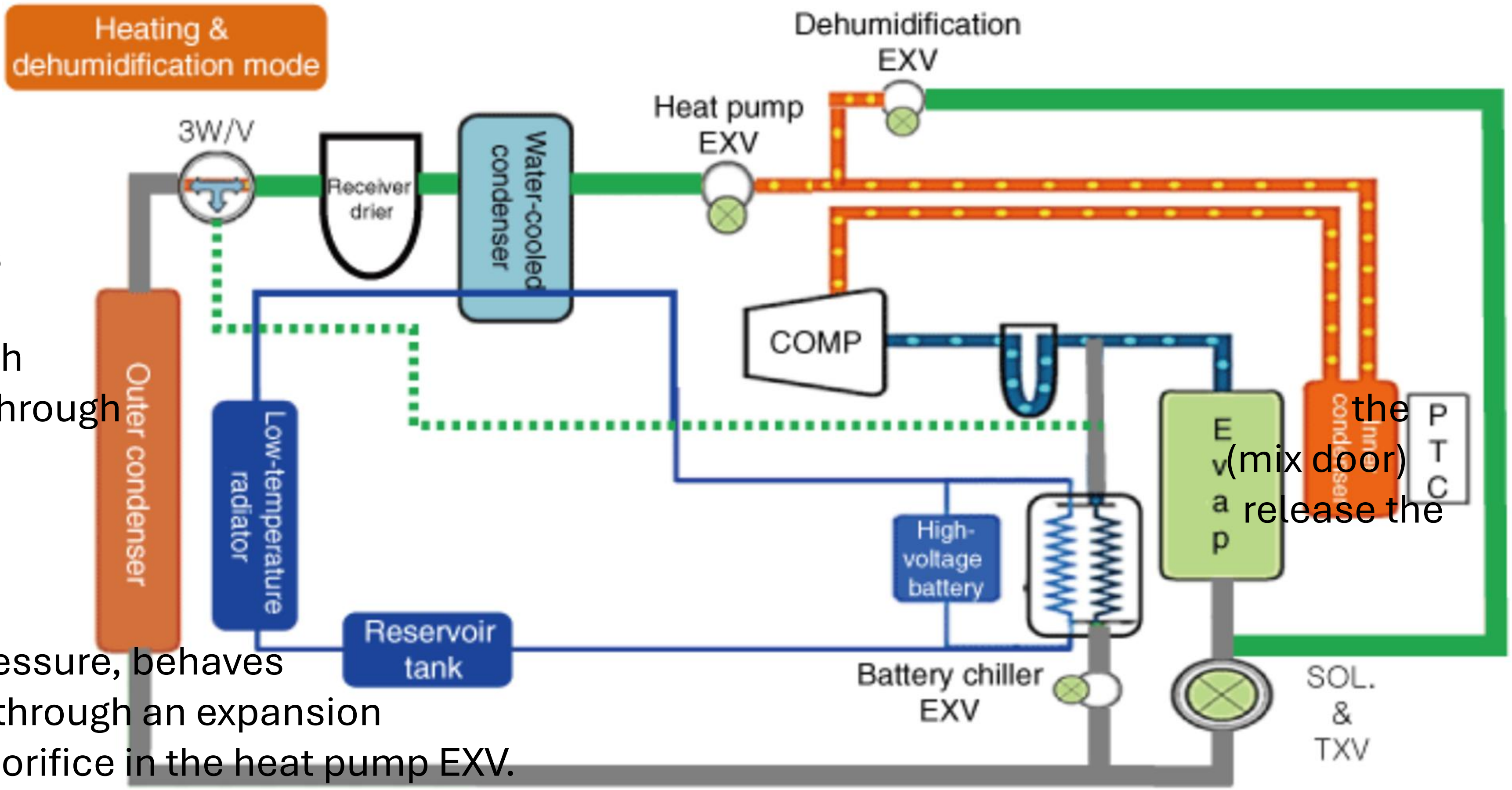
3.The compressed refrigerant, at high temperature and high pressure, behaves as if it were passing through an expansion device via a small orifice in the heat pump EXV. (the dehumidification EXV is closed)

4.The expanded refrigerant, at low temperature and low pressure, absorbs heat from the PE system in the water-cooled condenser.

5.The refrigerant 3-way valve opens in the bypass direction around the outside condenser, to minimize refrigerant heat losses.

6.As it flows through the outside condenser, the refrigerant rejects heat to the atmosphere and condenses further.

7.Having absorbed heat from the drive (PE) system, the refrigerant sends only gaseous refrigerant from the accumulator back to the electric compressor.



1.An electric compressor compresses the refrigerant.
2.As the compressed refrigerant at high temperature and high pressure flows through inside condenser, the blend door opens on the heating side to heat.
3.The compressed refrigerant, at high temperature and high pressure, behaves as if it were passing through an expansion device via a small orifice in the heat pump EXV.
4.The dehumidification EXV opens and part of the expanded refrigerant passes through the evaporator to remove moisture from the cabin air.
5.The expanded refrigerant, at low temperature and low pressure, absorbs heat from the PE system in the water-cooled condenser.
6.The refrigerant 3-way valve opens in the bypass direction around the outside condenser, to minimize refrigerant heat losses.
7.Having absorbed heat from the PE drive system, the refrigerant sends only gaseous refrigerant from the accumulator back to the electric compressor. (the refrigerant partly diverted to dehumidify the cabin air is recombined)

GM / Chevrolet – the Ultium platform

General Motors' next-generation EV platform

▶ Ultium models concerned:

– Chevrolet Equinox VÉ · Blazer VÉ · Silverado VÉ
– GMC Sierra VÉ · GMC Hummer VÉ
– Cadillac Lyriq · Honda Prologue

▶ Architecture thermique Ultium :

– Refrigerant: R-1234yf
– BCM, BECM, HVAC Module
– Refrigerant circuit with direct battery cooling (dedicated chiller)
– Waste heat recovery: OBC (onboard charger) + motor + inverter
– Chaufferette PTC auxiliaire (backup < -20 °C)

▶ GM diagnostic tool:

– GDS2 (Global Diagnostic System 2)
– Guided procedures in GDS2 for every thermal component
– EXV calibration available

Chevrolet Equinox VÉ

GM's affordable EV

► Thermal system (Ultium platform):

- Refrigerant: R-1234yf
- Heat pump standard on ALL trims
- Heat sources: outside air + OBC waste heat + motors
- Direct battery cooling by refrigerant (dedicated chiller)

► Specified heat pump performance:

- Effective in heat pump mode down to -15 °C (-25 °C with waste heat recovery)
- Pré-conditionnement disponible

► Architecture specific to the Equinox EV:

- 3 EXVs: cabin evaporator + battery chiller + outside exchanger
- Electric water pump dedicated to the battery circuit

► Specific diagnostics:

- Tool: GDS2 with MDI2
- Compressor replacement procedure: flush + mandatory GDS2 break-in procedure

Chevrolet Silverado VÉ · GMC Sierra VÉ · GMC Hummer VÉ

Electric trucks and utility vehicles — reinforced thermal architecture

▶ These vehicles have amplified thermal needs compared with cars

– Larger batteries (170–220 kWh on the Hummer EV) → more heat to manage
– High DCFC charging capacity: 350 kW (Silverado EV) → cooling is critical
– Use in extreme conditions

▶ Silverado VÉ / Sierra VÉ :

– Plateforme Ultium 400 V ou 800 V
– Twin battery chillers for greater cooling capacity
– Heat pump standard with motor + OBC heat recovery
– DCFC charging up to 350 kW (with preconditioning)

▶ GMC Hummer VÉ :

– Architecture 4 moteurs (Edition 1) ou 3 moteurs
– 'Ultium Drive' system with integrated thermal management
– Reinforced battery cooling for heavy-duty use
– Special modes: 'Crab Walk' (four-wheel steering) → more motor heat → heat pump called on

▶ Diagnostic points for Ultium trucks:

– Doubled cooling capacity → check both battery chillers
– GDS2: HD-specific procedures in the 'Thermal Management' section

What to do after a metal debris contamination

► **Components that MUST be REPLACED after metal contamination:**

- Compresseur
- All O-rings and seals
- Chiller
- Electronic expansion valves (EXV)
- Hoses that contain a muffler
- Accumulator/dryer with desiccant
- Refrigerant flow valves (RFV)
- Schrader valves on the service ports
- Check valves
- Heater Core

► **Components that MUST be FLUSHED after metal contamination:**

- All other metal lines that do not contain a muffler
- Condenseur
- Evaporator

► **Flushing removes all the POE oil from the A/C system. The A/C system must then be refilled with the correct quantity of POE oil.**

Mode 1: Maximum Cooling with reheat

This mode is when the vehicle needs to cool the propulsion battery while also both heating (reheat) and cooling the cabin. The heat rejection is significantly higher than what is needed for cabin comfort, so the front condenser is used to reject the heat to the environment.

COMPONENT	CONDITION
Air Conditioning With Motor Compressor	Low side control within NVH (noise, vibration, harshness) and pressure limits
Shutter / CRFM Fans	Open / On per airflow request
Evaporator and EXV	Open, control to evaporator out superheat
Drive Motor Battery Coolant Cooler and EXV	Open, control to drive motor battery coolant cooler out superheat
Air Conditioning Condenser & RFV	Open, control to air conditioning condenser sub-cool
Heater Core & RFV	Open, control to heater core out refrigerant temperature

Mode 2: Maximum Cooling

This mode is when the vehicle needs to cool the propulsion battery and the cabin. No reheat is needed so all the heat rejection is done by the front condenser.

COMPONENT	CONDITION
Air Conditioning With Motor Compressor	Low side control within NVH (noise, vibration, harshness) and pressure limits
Shutter / CRFM Fans	Open / On per airflow request
Evaporator and EXV	Open, control to evaporator out superheat
Drive Motor Battery Coolant Cooler and EXV	Open, control to drive motor battery coolant cooler out superheat
Air Conditioning Condenser & RFV	Open, control to air conditioning condenser sub-cool
Heater Core & RFV	Closed

Mode 3: Cabin Cooling with reheat

This mode is when the propulsion battery does not need to be cooled but the cabin requires both heating (reheat) and cooling. The heat rejection is significantly higher than what is needed for cabin comfort, so the front condenser is used to reject the heat to the environment.

COMPONENT	CONDITION
Air Conditioning With Motor Compressor	Low side control within NVH (noise, vibration, harshness) and pressure limits
Shutter / CRFM Fans	Open / On per airflow request
Evaporator and EXV	Open, control to evaporator out superheat
Drive Motor Battery Coolant Cooler and EXV	Closed
Air Conditioning Condenser & RFV	Open, control to air conditioning condenser sub-cool
Heater Core & RFV	Open, control to heater core out refrigerant temperature

Mode 4: Cabin Heating with dehumidification

This mode is when the propulsion battery does not need to be cooled but the cabin requires both heating and cooling (dehumidification). Since the cabin heating is greater than the heat supplied from the evaporator, the drive motor battery coolant cooler is used to supply heat.

COMPONENT	CONDITION
Air Conditioning With Motor Compressor	Low side control within NVH (noise, vibration, harshness) and pressure limits
Shutter / CRFM Fans	Closed / Off
Evaporator and EXV	Open, control to evaporator out superheat
Drive Motor Battery Coolant Cooler and EXV	Open, control to drive motor battery coolant cooler out superheat
Air Conditioning Condenser & RFV	Closed
Heater Core & RFV	Open, control to heater core out refrigerant temperature

Mode 5: Cabin Heating with Propulsion Cooling

This mode is when both the cabin needs to be heating and the propulsion battery needs cooling. The heat rejection from the propulsion battery is higher than what the cabin needs so the excess is rejected to the environment.

COMPONENT	CONDITION
Air Conditioning With Motor Compressor	Low side control within NVH (noise, vibration, harshness) and pressure limits
Shutter / CRFM Fans	Controlled per airflow request / On per airflow request
Evaporator and EXV	Closed
Drive Motor Battery Coolant Cooler and EXV	Open, control to drive motor battery coolant cooler out superheat
Air Conditioning Condenser & RFV	Open, control to air conditioning condenser sub-cool
Heater Core & RFV	Open, control to heater core out refrigerant temperature

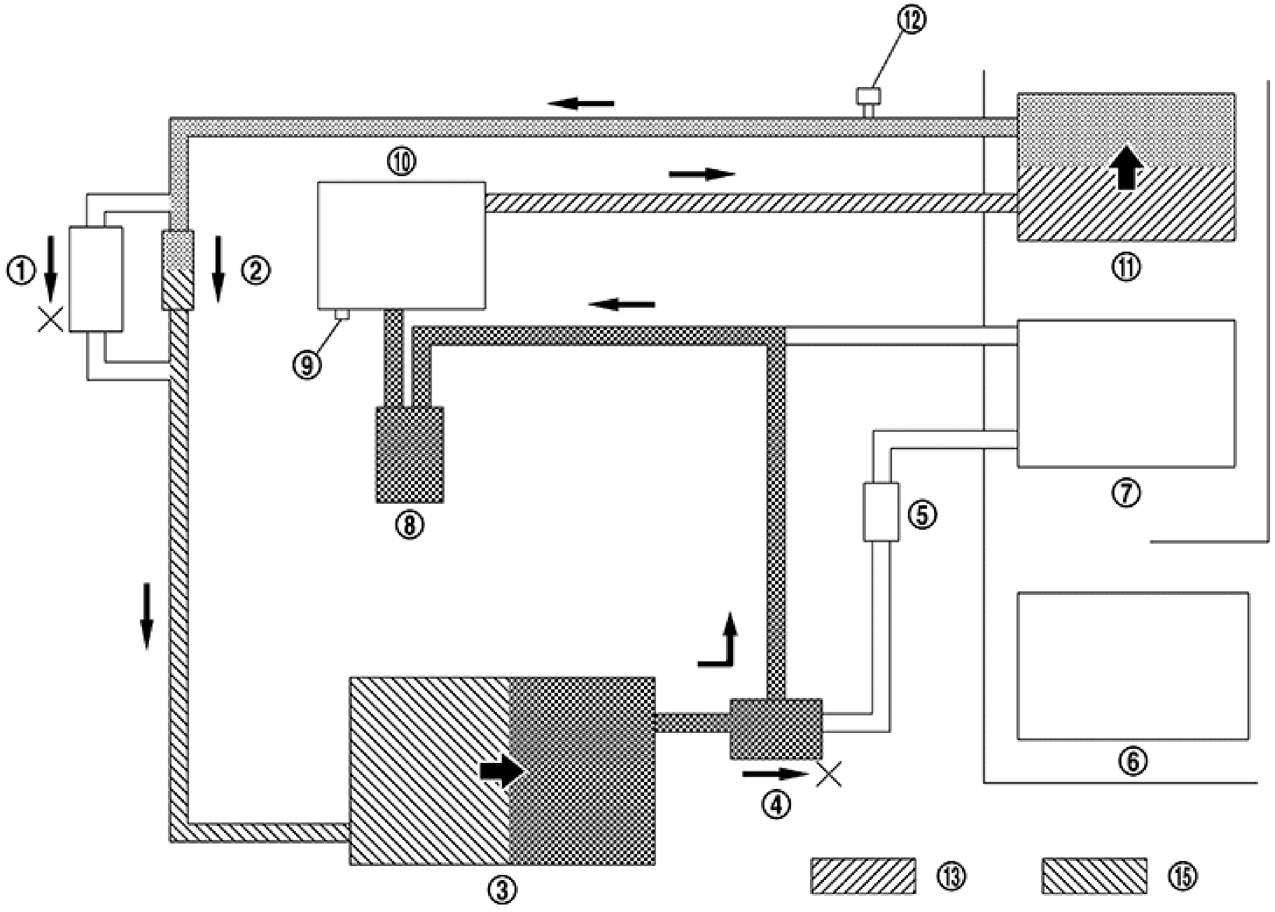
Mode 6: Cabin Heating

This mode is when the cabin needs heating and the propulsion system does not need cooling.

COMPONENT	CONDITION
Air Conditioning With Motor Compressor	High side control within NVH (noise, vibration, harshness) and pressure limits
Shutter / CRFM Fans	Closed / Off
Evaporator and EXV	Closed
Drive Motor Battery Coolant Cooler and EXV	Open, control to drive motor battery coolant cooler out superheat
Air Conditioning Condenser & RFV	Closed
Heater Core & RFV	Open, control to heater core out refrigerant temperature

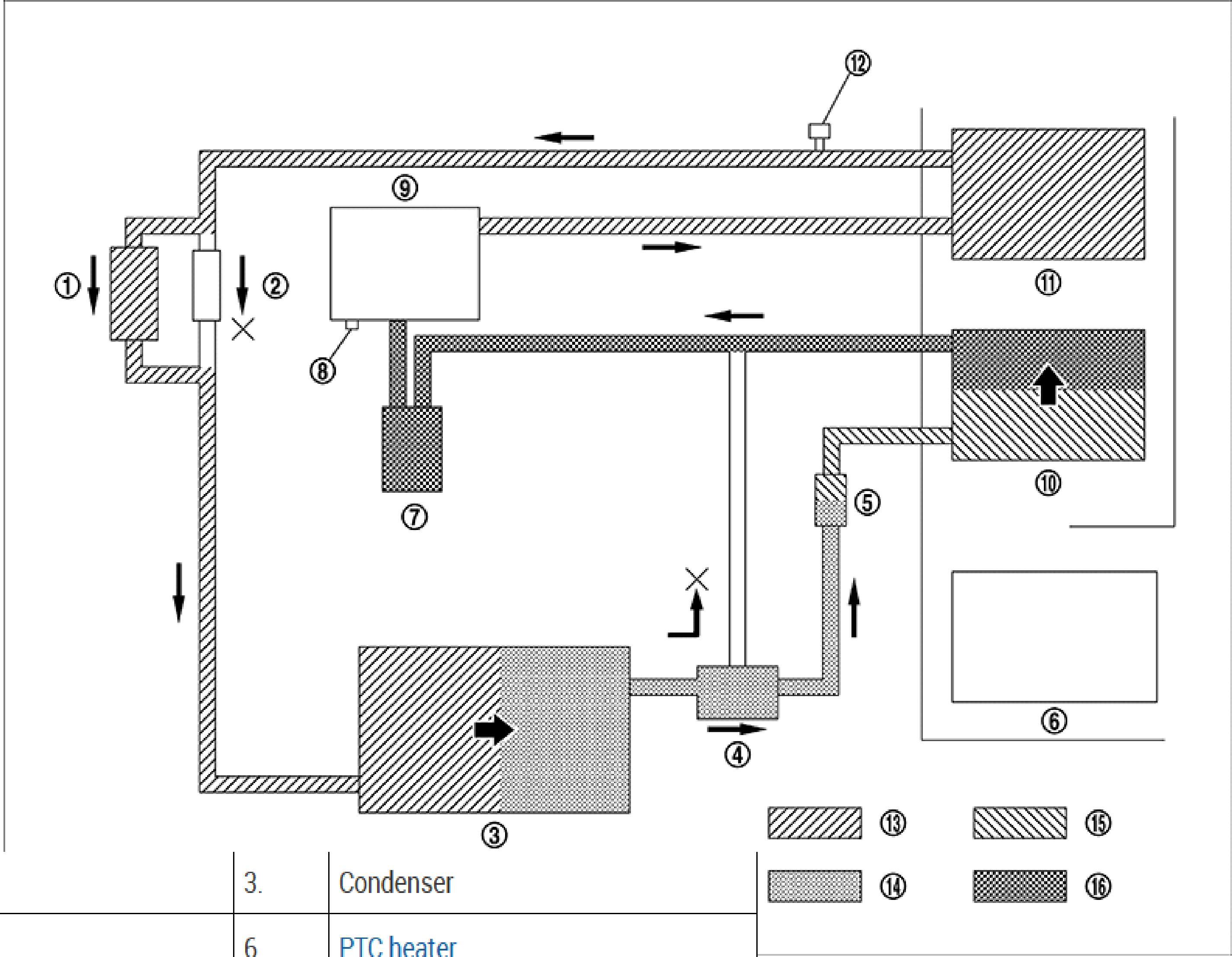
2e generation (2018-2025)

Cabin Heating



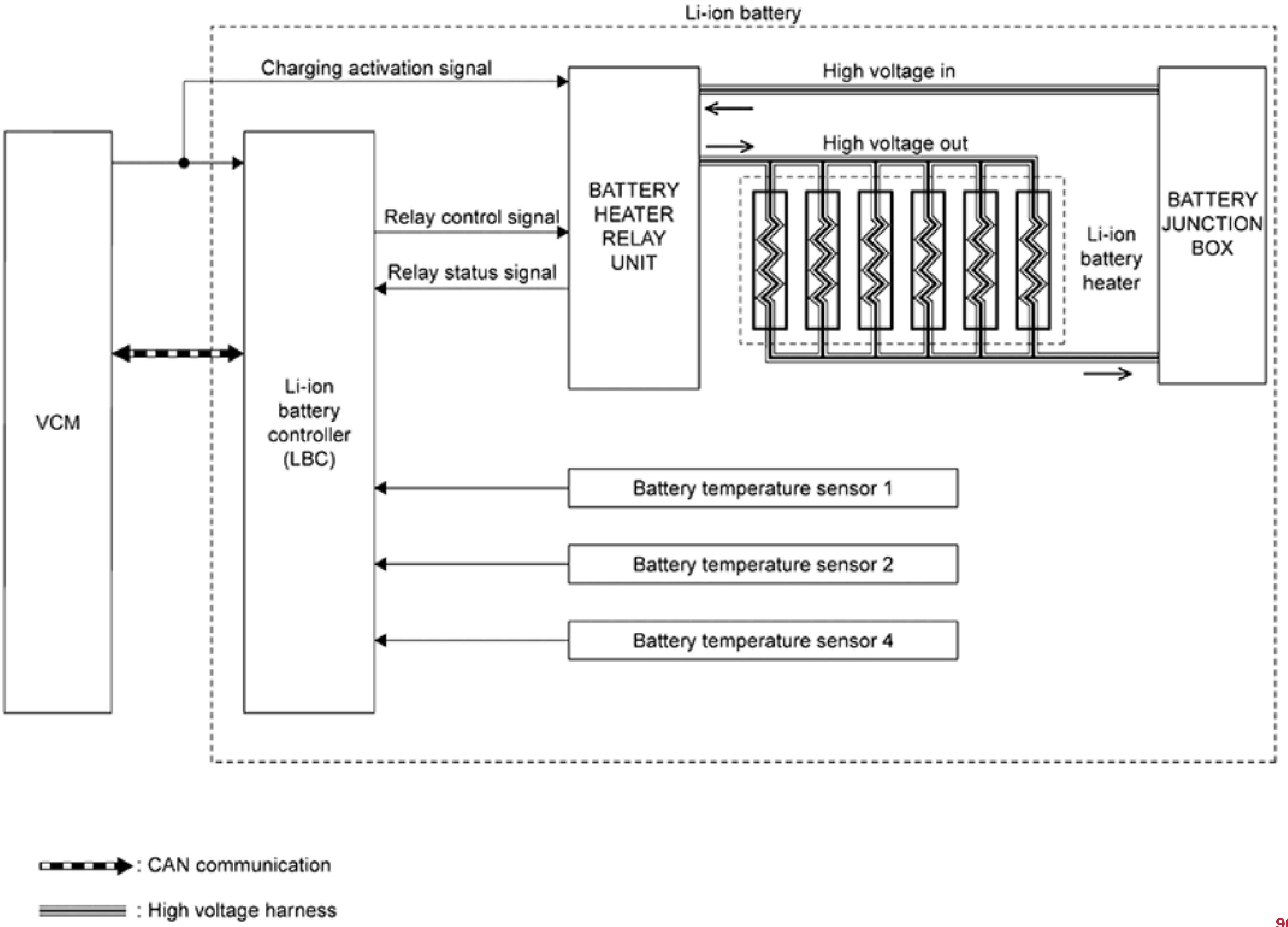
1.	2-way valve	2.	Orifice tube (Inner condenser)	3.	Condenser
4.	3-way valve	5.	Orifice tube (Evaporator)	6.	PTC heater
7.	Evaporator	8.	Accumulator	9.	Pressure relief valve
10.	Electric compressor	11.	Inner condenser	12.	Pressure sensor
13.	High-pressure gas	14.	High-pressure liquid	15.	Low-pressure liquid
16.	Low-pressure gas				

Cabin Cooling



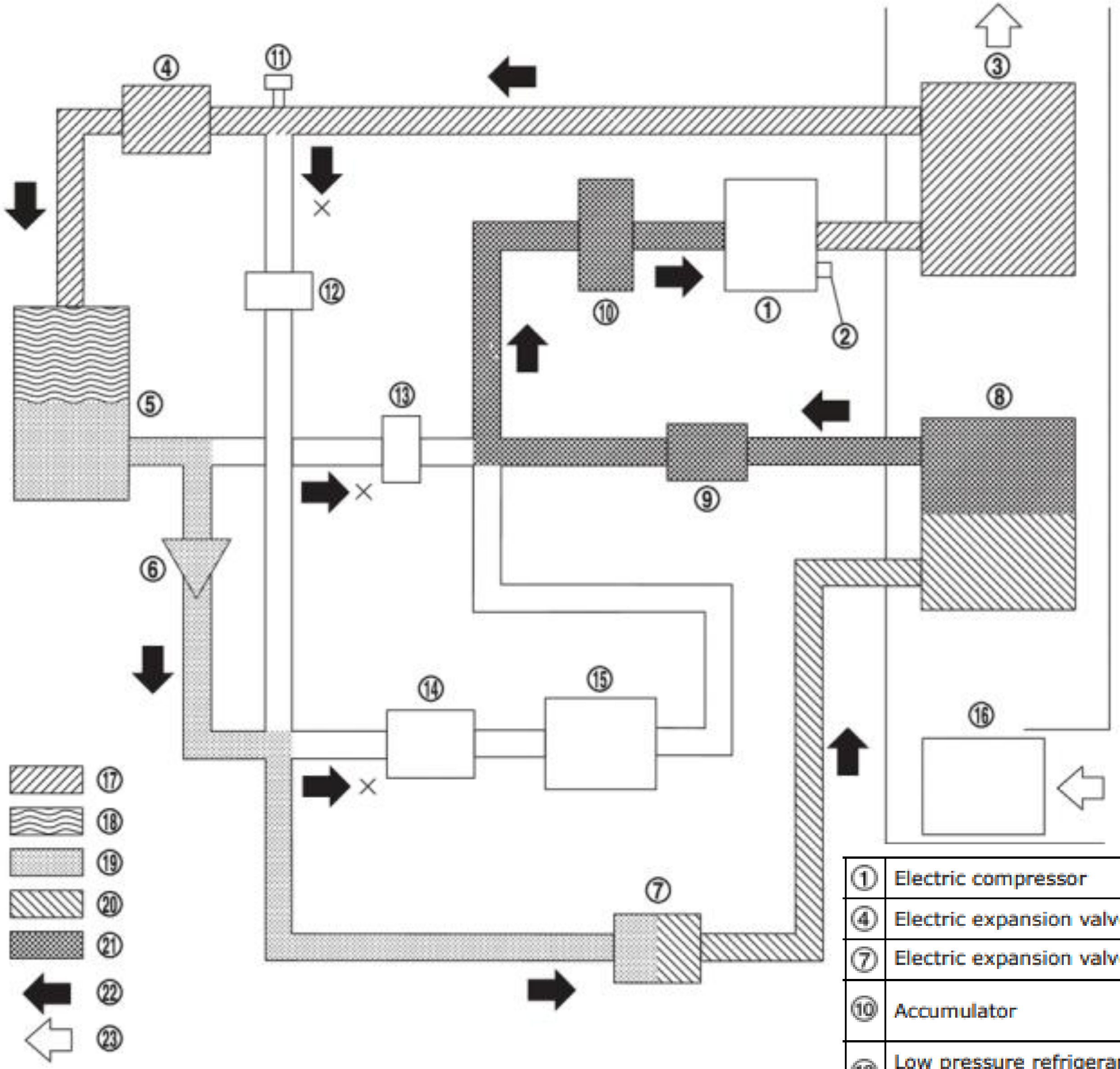
1.	2-way valve	2.	Orifice tube (Inner condenser)	3.	Condenser
4.	3-way valve	5.	Orifice tube (Evaporator)	6.	PTC heater
7.	Accumulator	8.	Pressure relief valve	9.	Electric compressor
10.	Evaporator	11.	Inner condenser	12.	Pressure sensor
13.	High-pressure gas	14.	High-pressure liquid	15.	Low-pressure liquid
16.	Low-pressure gas				

Battery Heating

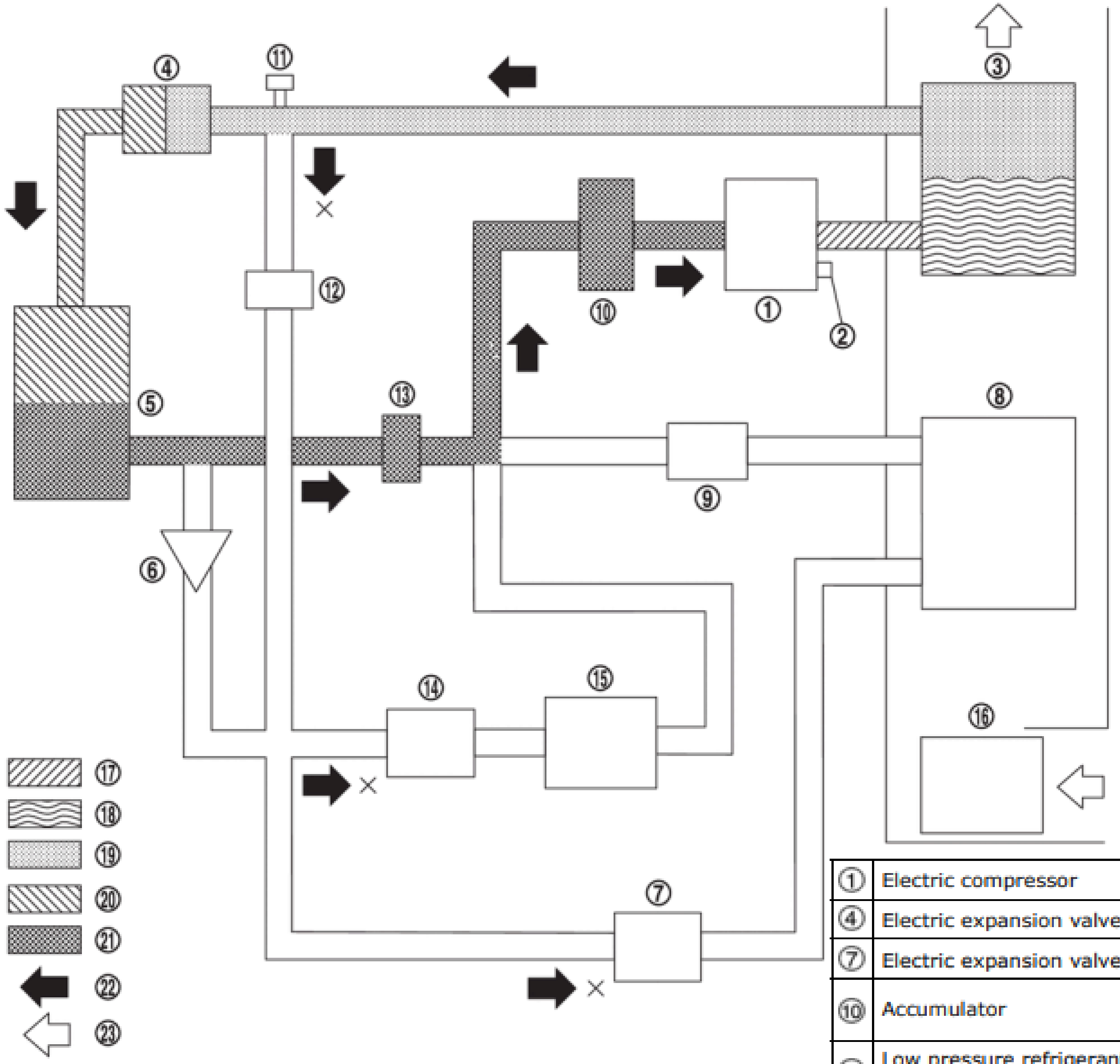


3^{e} generation (2026+)

COOLER MODE



①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
⑯	Blower motor	⑰	High-pressure gas	⑱	Gas-liquid two phase in high pressure
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		



HEATER MODE AND HEATER + THERMAL RECOVERY (CONDENSER) MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
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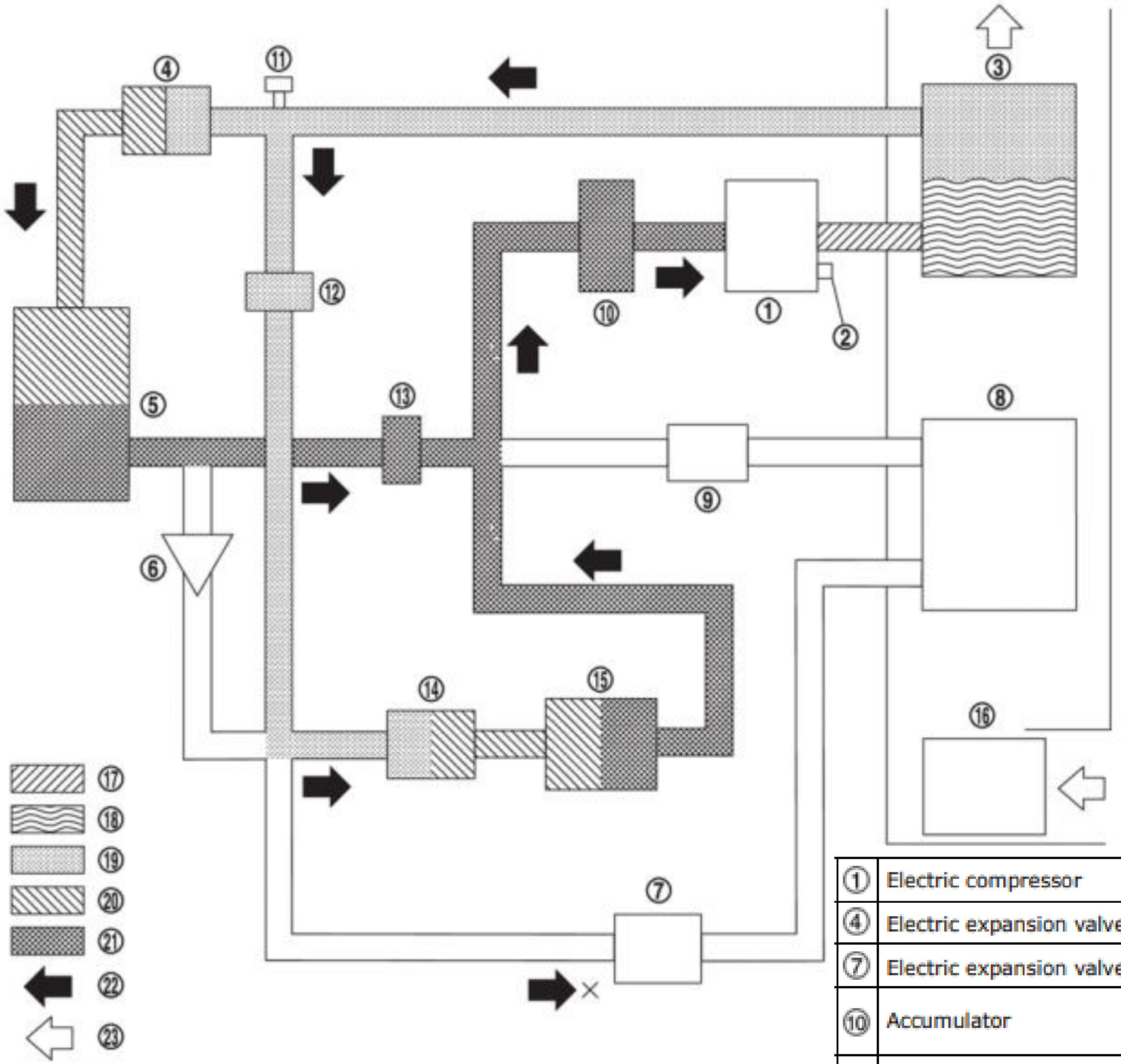
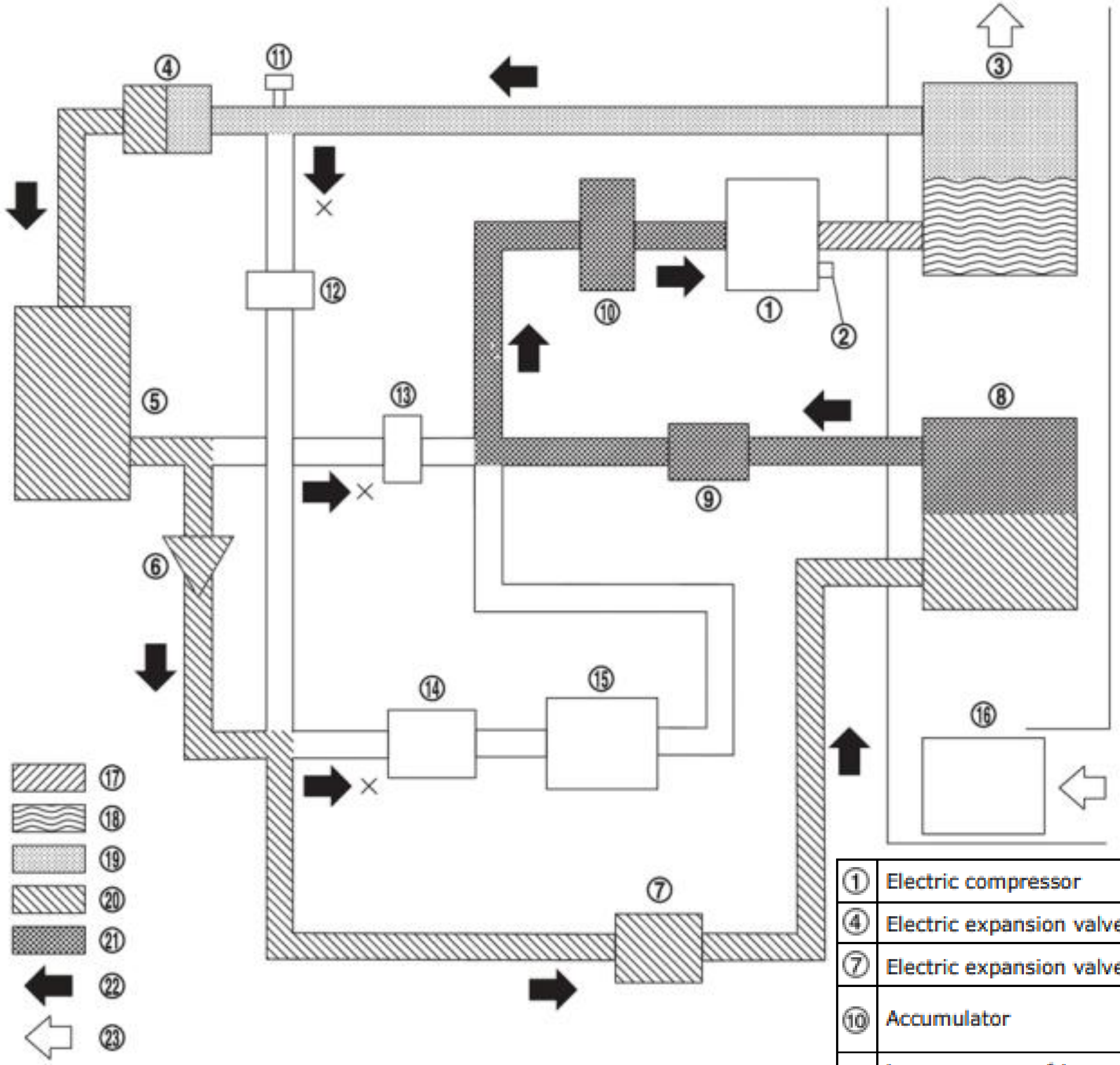
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		



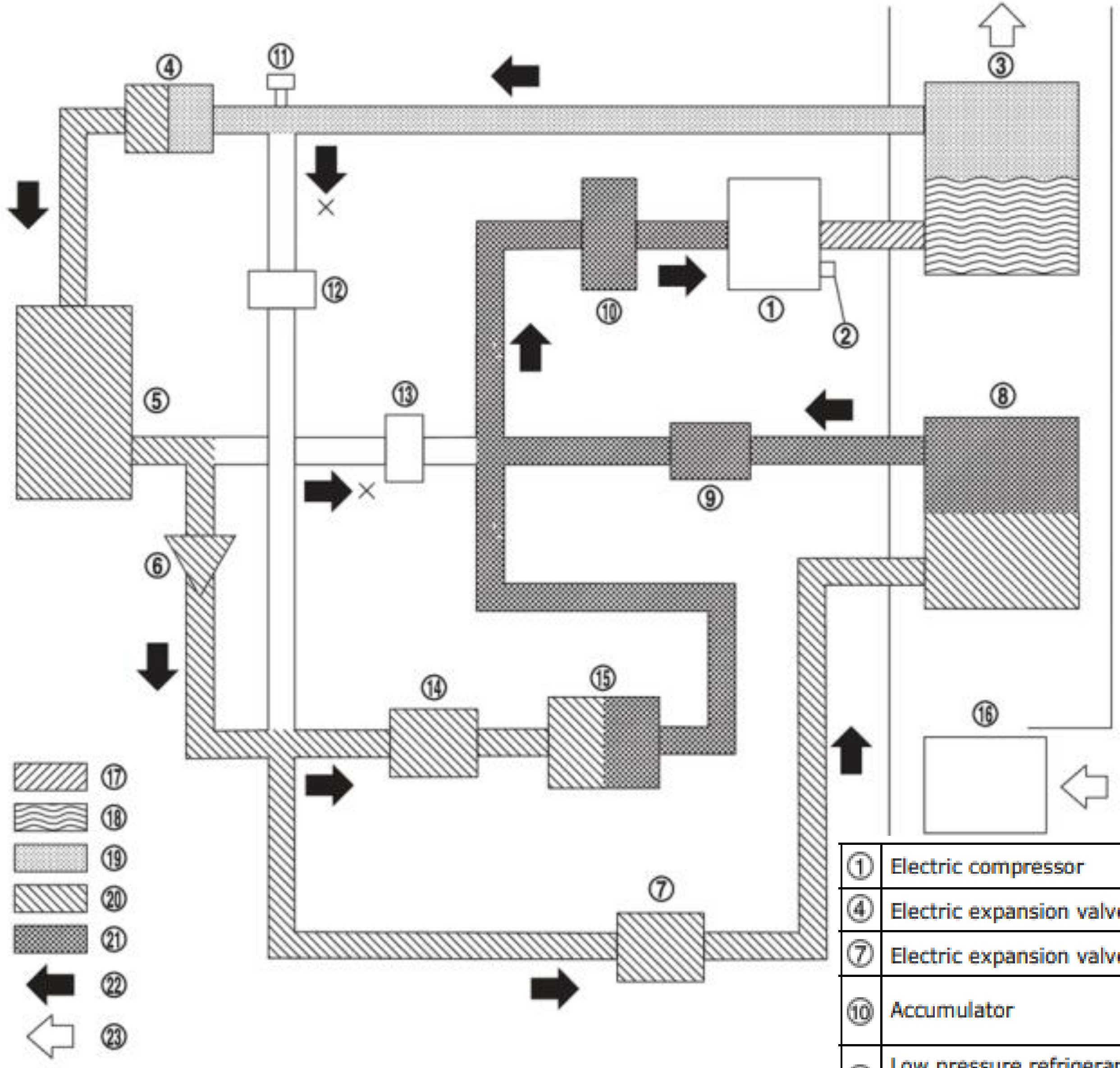
HEATER + THERMAL RECOVERY (CONDENSER & CHILLER) MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
⑯	Blower motor	⑰	High-pressure gas	⑱	Gas-liquid two phase in high pressure
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		



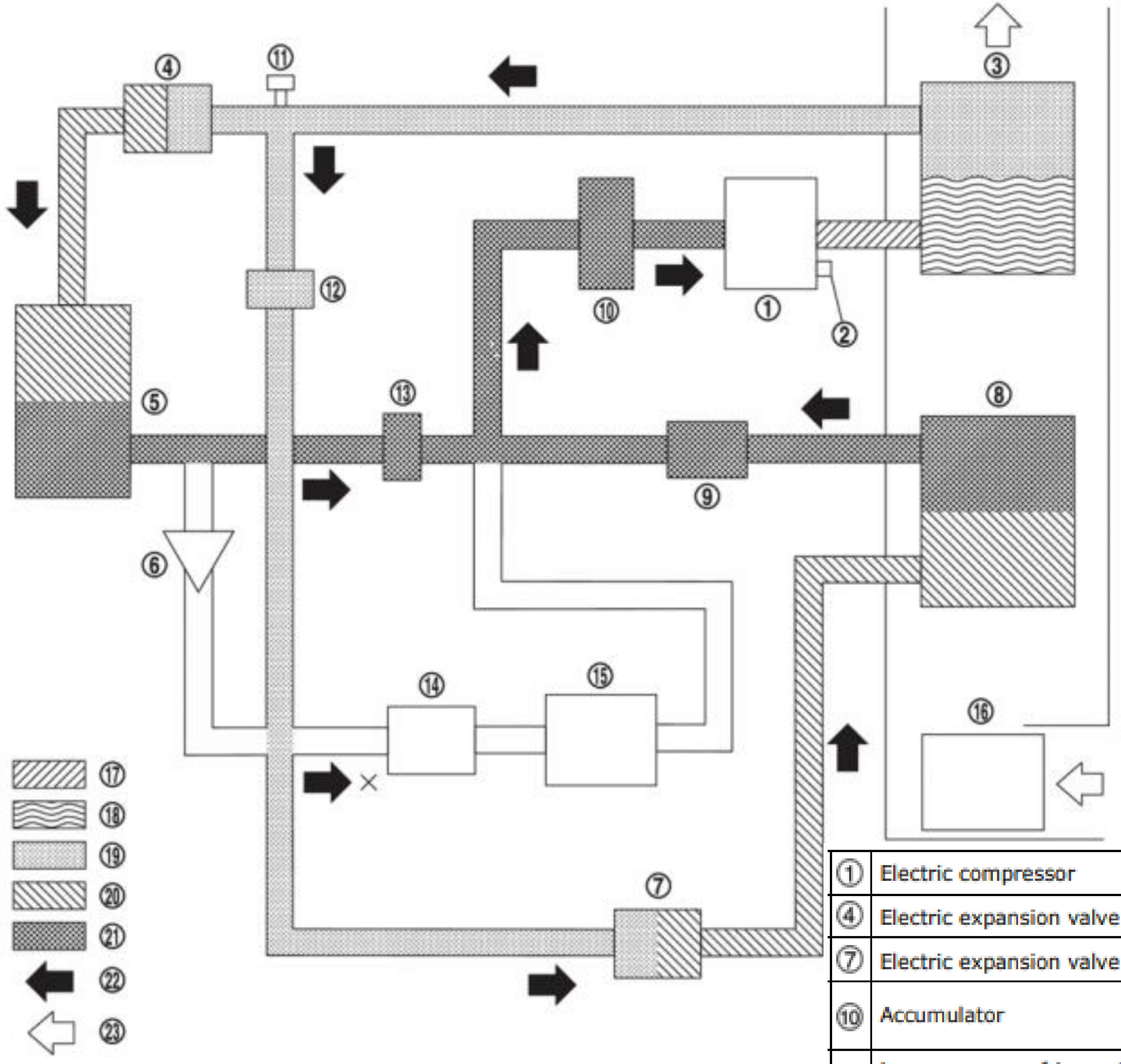
SERIES DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER) MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
⑯	Blower motor	⑰	High-pressure gas	⑱	Gas-liquid two phase in high pressure
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		



SERIES DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER & CHILLER) MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
⑯	Blower motor	⑰	High-pressure gas	⑱	Gas-liquid two phase in high pressure
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		



PARALLEL DEHUMIDIFICATION HEATING MODE AND PARALLEL DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER) MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
⑯	Blower motor	⑰	High-pressure gas	⑱	Gas-liquid two phase in high pressure
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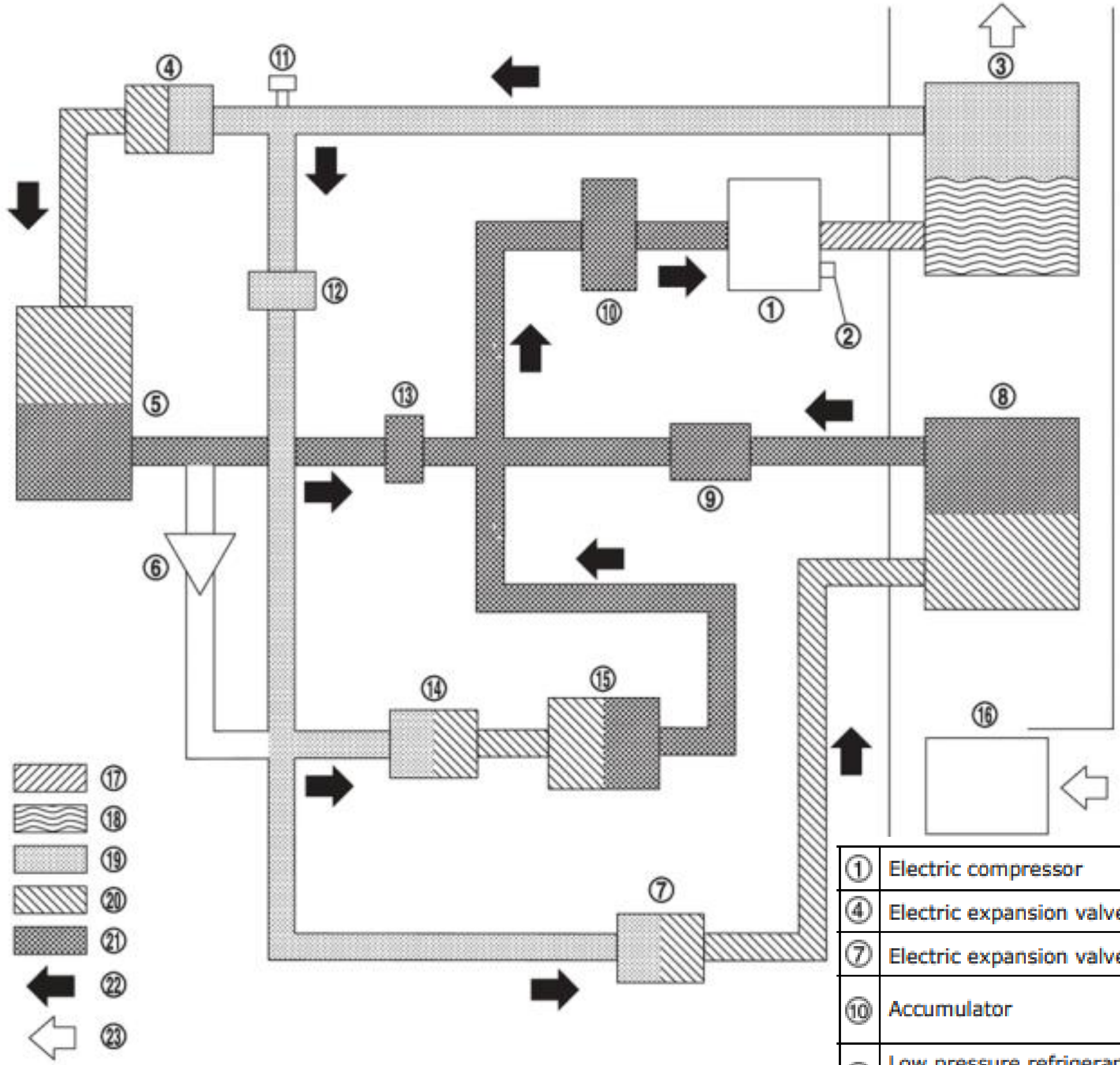
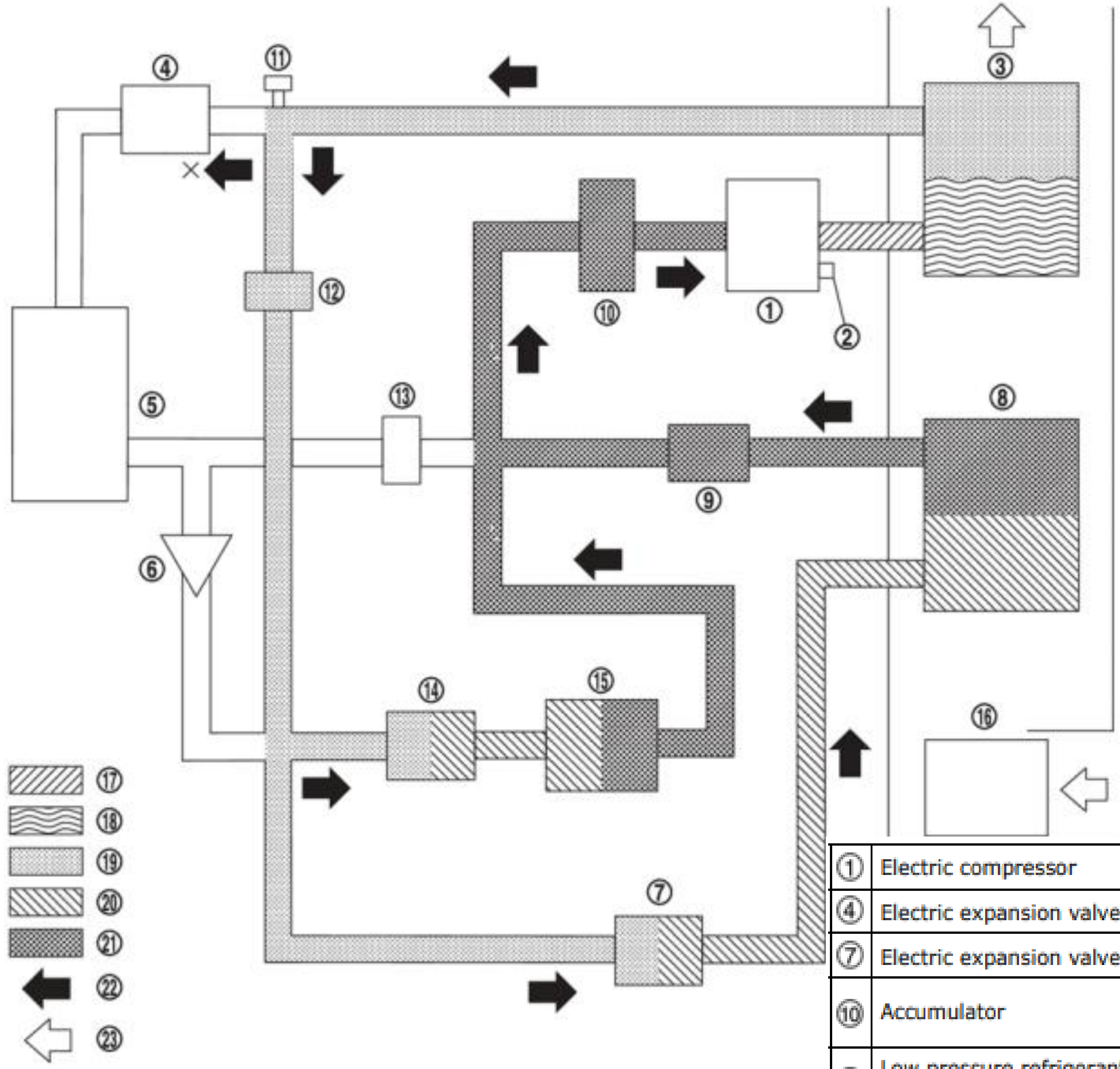
㉒	Refrigerant flow	㉓	Wind flow		



PARALLEL DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CONDENSER & CHILLER) MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
⑯	Blower motor	⑰	High-pressure gas	⑱	Gas-liquid two phase in high pressure
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		

PARALLEL DEHUMIDIFICATION HEATING + THERMAL RECOVERY (CHILLER) MODE



①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
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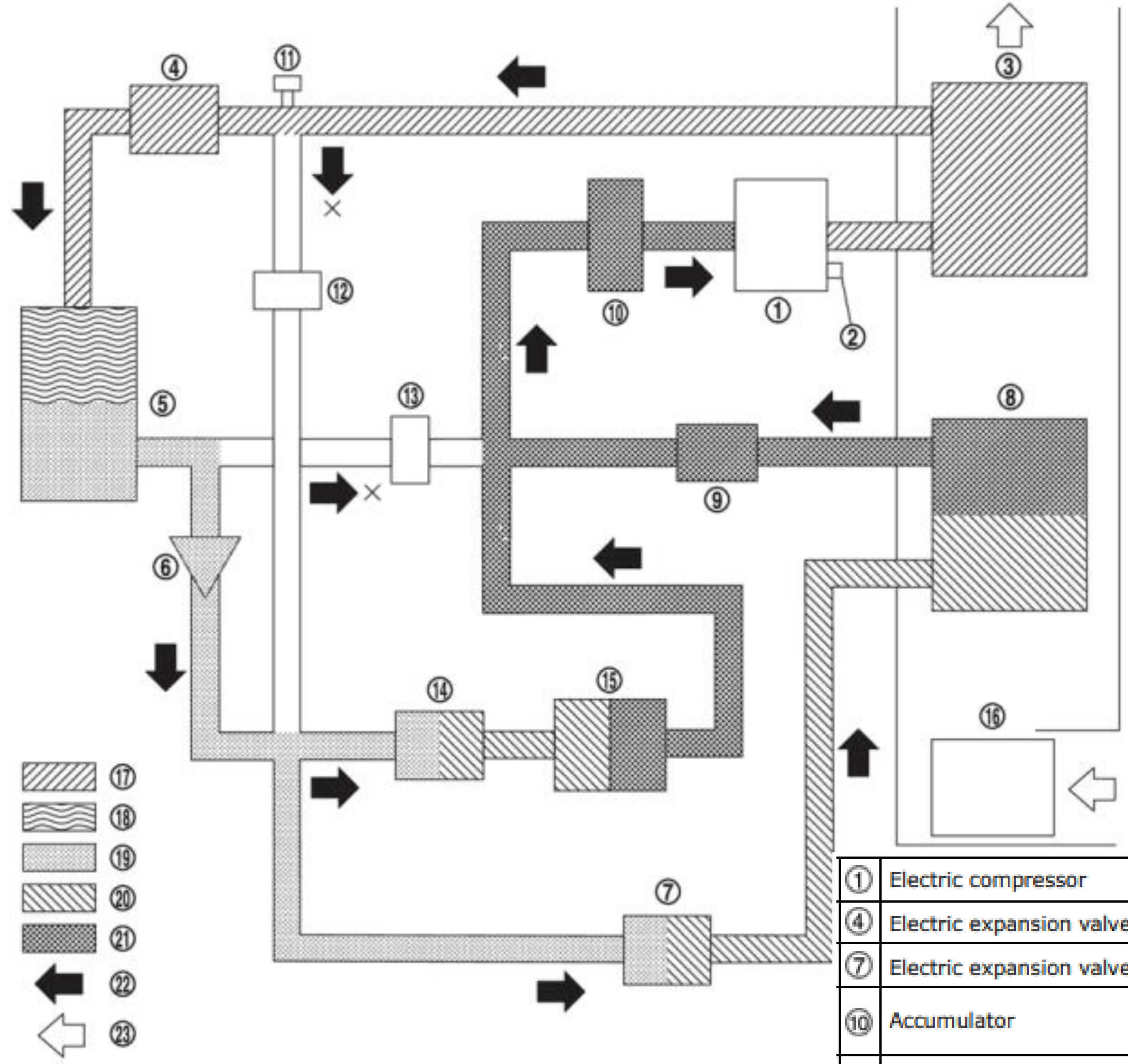
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		

COOLER + HIGH VOLTAGE BATTERY COOLING MODE



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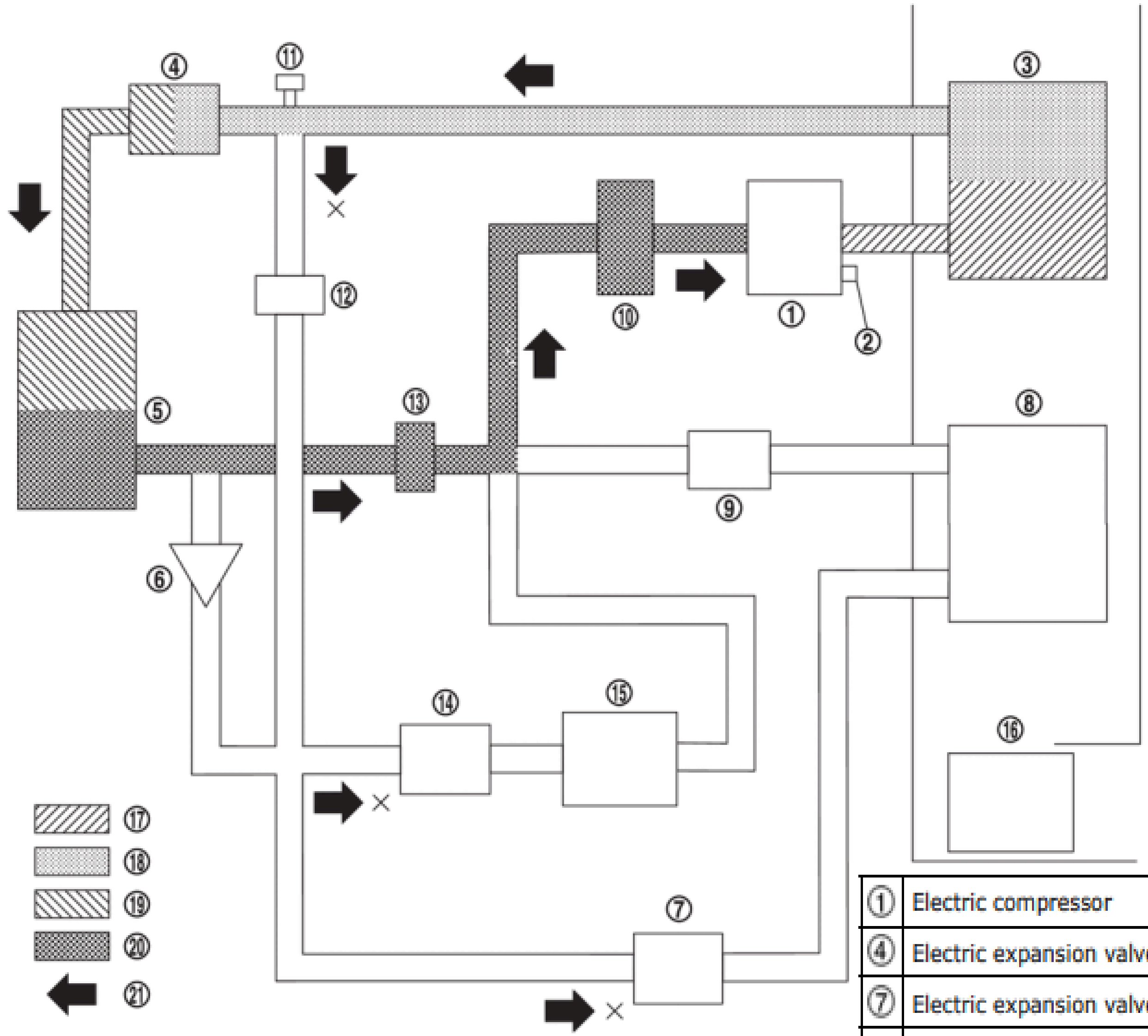
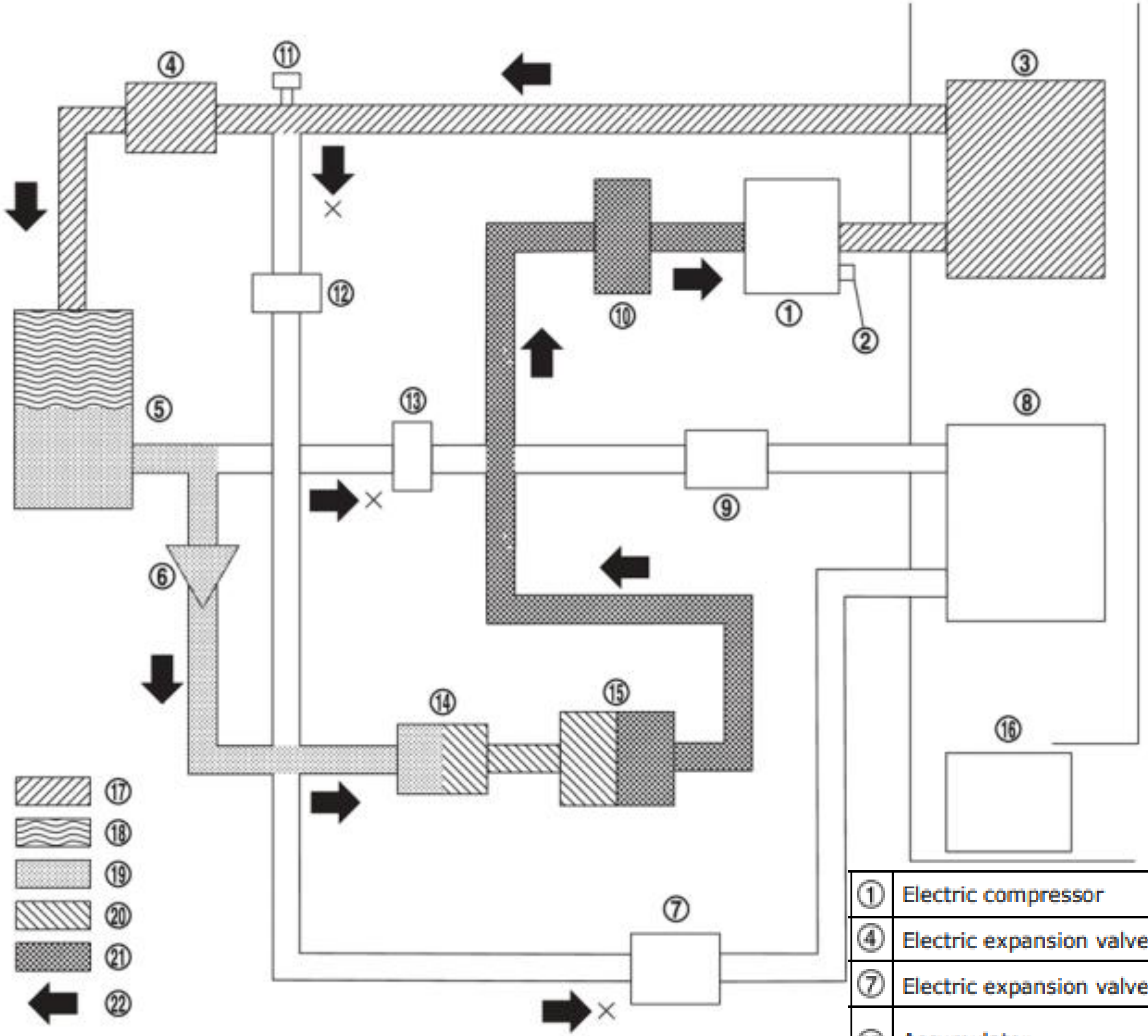
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow	㉓	Wind flow		



DEFROST MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
⑬	Low pressure refrigerant channel switching valve	⑭	Expansion valve (battery chiller)	⑮	Battery coolant chiller
⑯	Blower motor	⑰	High-pressure gas	⑱	High-pressure liquid
⑲	Gas-liquid two phase in low pressure	⑳	Low-pressure gas	㉑	Refrigerant flow



HIGH VOLTAGE BATTERY COOLING MODE

①	Electric compressor	②	Pressure relief valve	③	Inner condenser
④	Electric expansion valve (heater)	⑤	Condenser	⑥	One-way valve
⑦	Electric expansion valve (cooler)	⑧	Evaporator	⑨	Pressure regulator
⑩	Accumulator	⑪	Refrigerant pressure sensor	⑫	High pressure refrigerant channel switching valve
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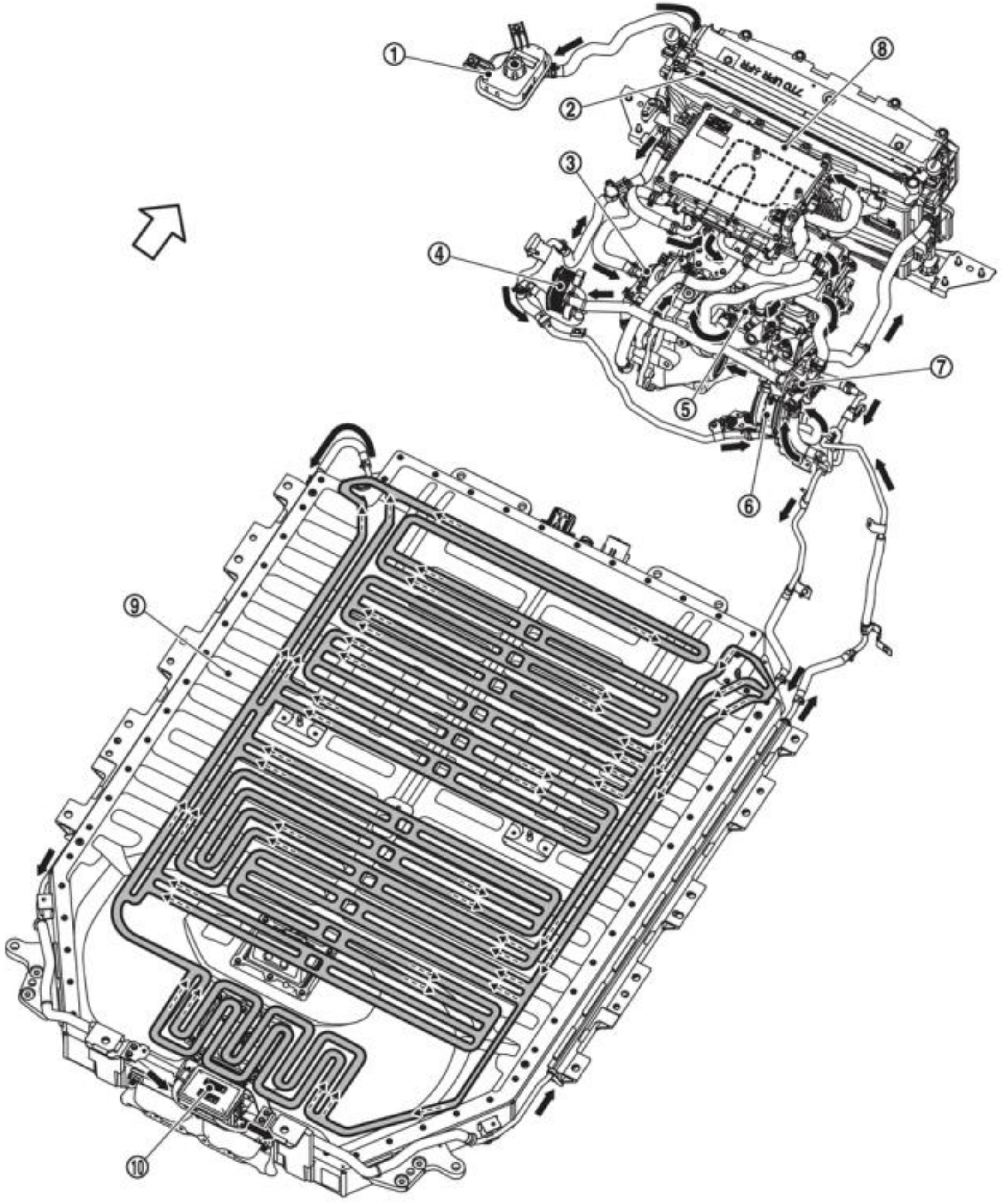
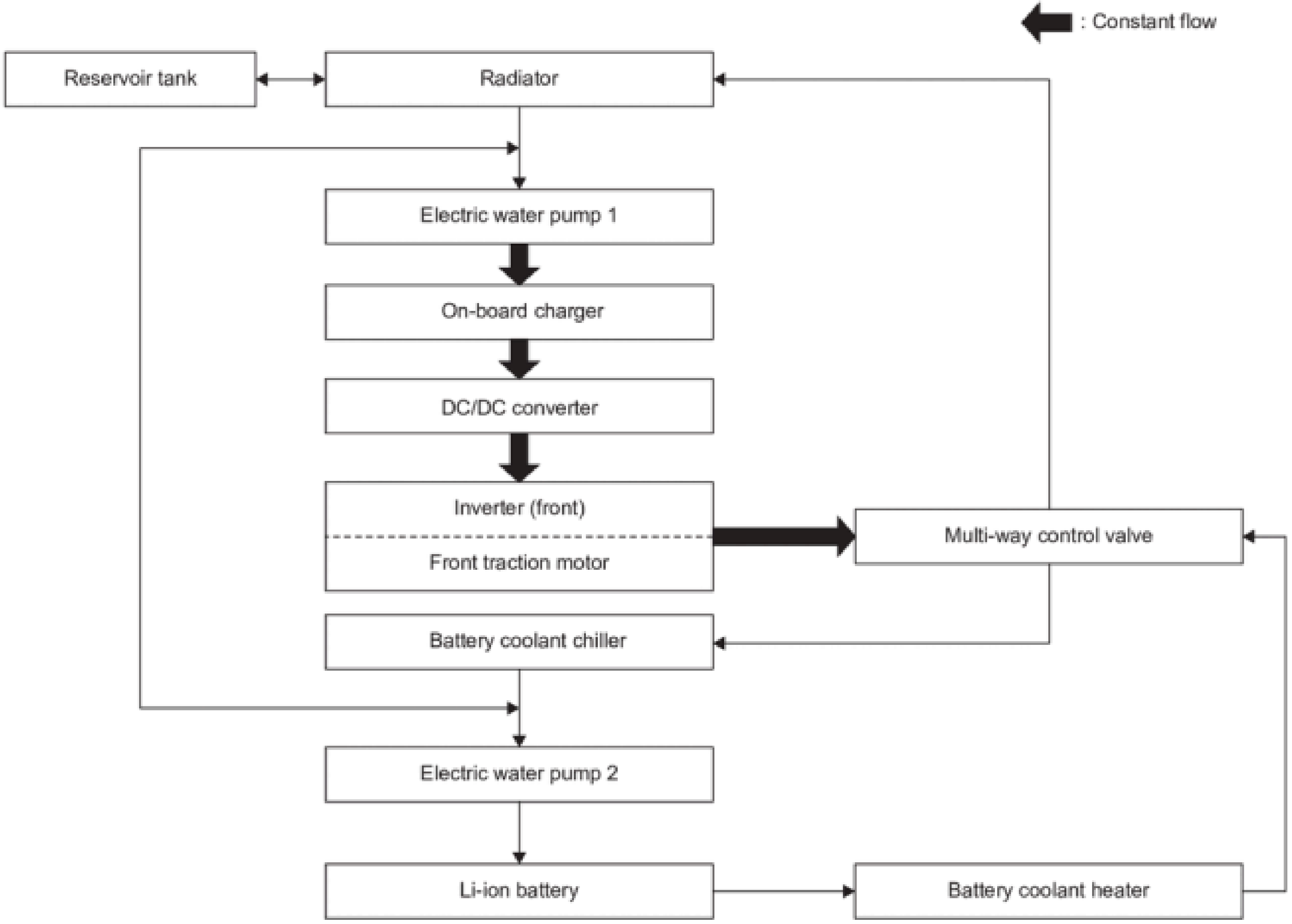
⑯	Blower motor	⑰	High-pressure gas	⑱	Gas-liquid two phase in high pressure
⑲	High-pressure liquid	⑳	Gas-liquid two phase in low pressure	㉑	Low-pressure gas
㉒	Refrigerant flow				

HIGH VOLTAGE BATTERY COOLING MODE






Ford

Ford – EV thermal management system

Mustang Mach-E · F-150 Lightning · E-Transit · Explorer VÉ

► **Ford is the second-largest NA manufacturer by EV volume (behind Tesla)**

– F-150 Lightning: the best-selling full-size electric truck in North America
– Mustang Mach-E: Ford's most popular electric SUV

► **Refrigerant: R-1234yf and R-134a**

► **Architecture commune Ford VÉ :**

– Thermal control modules HVAC, ACCM, HPVCM
– Communication : CAN FD
– Direct battery cooling by refrigerant (dedicated chiller)
– Waste heat recovery: motor + inverter + OBC

► **Ford diagnostic tool:**

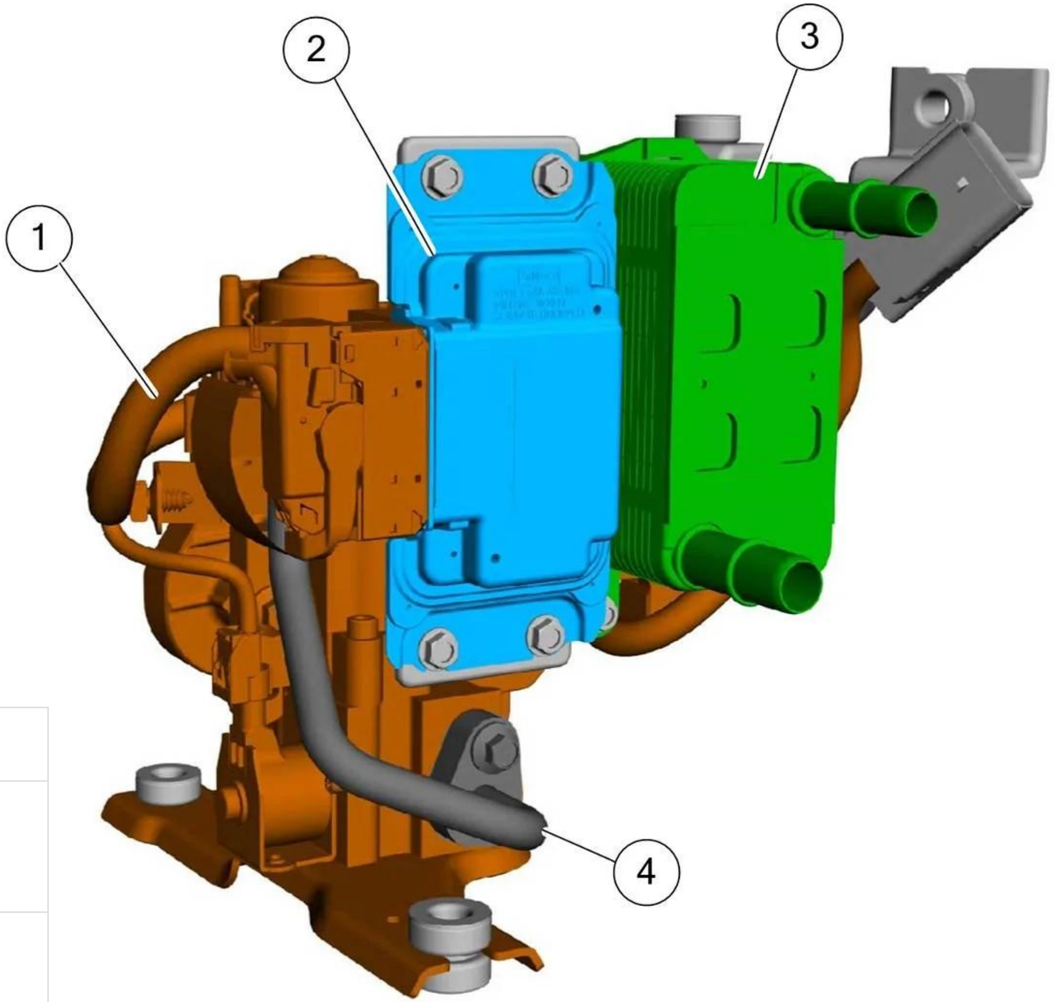
– FDRS (Ford Diagnostic and Repair System) — remplace IDS/FJDS depuis 2018
– Guided procedures available in FDRS for: refrigerant recharge, EXV calibration, remplacement compresseur, tests actionneurs thermiques

Ford F-150 Lightning

The #1 electric truck in North America

- **The F-150 Lightning is the best-selling electric truck in the NA truck segment**
  – Batterie Standard Range (98 kWh) ou Extended Range (131 kWh)
- **Thermal system (all trim levels):**
  – Heat pump standard on every version
  – Heat sources: outside air + motor + inverter + OBC
  – Direct battery cooling by refrigerant with a chiller
- **Unique feature: 'Pro Power Onboard'**
  – The Lightning can export up to 9.6 kW to external devices
  – Under heavy Pro Power Onboard use: more heat → the heat pump is called on to cool the DCACA
- **Performance PAC en conditions froides :**
  – At -20 °C: WLTP range ~320 km → real-world range ~250 km (heat pump active)
  – Without a heat pump at -20 °C: range down to ~150–180 km (PTC alone)
- **Specific diagnostics:**
  – FDRS + VCM3 obligatoire
  – Thermal fault tree in FDRS: 'Heat Pump / Thermal Management Diagnosis'

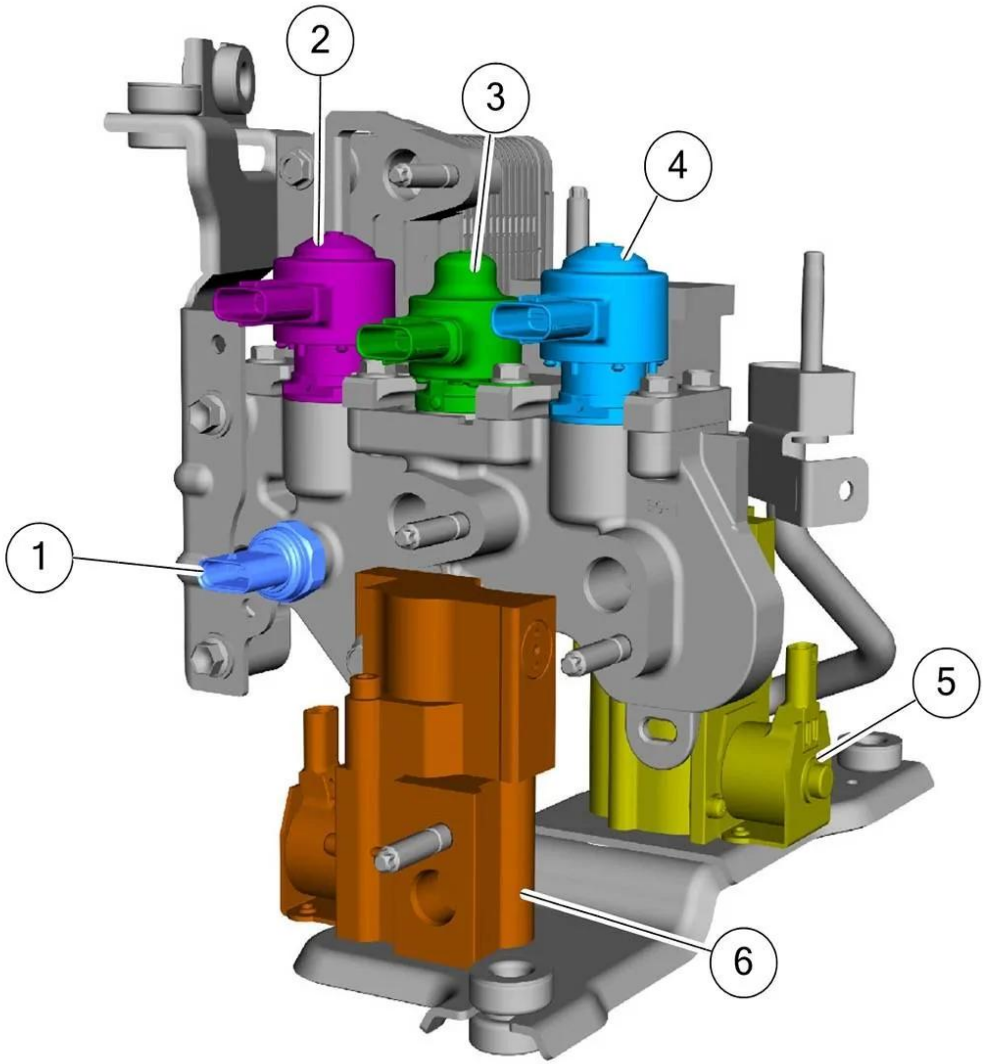
Ford F-150 Lightning



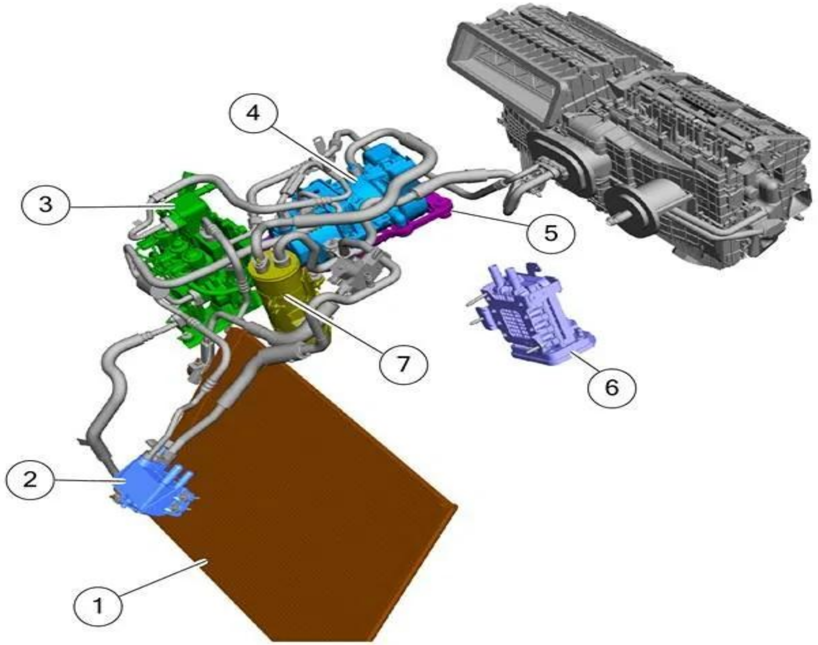
1	Heat pump valve unit
2	Heat pump valve control module (HPVCM)
3	Water cooled condenser (WCC)
4	Heat pump valve unit to evaporator line

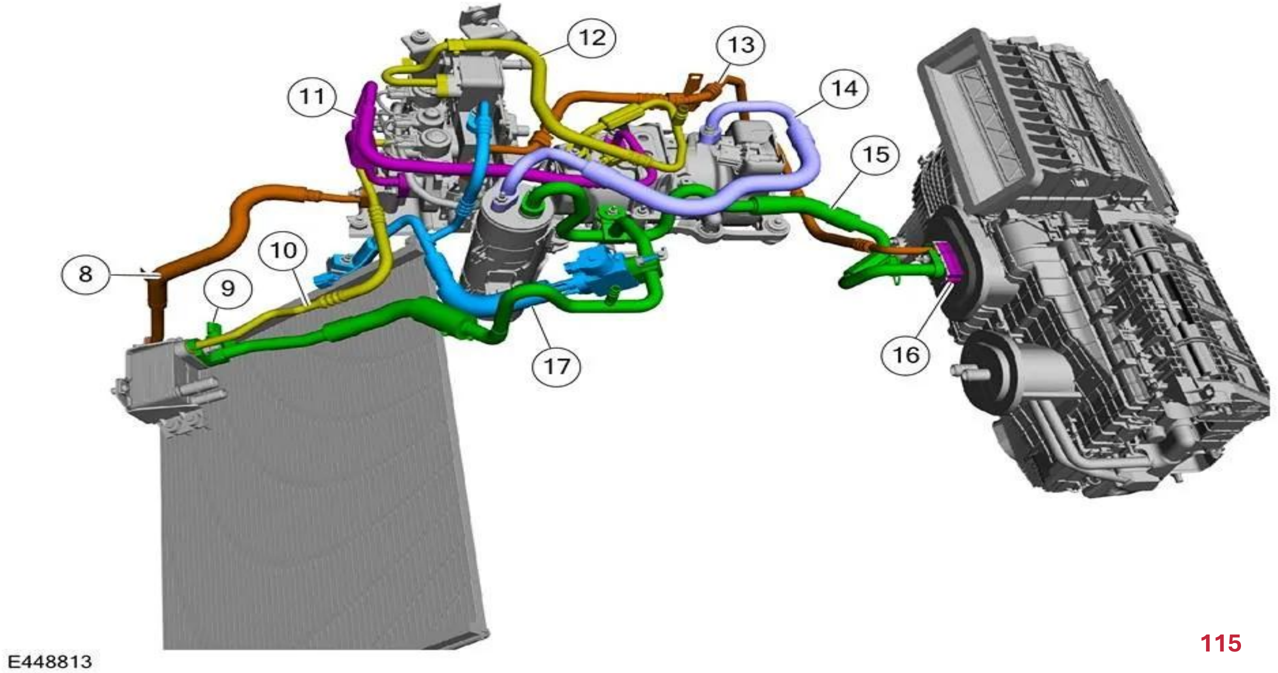
Ford F-150 Lightning

1	Refrigerant pressure and temperature sensor A
2	Vapor injection heat pump heating electronic expansion valve (HEXV)
3	Vapor injection heat pump battery chiller electronic expansion valve (BCEXV)
4	Vapor injection heat pump cooling electronic expansion valve (CEXV)
5	Vapor injection heat pump cooling liquid gas seperator (CLGS)
6	Vapor injection heat pump heating liquid gas seperator (HLGS)



1	Condenser
2	High voltage battery coolant cooler
3	Heat pump assembly
4	Refrigerant compressor
5	Refrigerant compressor bracket
6	Cabin coolant heater
7	Accumulator
8	Condenser inlet line
9	Refrigerant pressure and temperature sensor c
10	High voltage battery coolant cooler liquid line
11	Refrigerant vapor injection line
12	Refrigerant compressor outlet line
13	Evaporator inlet line
14	Refrigerant compressor to accumulator line
15	Evaporator outlet line
16	Thermostatic expansion valve
17	Condenser outlet line

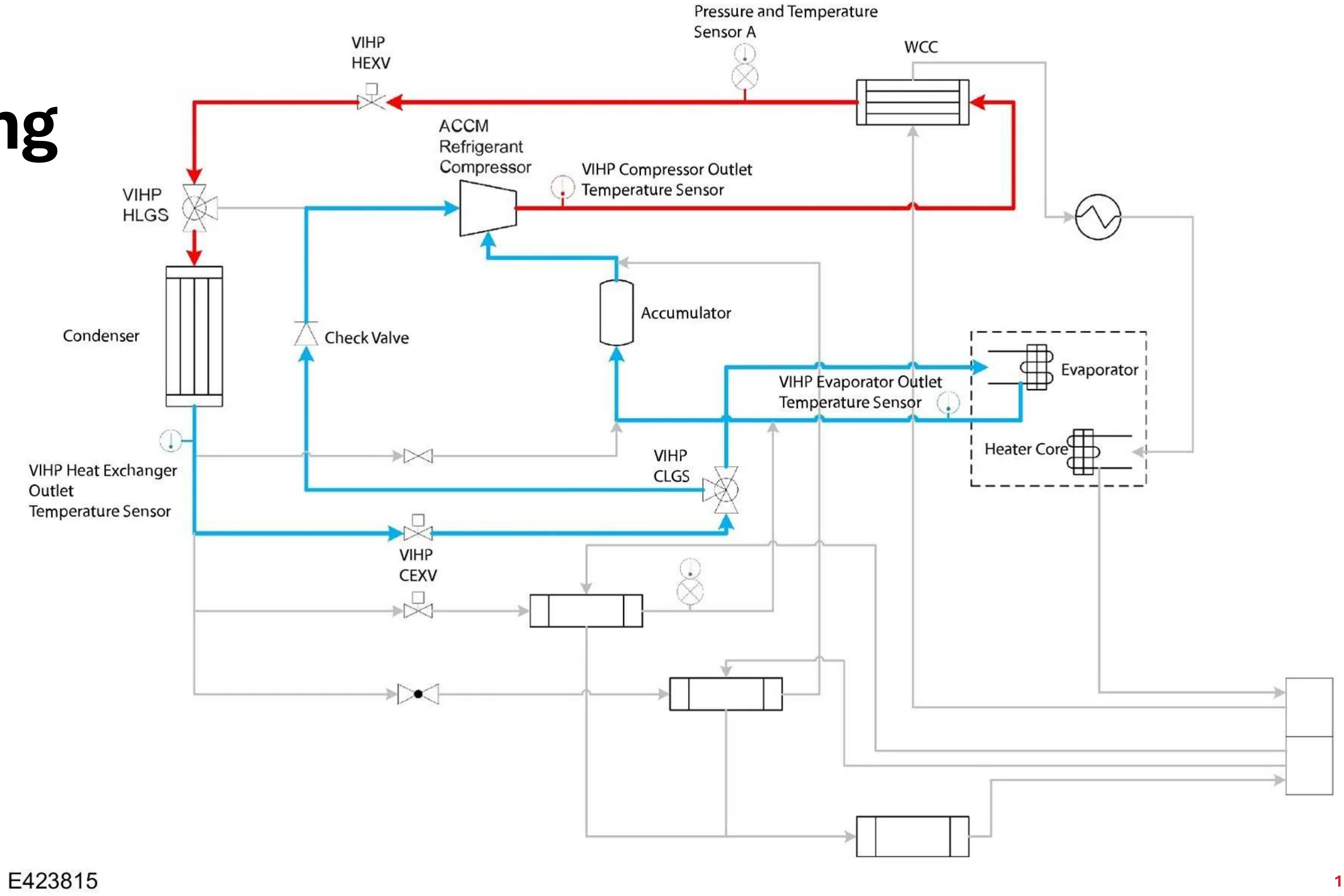




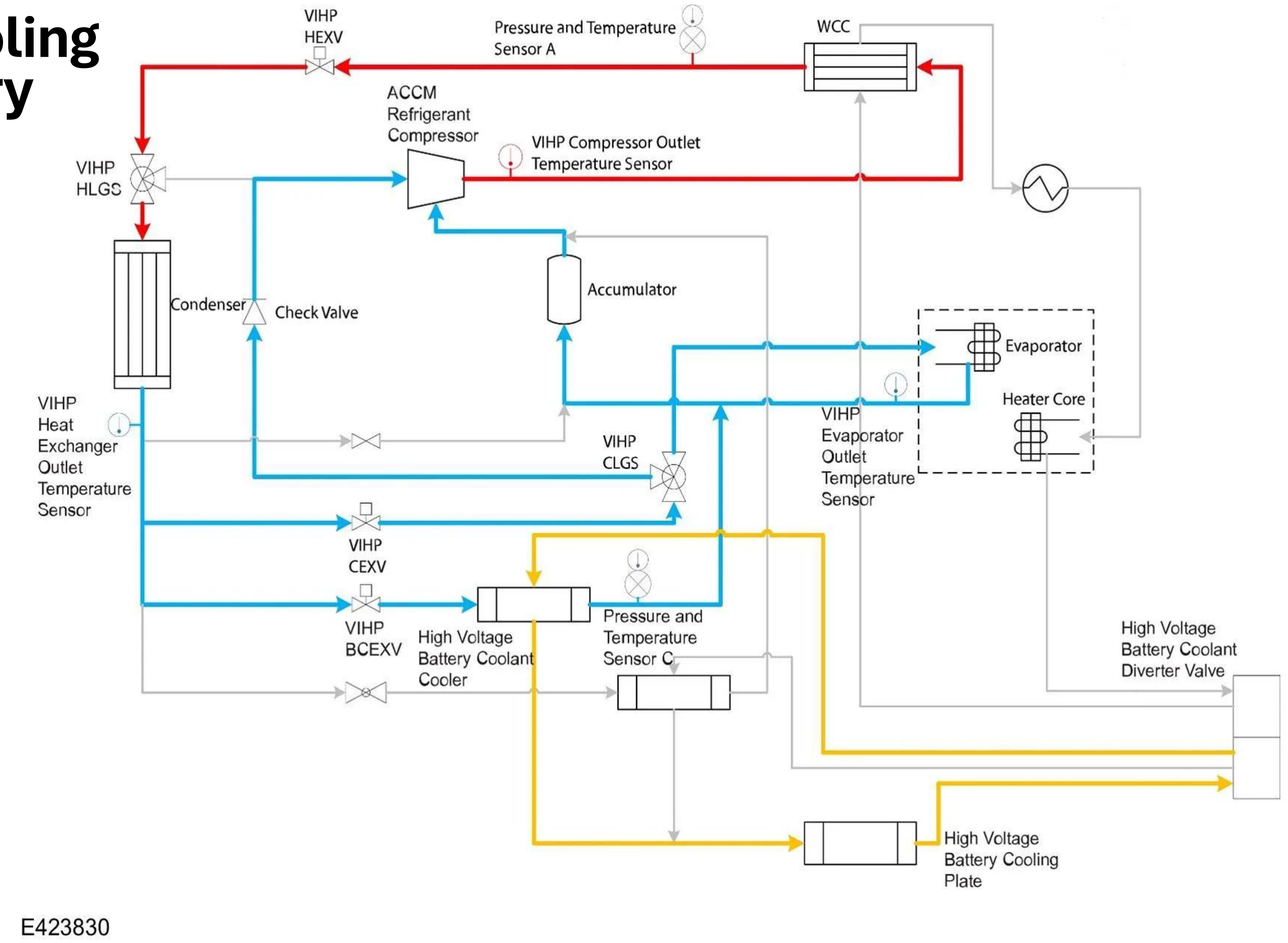
Ford Mustang Mach-E

- **The Mach-E was the first Ford EV to adopt an integrated heat pump**
- **Versions et PAC :**
  – Standard Range (75 kWh): heat pump optional 2021–2022; standard 2023+
  – Extended Range (99 kWh): heat pump standard on every version
- **Architecture thermique :**
  – 3 EXVs: cabin evaporator · battery chiller · outside exchanger
  – 5.4 kW PTC heater (backup below -15 °C)
- **Performances PAC Mach-E :**
  – Winter range gain vs PTC alone: +30–40 km at 0 °C
  – COP measured at 0 °C: 2.4–2.8
- **Known issues:**
  – Refrigerant leak at an O-ring fitting (battery chiller side)
- **Mach-E specific procedures:**
  – Huile compresseur : Motorcraft POE YN50

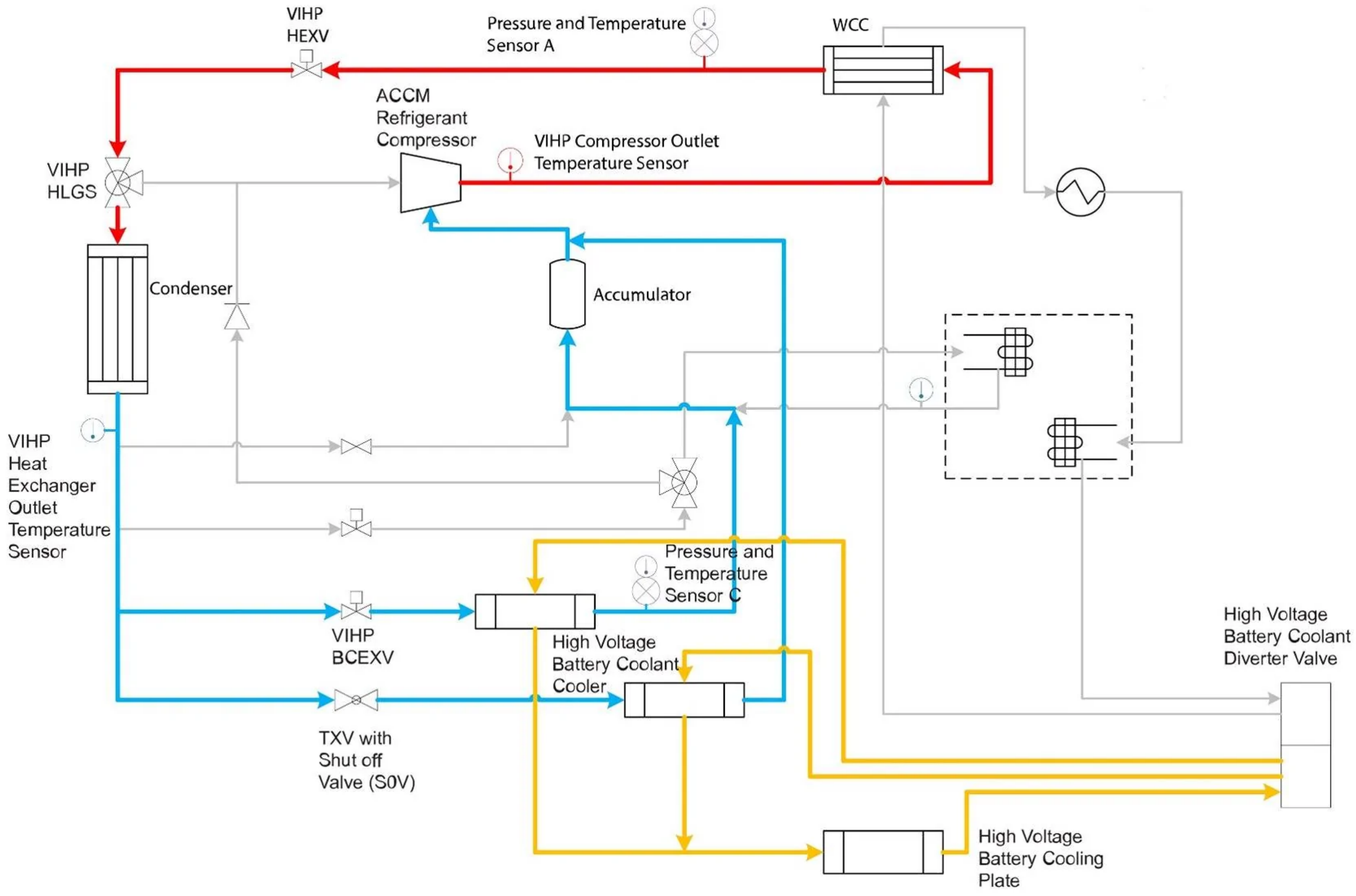
Cabin Cooling



Cabin Cooling and Battery Cooling



Battery Cooling



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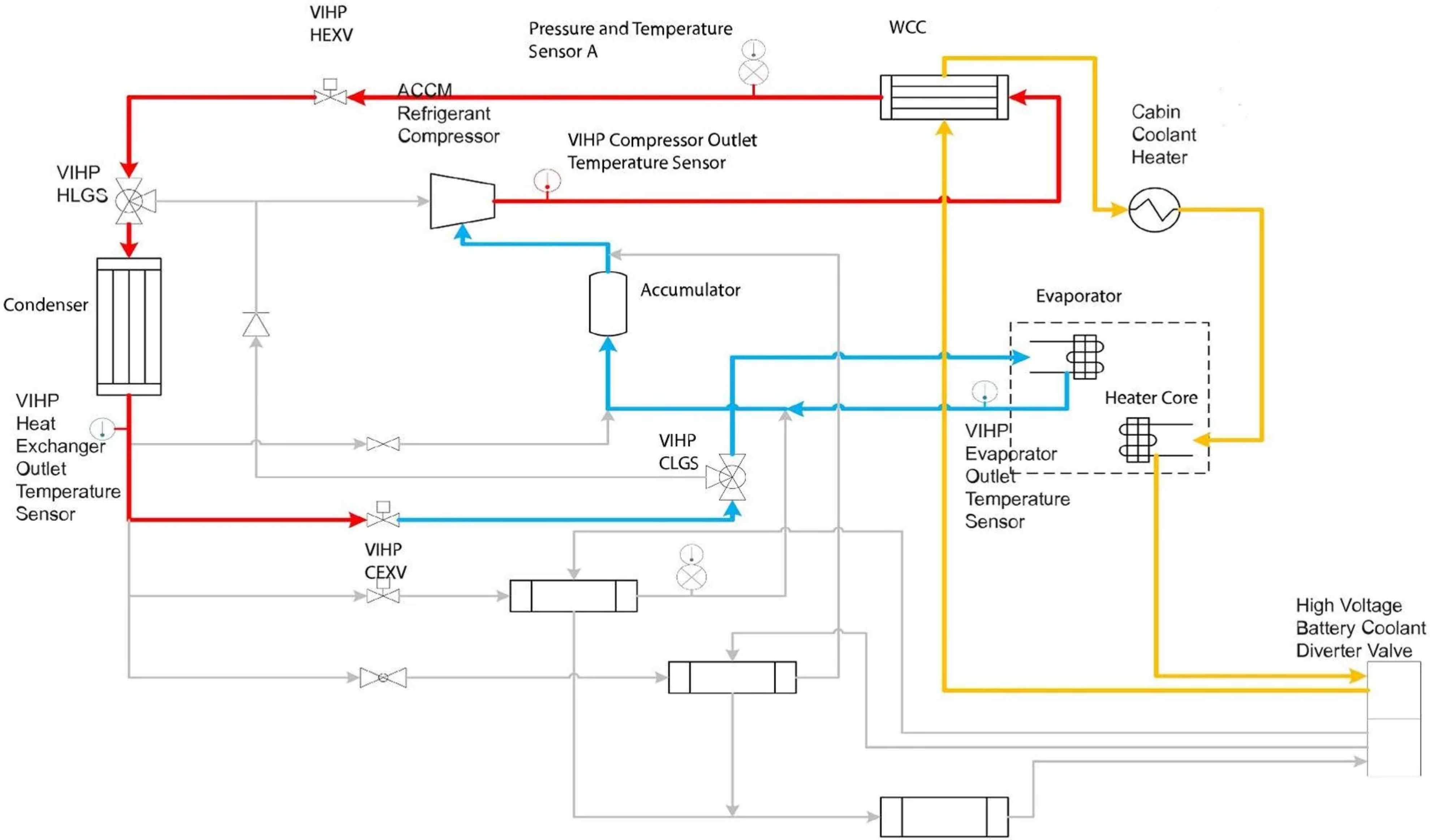
Cabin Reheat



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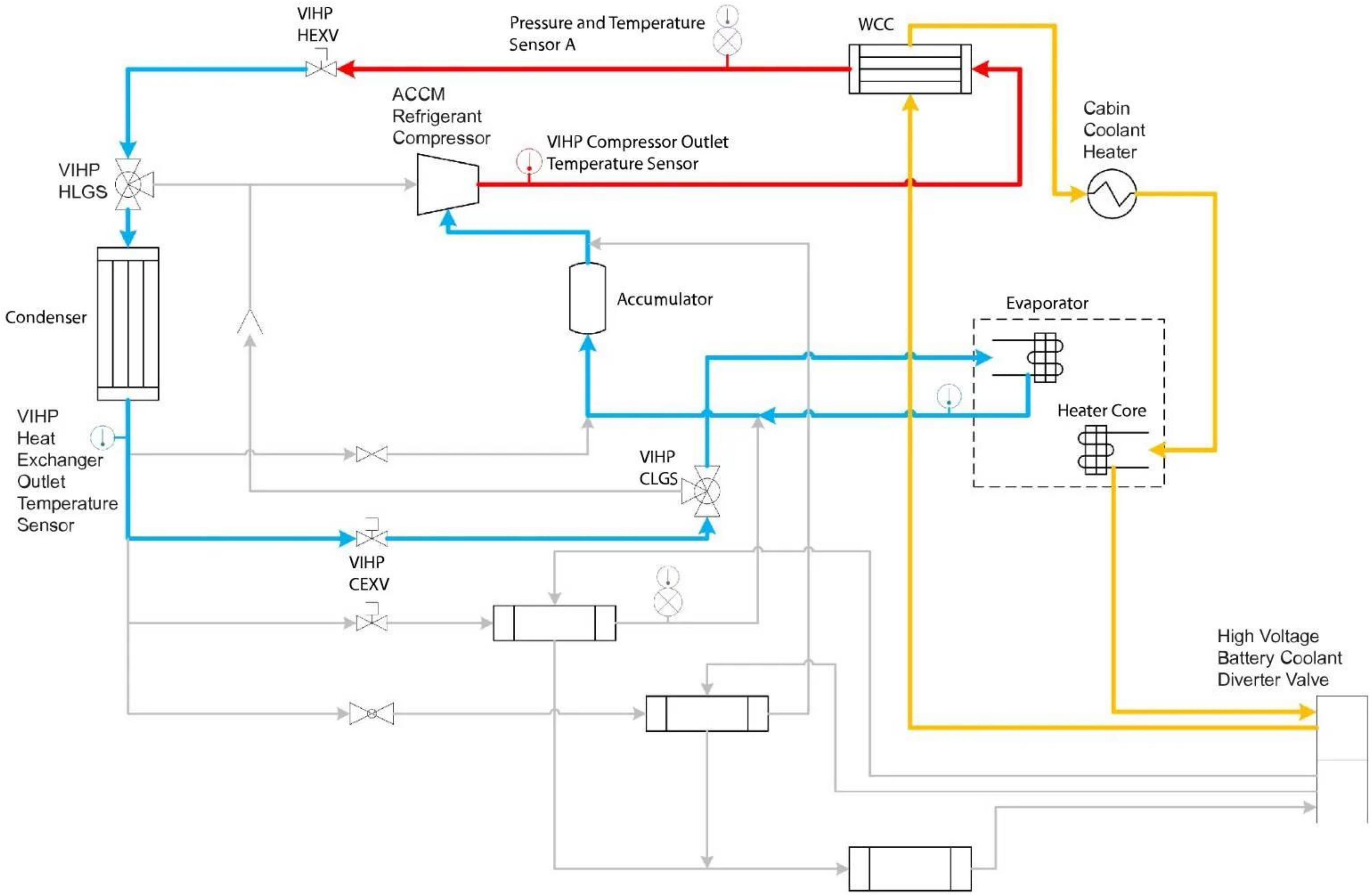
Cabin Dehumidification



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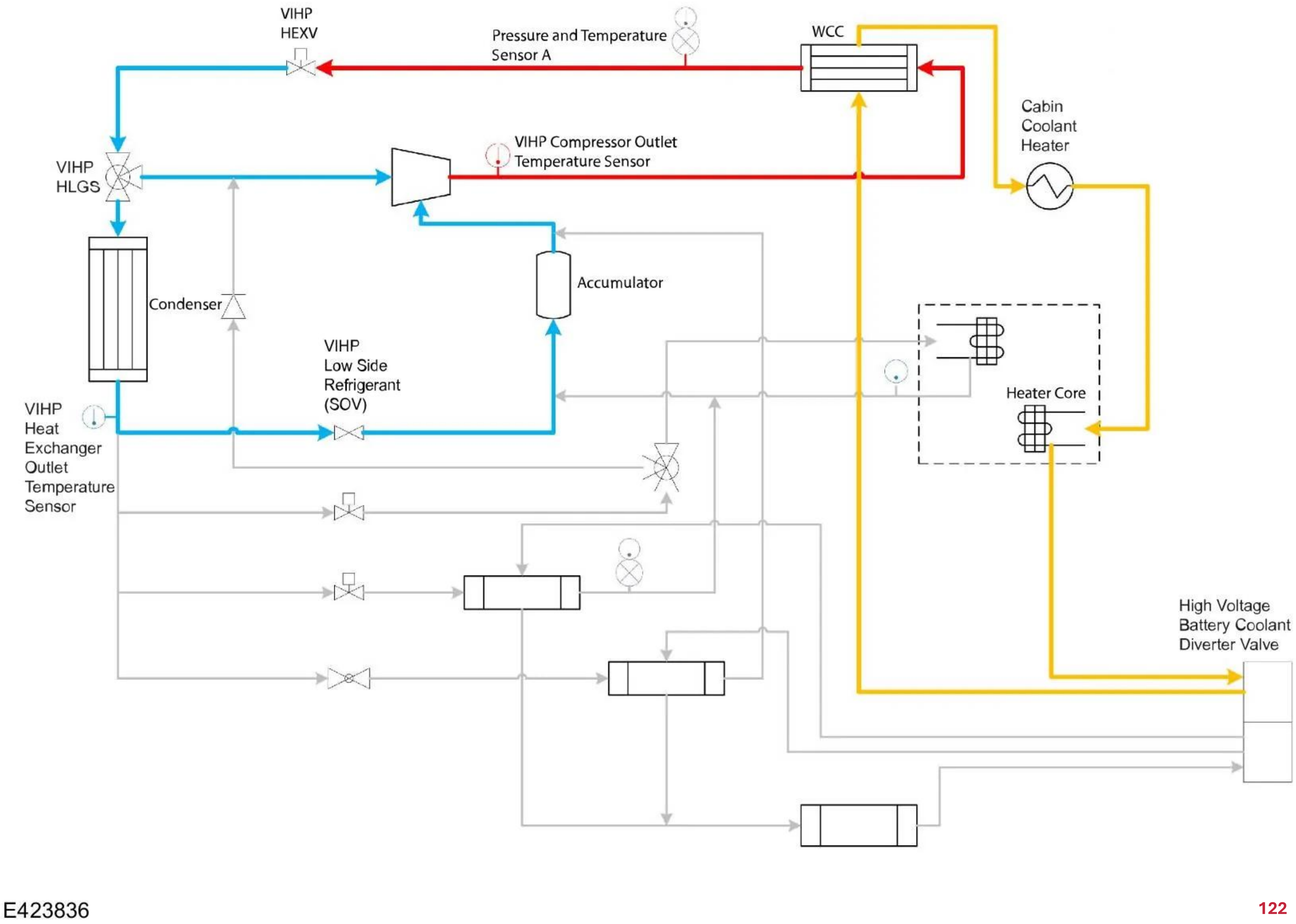
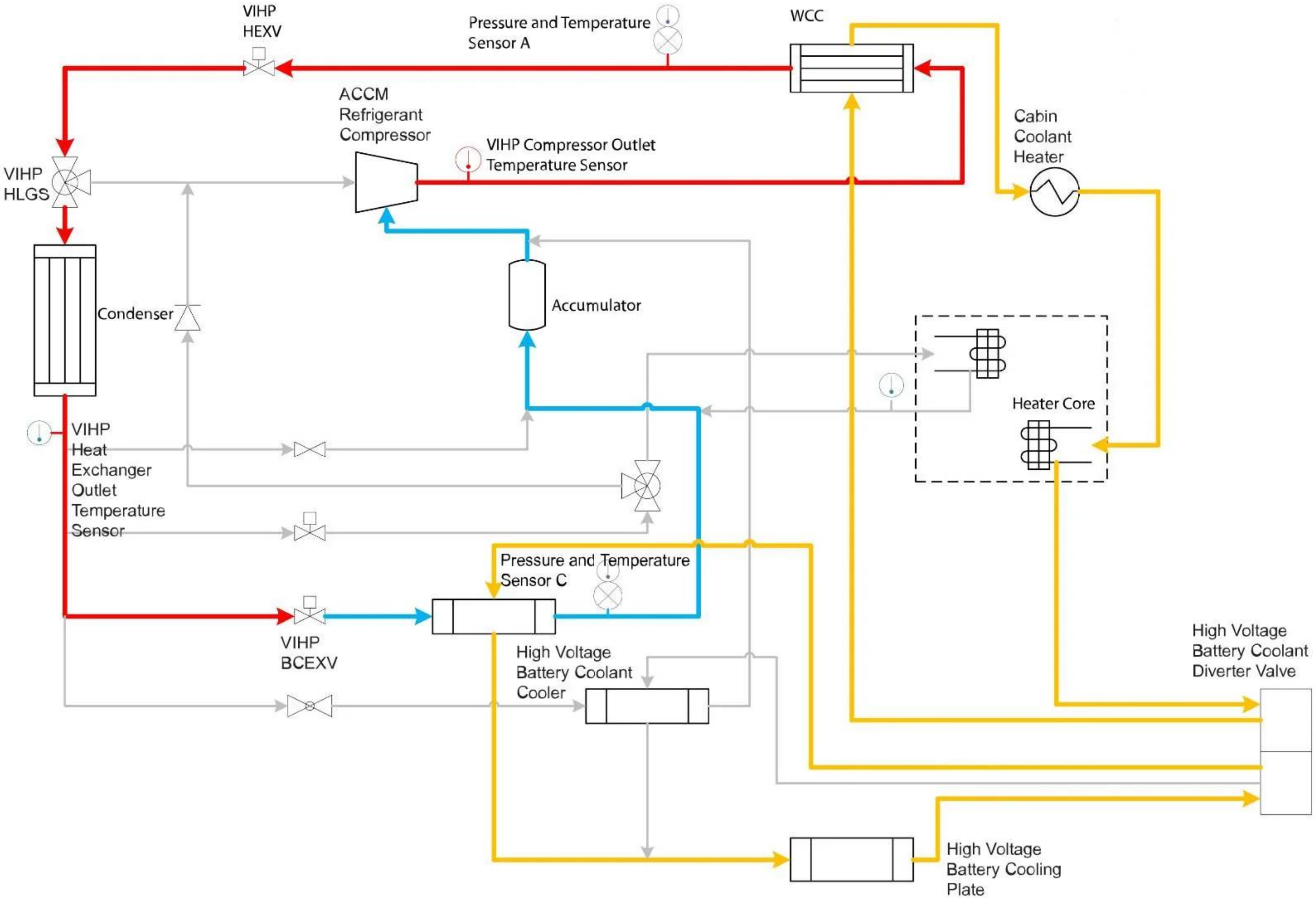
Cabin Heating



Cabin Heating Plus Battery Cooling



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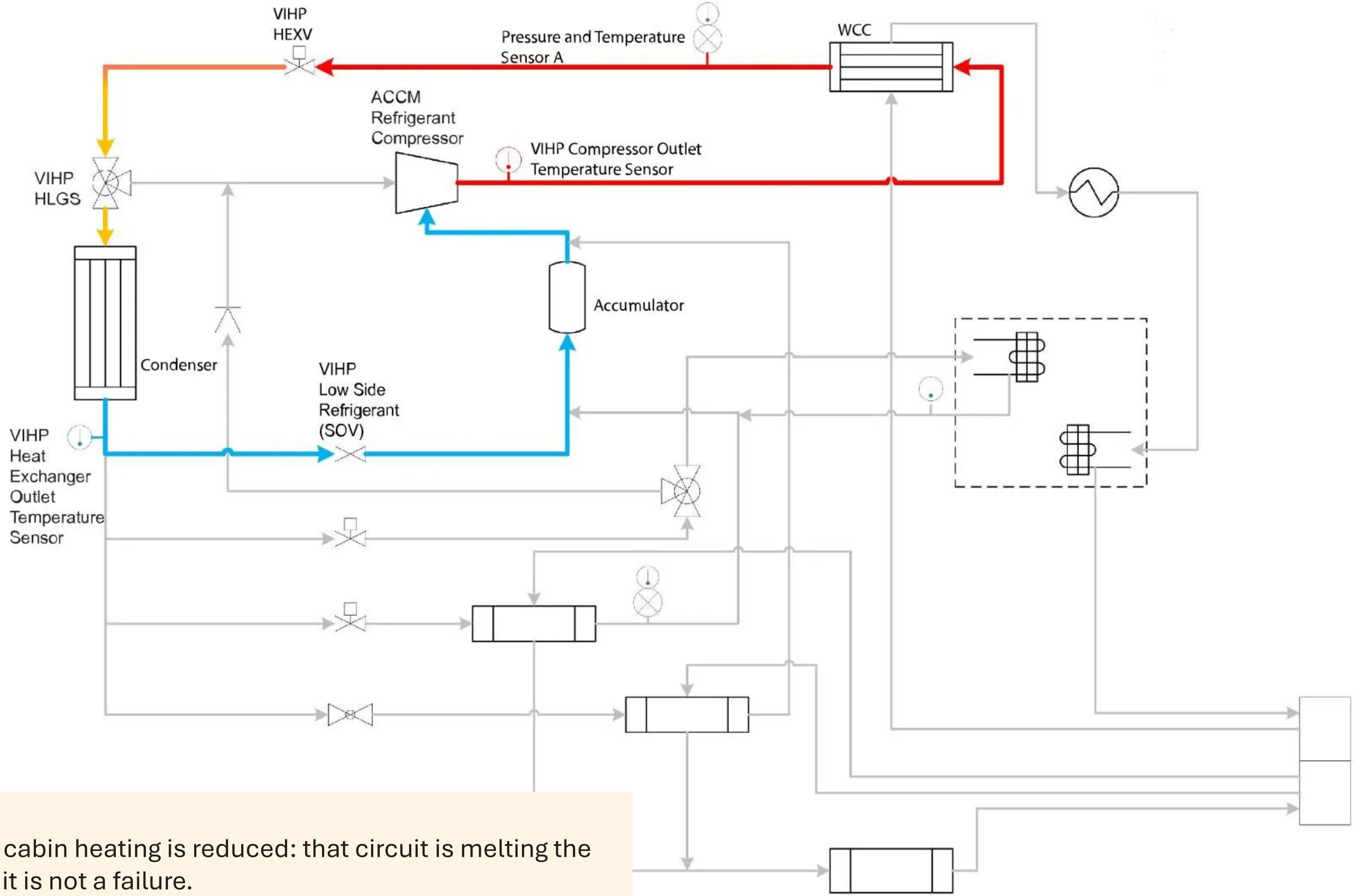
Deicing



⚠ **ATTENTION** During deicing, cabin heating is reduced: that circuit is melting the frost off the outside exchanger, it is not a failure.

Electric compressor replacement procedure

Complete steps — general application

- **Prerequisites:**
  - Full HV shutdown procedure completed
  - SAE J2843 certified recovery station available
  - New compressor + receiver-drier + O-rings + the oil quantity specified by the manufacturer
- **Step 1: Refrigerant recovery**
  - Connect the station to the HP and LP valves · Start recovery
  - Wait until pressure falls below 1 psi
- **Step 2: HV shutdown and compressor removal**
  - Disconnect the HV connector (ALWAYS with the full HV shutdown procedure)
  - Disconnect the refrigerant fittings (HP and LP) — cap them immediately
  - Remove the brackets and mounting bolts
- **Step 3: Flushing the system (if debris was found in the recovered refrigerant)**
  - Flush the lines with clean refrigerant or an approved flushing agent
  - DO NOT reassemble without flushing if contamination is confirmed
- **Step 4: Installing the new compressor**
  - Add the specified quantity of oil (PAG yf or POE depending on the manufacturer)
  - Install new O-rings lubricated with the correct oil
  - Replace the receiver-drier · Reconnect the lines · Plug in the HV connector
- **Step 5: Evacuation + recharge + break-in procedure (if required)**

System flushing and evacuation procedure

Critical steps to make the components last

▶ **When should the system be flushed?**

– Compressor replaced after a mechanical failure (debris in the refrigerant)
– Refrigerant contamination found (wrong oil, excessive moisture)
– After a collision that ruptured lines

▶ **Flushing procedure:**

– Approved flushing agents: virgin R-1234yf refrigerant (the cleanest method but $$$) or a dedicated A/C flushing agent
– DO NOT use nitrogen alone to flush the exchangers
– Flush each circuit separately; collect the flush in a clean container
– Inspect the flushing fluid: any trace of debris → flush again

▶ **Evacuation procedure (pulling a vacuum):**

– Connect the vacuum pump to the recovery/recharge station
– Minimum duration: 30 minutes
– Target vacuum level: < 500 microns of mercury (< 500 μmHg / < 0.67 mbar)
– Use a digital vacuum gauge (more accurate than an analog gauge)

▶ **Vacuum hold test:**

– After 30 min of evacuation: stop the pump, close the valves, wait 15 min
– If pressure rises by more than 500 μmHg → a leak in the circuit or residual moisture
– If it rises quickly (> 1,000 μmHg in 5 min) → active leak → find it and repair it
– If stable → go ahead with the recharge

Compressor oil types and quantities

The right type, the right amount — no compromise

▶ Why is compressor oil so critical?

– It lubricates the scroll mechanism and the compressor seals
– It circulates through the whole refrigerant circuit (mixed with the refrigerant)
– Wrong type or wrong amount = premature wear or compressor damage

▶ Oil types and compatibility (NA market):

– PAG 46 / PAG 100 / PAG 150: for R-134a systems
– PAG 46yf / PAG 100yf: SPECIFIC to R-1234yf (do not substitute standard PAG)
– POE 68: polyester oil; compatible with R-1234yf on some models (check the TSB)
– POE 100: some high-pressure compressors
– PVE: some high-pressure compressors

⚠ **WARNING** Never mix PAG and POE: the mixture forms a sticky gel that plugs the EXVs and damages the system. The oil is highly hygroscopic: keep the containers closed, never leave them open.

Common repairs – Refrigerant leaks

The #1 cause of heat pump failure

▶ Why are leaks so common in Quebec?

- Poor road conditions (gravel, ruts) → vibration and chafing on the lines
- Road de-icing salt (Ontario, Quebec, the American Midwest) → corrosion of aluminum fittings
- Frequent collisions (parking lot bumps, minor front impacts) → damaged refrigerant lines

▶ Common leak locations:

- Quick-connect fittings: the most frequent
- Compressor body seal
- Flexible lines (abrasion on the brackets)
- Micro-cracks in rigid aluminum lines (road salt corrosion)

▶ Detection procedure:

- Step 1: Visual (oil traces) ·
- Step 2: Electronic detector
- Step 3: UV dye if the leak has not been located
- Step 4: Nitrogen pressure test

▶ Repair:

- O-rings: replace ALL the O-rings in the area worked on (not just the one that leaked)
- Use R-1234yf certified O-rings
- Damaged lines: replace the complete part
- After the repair: leak test + evacuation + recharge

Quality control after the repair

Final checks before handing the vehicle back

▶ After any repair on the refrigerant circuit:
– Reconnect the HV system (MSM service connector)

▶ Checks to perform:
– ✓ HP and LP pressures within normal ranges (compare with the P-T chart)
– ✓ Compressor speed in the expected range for the demand
– ✓ Discharge air temperature in heat pump heating: > 40 °C at 0 °C ambient
– ✓ Discharge air temperature in A/C: < 15 °C at 25 °C ambient
– ✓ No new DTC after the road test
– ✓ No refrigerant leak detected (re-check with the electronic detector)
– ✓ Defrost cycle works

▶ Documenting the repair:
– Record the HP/LP pressures measured after the recharge
– Quantity of refrigerant recovered + quantity recharged
– Type and quantity of oil added
– DTCs present at diagnosis + codes cleared
– Technician certification (required for the file)

▶ Talking to the customer:
– Explain the symptoms, the cause and the repair performed
– Tell them about winter preconditioning to maximize range

Condensers

To check a condenser, we can measure the inlet temperature against the outlet temperature:

Spec: the difference must be between 20 °F and 50 °F.

Condenser inlet (vapour)

Condenser outlet (liquid)

Reading the results (sensible heat)

If below 20 °F

1. System overcharged
2. Faulty condenser
3. Poor air flow across the condenser
4. Contaminated refrigerant
5. Faulty compressor
6. Air in the system

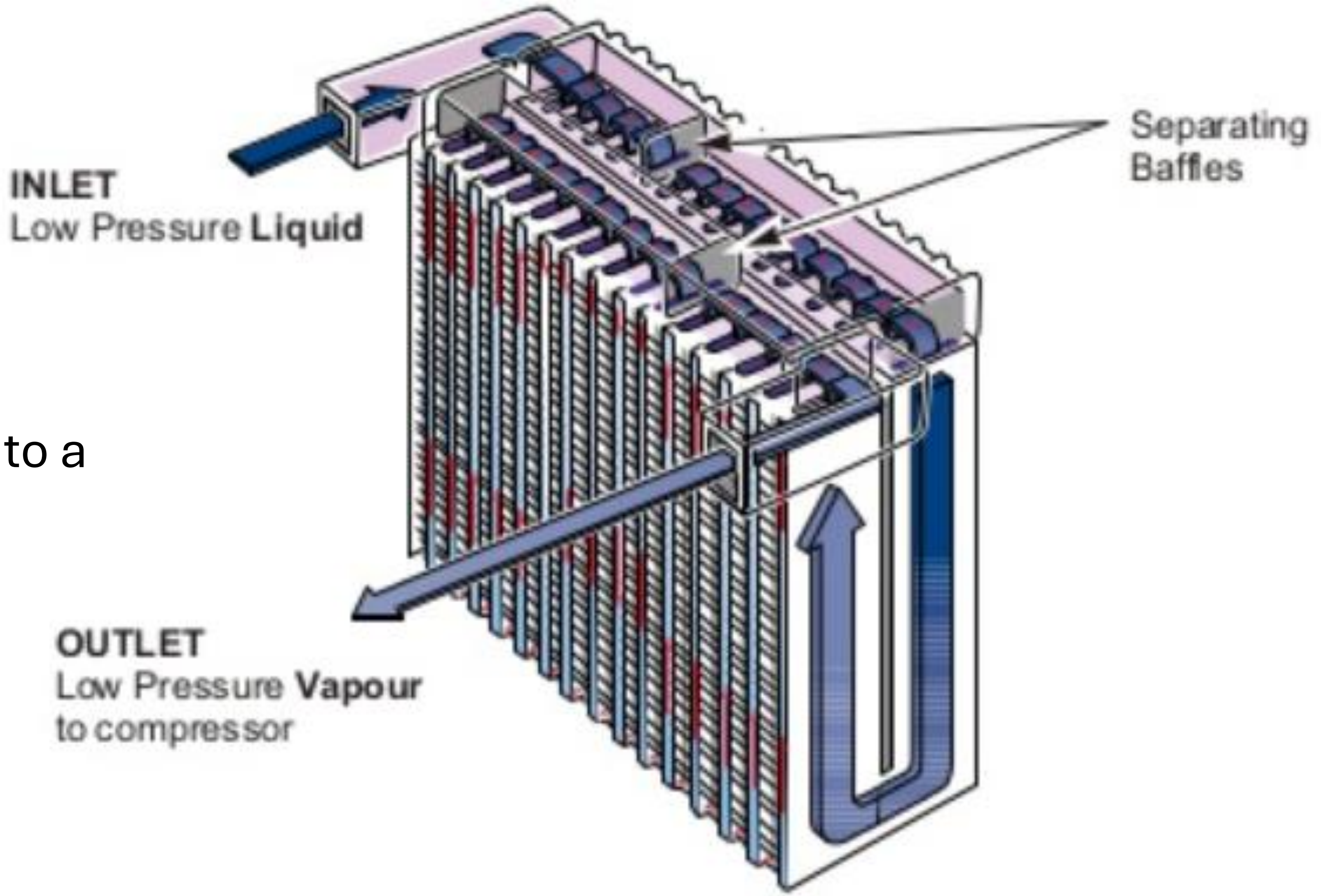
If above 50 °F

1. System undercharged
2. Blocked condenser or other restriction

Evaporator (with expansion valve)

For superheat:
(LATENT HEAT)

- Performance test;
- Measure the evaporator outlet temperature;
- Convert the low-side pressure to a temperature;



SPEC: EVAP OUTLET – LP (TEMP) = 4 °F TO 16 °F

Reading the results

If the difference is below 4 °F:

- Expansion valve stuck open
- System overcharged

If the difference is above 16 °F:

- Expansion valve stuck closed
- System undercharged

Gauge diagnostics: the quick reference

Convert the pressure you read into a temperature (P-T chart), then compare it with the temperatures measured on the lines

Check	Calculation	Target	Out of range
Subcooling	Condensing temp (converted HP) – temp at the condenser outlet	5.5 to 14 °C (10 to 25 °F)	Low: low refrigerant charge. High: overcharge; restriction between the condenser and the expansion valve; expansion valve stuck closed
Condenser efficiency (TD)	Condensing temp – ambient temp	Gas: < 22 °C (40 °F) EV: < 17 °C (30 °F)	High: ineffective condenser fan; fouled or corroded fins; air deflector missing or damaged
Compression ratio	(HP + 14.7) ÷ (LP + 14.7), in absolute psi	< 7	High: inefficient compression or a restriction in the circuit

▶ **Measuring properly:**

– Clean probe, placed as close as possible to the exchanger

– Pressures stabilized before drawing conclusions (system at steady state)

The pair that sums it all up: subcooling and superheat

Worked example and combined reading

▶ Example: checking subcooling

– HP reading: 142 psi → condensing temperature: 42 °C (P-T chart)
– Probe at the condenser outlet: 34 °C
– Subcooling = 42 − 34 = 8 °C → within range (5.5 to 14 °C)
– Conclusion: the refrigerant leaves the condenser fully liquid, the charge is good

▶ Completing the picture:

– The efficiency test built into the scan tool: a good starting point before taking anything apart
– Condenser test (20–50 °F) and evaporator test (superheat 4–16 °F) from the previous sections

🔑 **KEY POINT** SUBCOOLING answers "is there the right amount of refrigerant?"; SUPERHEAT answers "is the expansion valve metering correctly?". The two together, with stabilized pressures, give a reliable diagnosis without taking anything apart.

Efficiency test

- An efficiency test is often available on the scan tool for modern air conditioning systems
- This check tells you whether the air conditioning system is running with the recommended refrigerant charge

Example of a vehicle with an A/C-cooled high-voltage battery

High-voltage battery unit with heat exchanger

The high-voltage battery unit is cooled by refrigerant. For this reason the air conditioning refrigerant circuit is extended to include the high-voltage battery unit. A heat exchanger is located in the high-voltage battery unit and is also connected to the refrigerant circuit of the Air conditioning.

Subaru and Toyota

High-voltage compressor failures where debris can enter the high-voltage battery and plug the heat exchanger inside the battery

Questions & Comments

Thank you all for taking part!

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